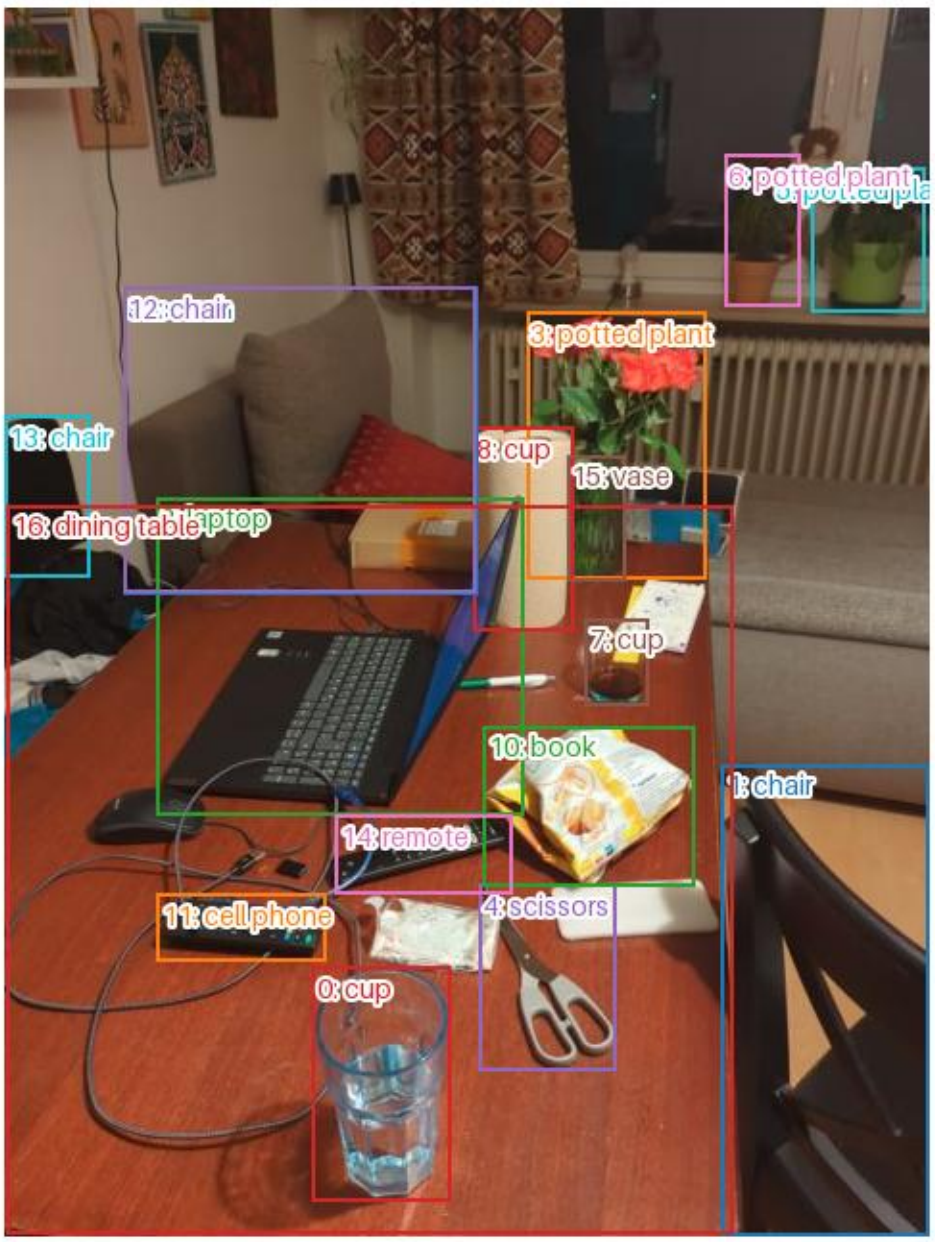
# bench_photos/blocked_1.jpg -- Page 1 / 61

Frame: 482x640 | Detections: 17 (M2 boxes: 17) | M2 2x identical: True




Shown M2 text (relate.context):

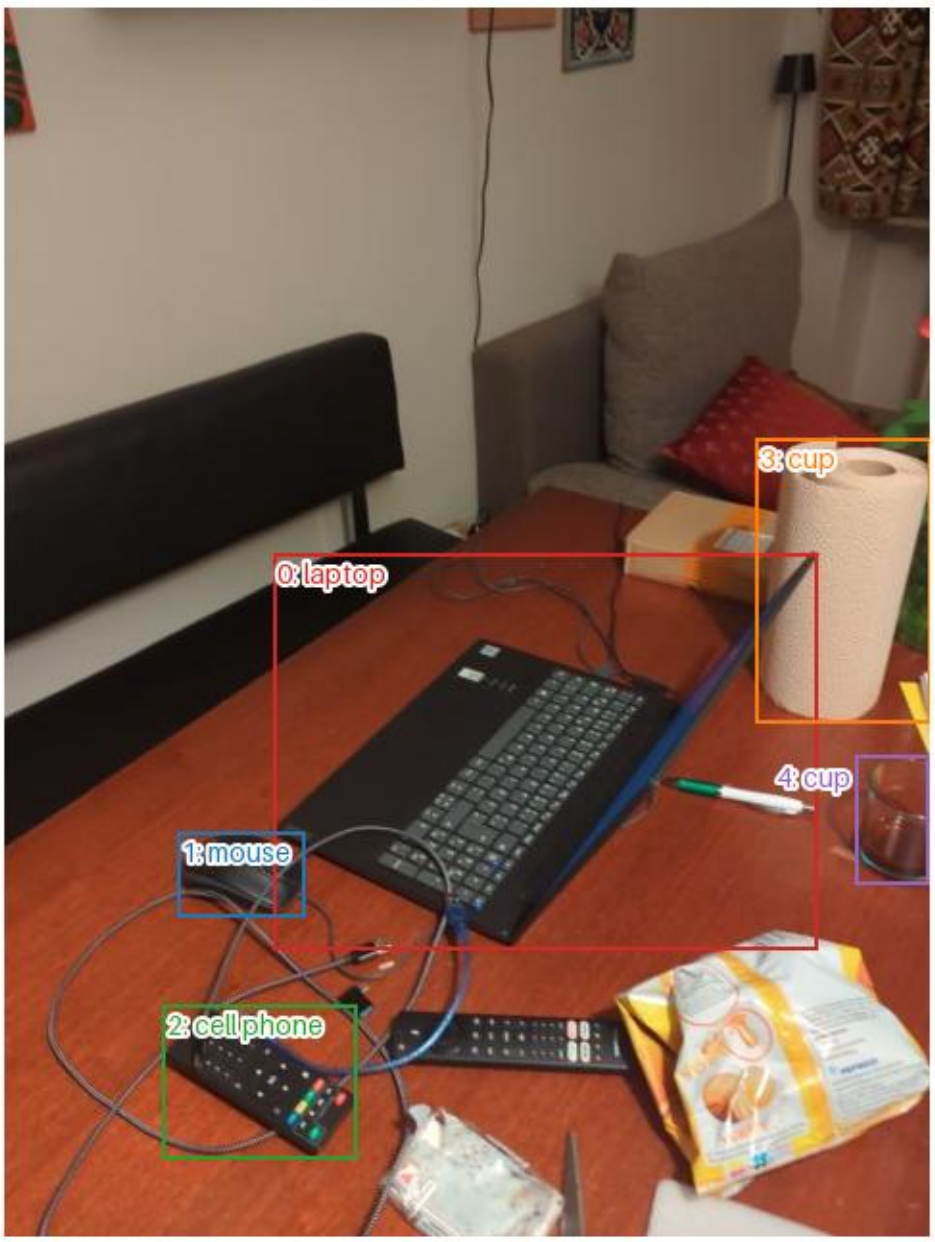
M2 relations:
- potted plant [3] contained in vase [15].
- laptop [2] resting on dining table [16].
- scissors [4] resting on dining table [16].
- remote [14] resting on dining table [16].
- cup [0] resting on dining table [16].

Raw top-10 M2 rows (score / threshold / pass):

1. potted plant [3] --(contained in)--> vase [15] score=0.7035 thr=0.5697 pass=True
2. laptop [2] --(resting on)--> dining table [16] score=0.6914 thr=0.5001 pass=True
3. scissors [4] --(resting on)--> dining table [16] score=0.6911 thr=0.5001 pass=True
4. remote [14] --(resting on)--> dining table [16] score=0.6790 thr=0.5001 pass=True
5. cup [0] --(resting on)--> dining table [16] score=0.6550 thr=0.5001 pass=True
6. book [10] --(resting on)--> dining table [16] score=0.6543 thr=0.5001 pass=True
7. cup [7] --(resting on)--> dining table [16] score=0.6418 thr=0.5001 pass=True
8. cup [8] --(resting on)--> dining table [16] score=0.6393 thr=0.5001 pass=True
9. cell phone [11] --(resting on)--> dining table [16] score=0.5965 thr=0.5001 pass=True
10. chair [1] --(beside)--> dining table [16] score=0.5729 thr=0.5001 pass=True

# bench_photos/blocked_2.jpg -- Page 2 / 61

Frame: 482x640 | Detections: 5 (M2 boxes: 5) | M2 2x identical: True




Shown M2 text (relate.context):

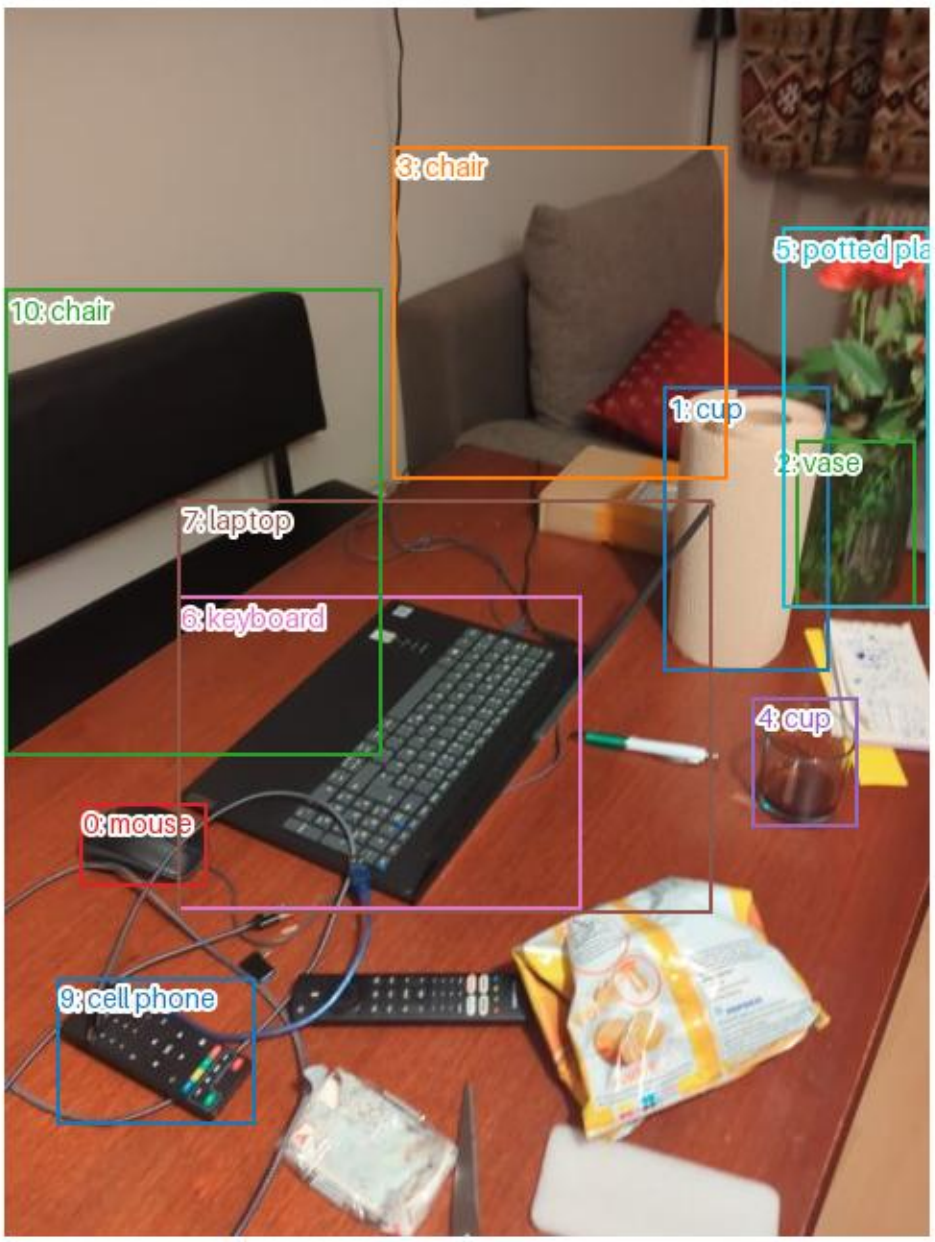
M2 relations:
- laptop [0] behind cell phone [2].
- cup [3] above cup [4].
- cell phone [2] in front of laptop [0].
- laptop [0] to the right of mouse [1].

Raw top-10 M2 rows (score / threshold / pass):

1. cup [3] --(beside)--> laptop [0] score=0.4261 thr=0.5001 pass=False
2. laptop [0] --(behind)--> cell phone [2] score=0.4144 thr=0.3004 pass=True
3. laptop [0] --(resting on)--> cup [3] score=0.3995 thr=0.5001 pass=False
4. cup [4] --(beside)--> cup [3] score=0.3889 thr=0.5001 pass=False
5. cup [3] --(above)--> cup [4] score=0.3792 thr=0.3064 pass=True
6. cup [4] --(beside)--> laptop [0] score=0.3678 thr=0.5001 pass=False
7. mouse [1] --(beside)--> laptop [0] score=0.3220 thr=0.5001 pass=False
8. cell phone [2] --(in front of)--> laptop [0] score=0.3196 thr=0.2748 pass=True
9. mouse [1] --(above)--> cell phone [2] score=0.2921 thr=0.3064 pass=False
10. laptop [0] --(to the right of)--> mouse [1] score=0.2878 thr=0.2703 pass=True

# bench_photos/blocked_3.jpg -- Page 3 / 61

Frame: 482x640 | Detections: 11 (M2 boxes: 11) | M2 2x identical: True




Shown M2 text (relate.context):

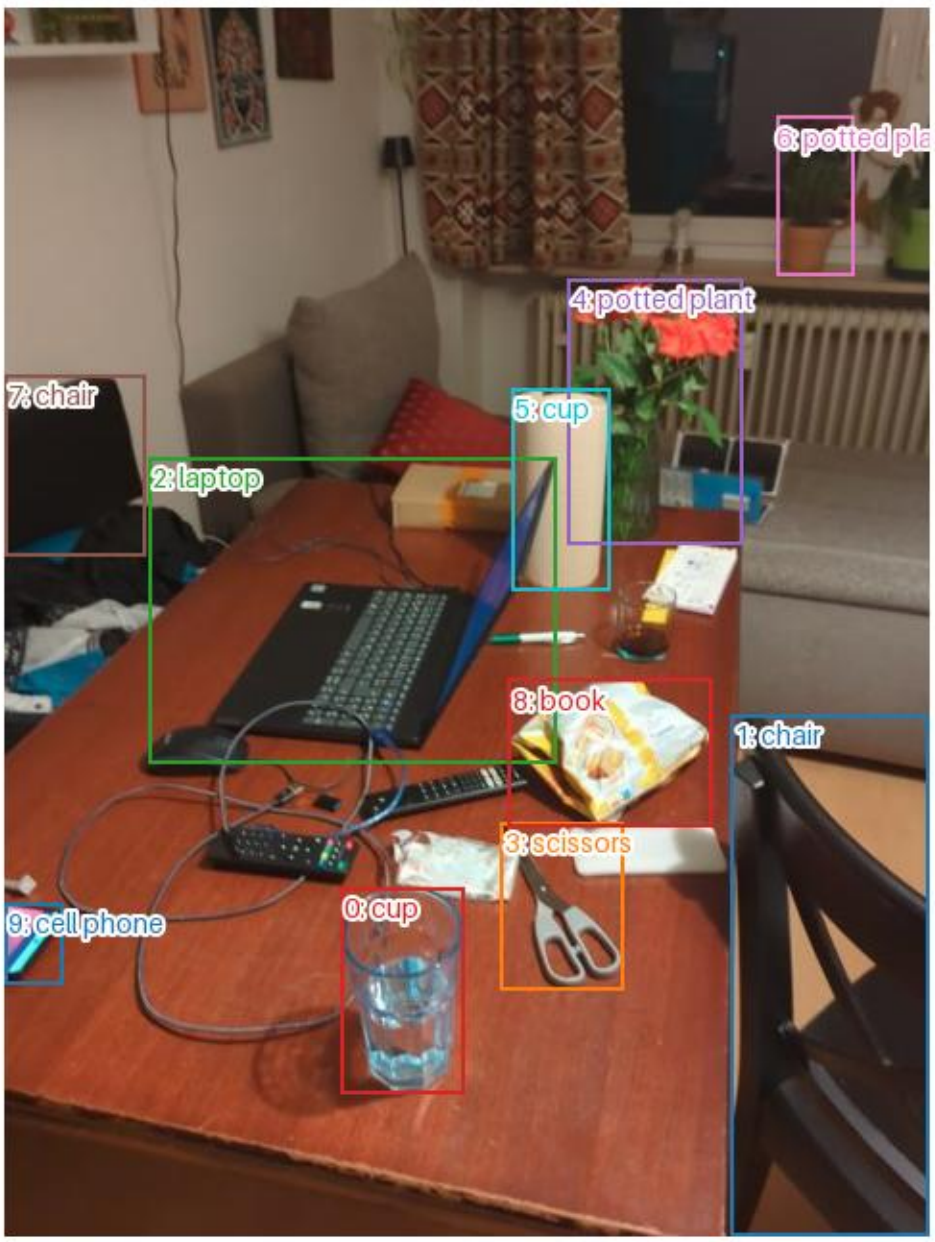
M2 relations:
- potted plant [5] contained in vase [2].
- vase [2] holding potted plant [5].
- keyboard [6] attached to laptop [7].
- laptop [7] resting on keyboard [6].

Raw top-10 M2 rows (score / threshold / pass):

1. potted plant [5] --(contained in)--> vase [2] score=0.7166 thr=0.5697 pass=True
2. vase [2] --(holding)--> potted plant [5] score=0.6109 thr=0.5001 pass=True
3. keyboard [6] --(attached to)--> laptop [7] score=0.5512 thr=0.4749 pass=True
4. laptop [7] --(resting on)--> keyboard [6] score=0.5080 thr=0.5001 pass=True
5. cup [1] --(resting on)--> vase [2] score=0.4418 thr=0.5001 pass=False
6. cup [1] --(holding)--> potted plant [5] score=0.3568 thr=0.5001 pass=False
7. cell phone [9] --(attached to)--> remote [8] score=0.3233 thr=0.4749 pass=False
8. cup [1] --(resting on)--> laptop [7] score=0.2989 thr=0.5001 pass=False
9. mouse [0] --(attached to)--> keyboard [6] score=0.2983 thr=0.4749 pass=False
10. remote [8] --(attached to)--> cell phone [9] score=0.2976 thr=0.4749 pass=False

bench_photos/blocked_4.jpg -- Page 4 / 61

Frame: 482x640 | Detections: 10 (M2 boxes: 10) | M2 2x identical: True




Shown M2 text (relate.context):

M2 relations:
- cup [0] in front of laptop [2].

Raw top-10 M2 rows (score / threshold / pass):

1. cup [5] --(holding)--> potted plant [4] score=0.4935 thr=0.5001 pass=False
2. scissors [3] --(resting on)--> book [8] score=0.4330 thr=0.5001 pass=False
3. potted plant [4] --(beside)--> cup [5] score=0.4066 thr=0.5001 pass=False
4. cup [5] --(resting on)--> laptop [2] score=0.3973 thr=0.5001 pass=False
5. laptop [2] --(resting on)--> book [8] score=0.3971 thr=0.5001 pass=False
6. book [8] --(resting on)--> laptop [2] score=0.3587 thr=0.5001 pass=False
7. cup [0] --(in front of)--> laptop [2] score=0.2911 thr=0.2748 pass=True
8. scissors [3] --(beside)--> cup [0] score=0.2747 thr=0.5001 pass=False
9. chair [1] --(surrounding)--> laptop [2] score=0.2720 thr=0.6228 pass=False
10. laptop [2] --(resting on)--> cup [5] score=0.2689 thr=0.5001 pass=False

# bench_photos/empty_1.jpg -- Page 5 / 61

Frame: 482x640 | Detections: 3 (M2 boxes: 3) | M2 2x identical: True



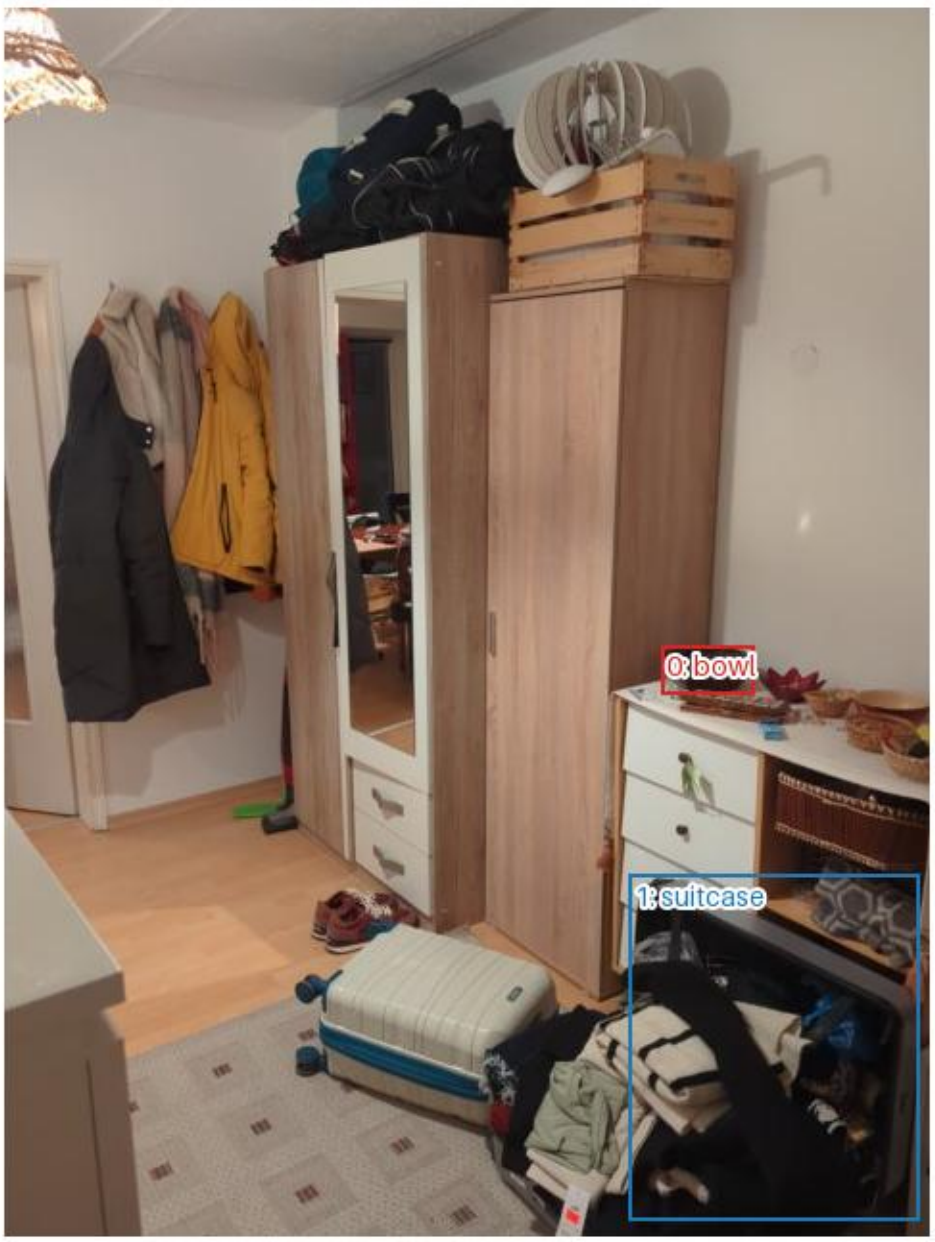

Shown M2 text (relate.context):

M2 relations:
- bottle [1] resting on bottle [2].
- bottle [1] to the left of toilet [0].
- bottle [2] to the left of toilet [0].
- toilet [0] to the right of bottle [2].
- toilet [0] to the right of bottle [1].

Raw top-10 M2 rows (score / threshold / pass):

1. bottle [1] --(resting on)--> bottle [2] score=0.6340 thr=0.5001 pass=True
2. bottle [2] --(contained in)--> bottle [1] score=0.4028 thr=0.5697 pass=False
3. bottle [1] --(to the left of)--> toilet [0] score=0.3909 thr=0.2703 pass=True
4. bottle [2] --(to the left of)--> toilet [0] score=0.3870 thr=0.2703 pass=True
5. toilet [0] --(to the right of)--> bottle [2] score=0.3749 thr=0.2703 pass=True
6. toilet [0] --(to the right of)--> bottle [1] score=0.3325 thr=0.2703 pass=True

Frame: 482x640 | Detections: 2 (M2 boxes: 2) | M2 2x identical: True




Shown M2 text (relate.context):

M2 relations:
- none above threshold.

Raw top-10 M2 rows (score / threshold / pass):

1. bowl [0] --(contained in)--> suitcase [1] score=0.2727 thr=0.5697 pass=False
2. suitcase [1] --(contained in)--> bowl [0] score=0.2683 thr=0.5697 pass=False

# bench_photos/empty_3.jpg -- Page 7 / 61

Frame: 482x640 | Detections: 5 (M2 boxes: 5) | M2 2x identical: True



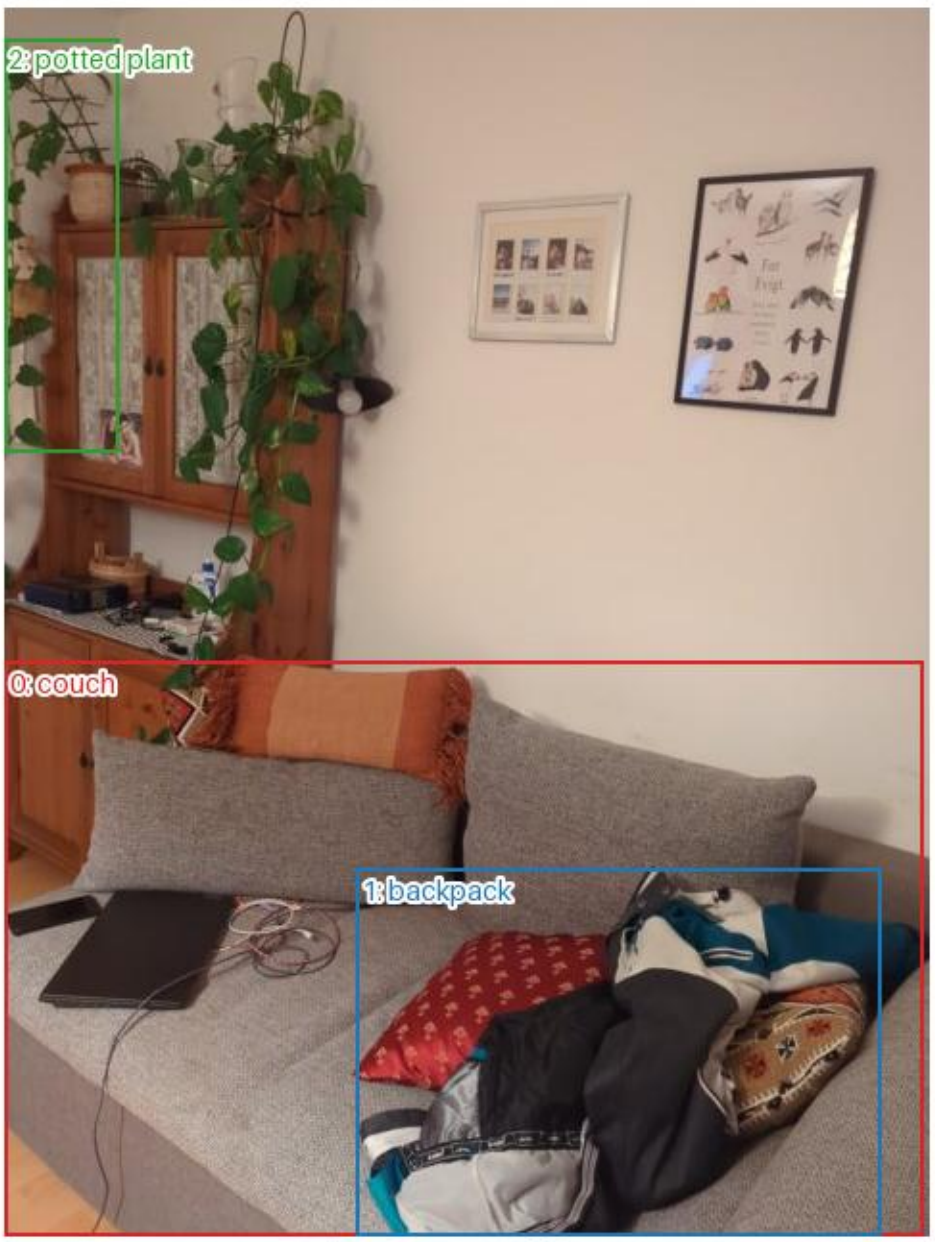

Shown M2 text (relate.context):

M2 relations:
- potted plant [1] behind oven [4].

Raw top-10 M2 rows (score / threshold / pass):

1. potted plant [2] --(resting on)--> potted plant [1] score=0.3794 thr=0.5001 pass=False
2. bottle [3] --(resting on)--> bottle [0] score=0.3249 thr=0.5001 pass=False
3. bottle [0] --(contained in)--> bottle [3] score=0.3230 thr=0.5697 pass=False
4. potted plant [1] --(beside)--> potted plant [2] score=0.3075 thr=0.5001 pass=False
5. potted plant [1] --(behind)--> oven [4] score=0.3008 thr=0.3004 pass=True
6. bottle [3] --(to the right of)--> potted plant [2] score=0.2112 thr=0.2703 pass=False
7. potted plant [2] --(to the left of)--> bottle [3] score=0.2107 thr=0.2703 pass=False
8. bottle [0] --(to the right of)--> potted plant [2] score=0.1747 thr=0.2703 pass=False
9. potted plant [2] --(behind)--> oven [4] score=0.1712 thr=0.3004 pass=False
10. potted plant [2] --(to the left of)--> bottle [0] score=0.1690 thr=0.2703 pass=False

Frame: 482x640 | Detections: 3 (M2 boxes: 3) | M2 2x identical: True



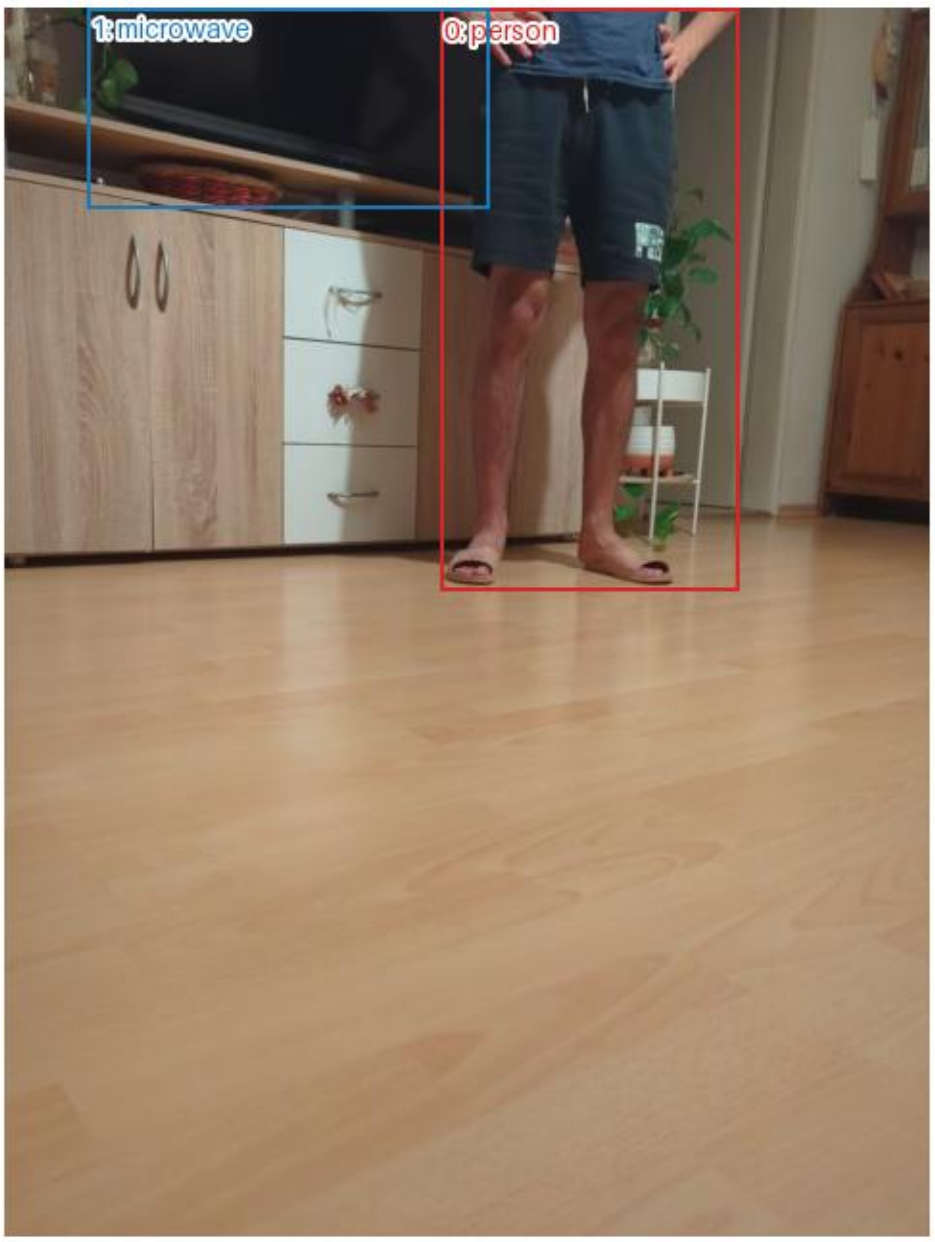

Shown M2 text (relate.context):

M2 relations:
- backpack [1] resting on couch [0].

Raw top-10 M2 rows (score / threshold / pass):

1. backpack [1] --(resting on)--> couch [0] score=0.7141 thr=0.5001 pass=True
2. couch [0] --(holding)--> backpack [1] score=0.4955 thr=0.5001 pass=False
3. potted plant [2] --(above)--> couch [0] score=0.2900 thr=0.3064 pass=False
4. couch [0] --(below)--> potted plant [2] score=0.2676 thr=0.3004 pass=False
5. potted plant [2] --(behind)--> backpack [1] score=0.2621 thr=0.3004 pass=False
6. backpack [1] --(in front of)--> potted plant [2] score=0.2229 thr=0.2748 pass=False

Frame: 482x640 | Detections: 2 (M2 boxes: 2) | M2 2x identical: True



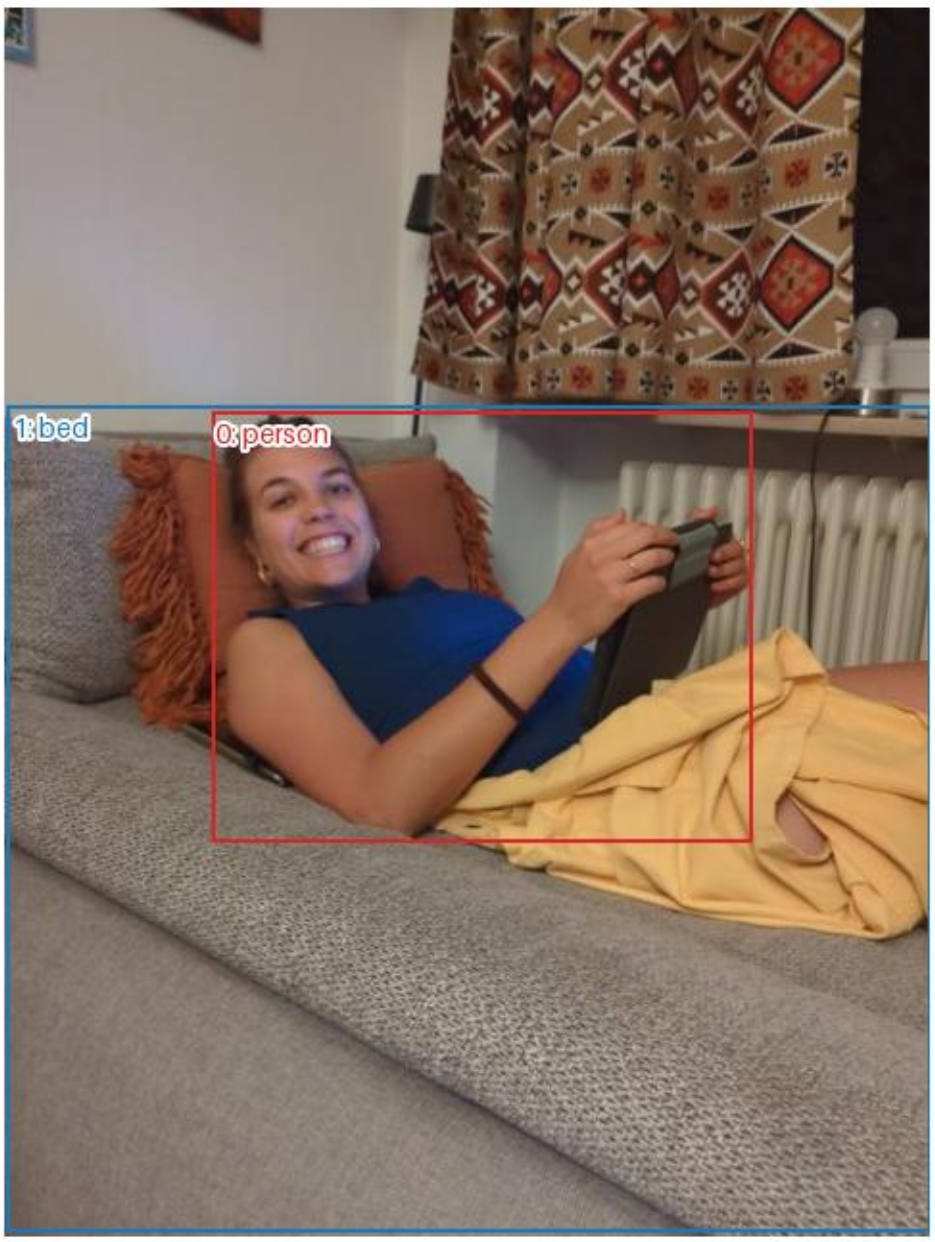

Shown M2 text (relate.context):

M2 relations:
- none above threshold.

Raw top-10 M2 rows (score / threshold / pass):

1. person [0] --(holding)--> microwave [1] score=0.3342 thr=0.5001 pass=False
2. microwave [1] --(behind)--> person [0] score=0.2027 thr=0.3004 pass=False

Frame: 482x640 | Detections: 2 (M2 boxes: 2) | M2 2x identical: True




Shown M2 text (relate.context):

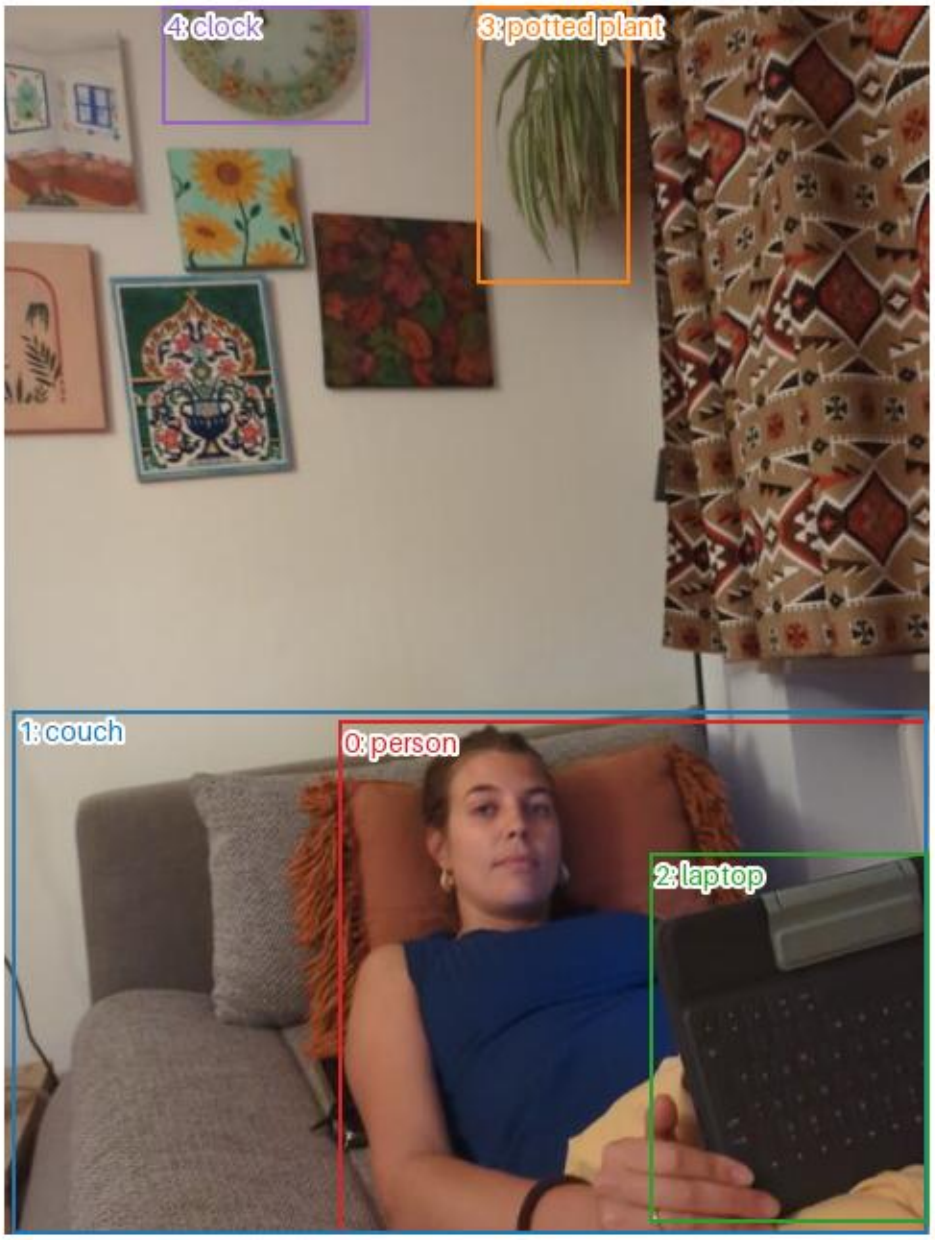
M2 relations:
- none above threshold.

Raw top-10 M2 rows (score / threshold / pass):

1. person [0] --(resting on)--> bed [1] score=0.4722 thr=0.5001 pass=False
2. bed [1] --(behind)--> person [0] score=0.0127 thr=0.3004 pass=False

bench_photos/person_facing_150cm.jpg -- Page 11 / 61

Frame: 482x640 | Detections: 5 (M2 boxes: 5) | M2 2x identical: True



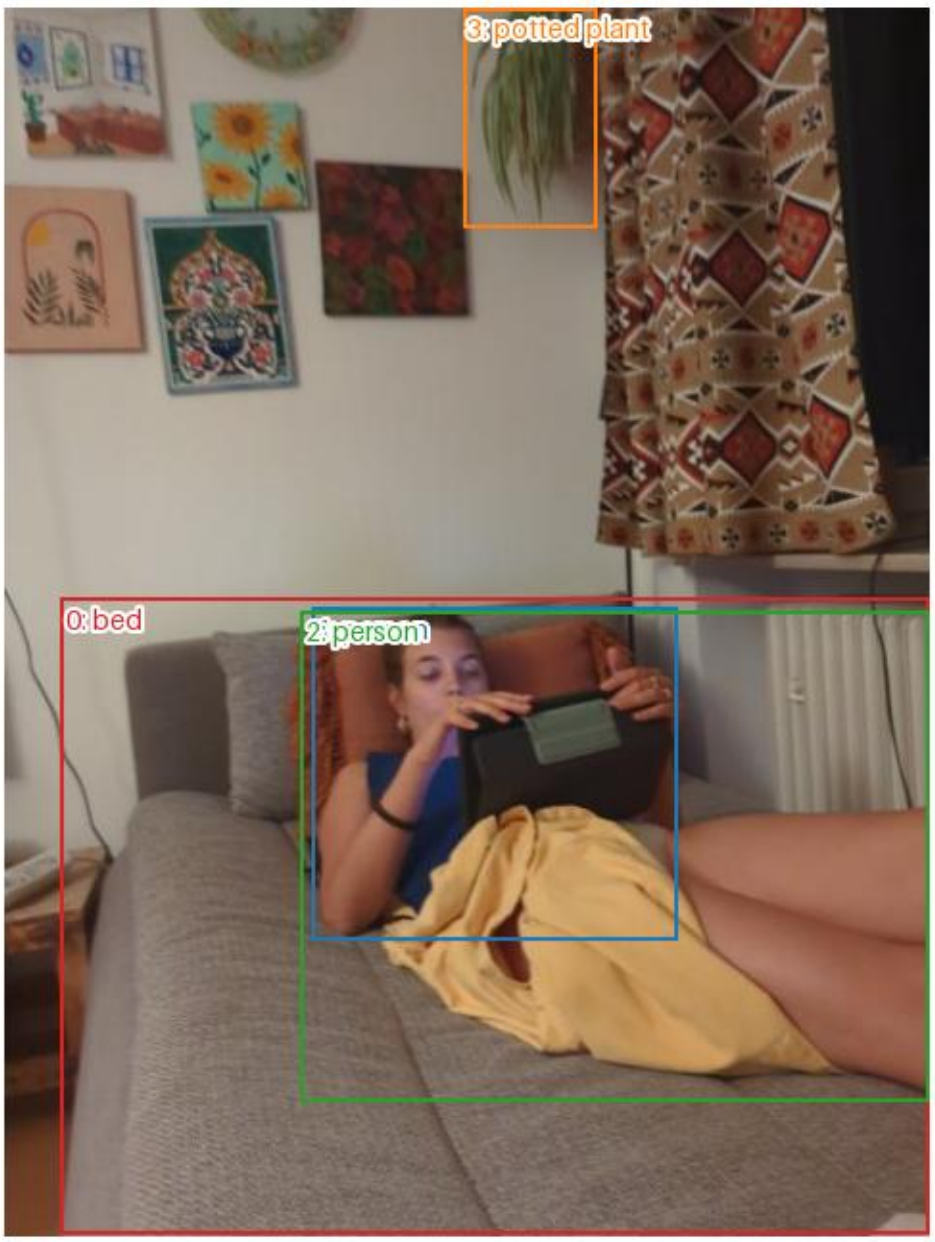

Shown M2 text (relate.context):

M2 relations:
- person [0] using laptop [2].
- person [0] sitting on couch [1].
- laptop [2] resting on person [0].
- couch [1] in front of laptop [2].
- potted plant [3] above couch [1].

Raw top-10 M2 rows (score / threshold / pass):

1. person [0] --(using)--> laptop [2] score=0.8066 thr=0.5697 pass=True
2. person [0] --(sitting on)--> couch [1] score=0.7072 thr=0.5001 pass=True
3. laptop [2] --(resting on)--> person [0] score=0.5591 thr=0.5001 pass=True
4. laptop [2] --(resting on)--> couch [1] score=0.4438 thr=0.5001 pass=False
5. potted plant [3] --(beside)--> clock [4] score=0.4046 thr=0.5001 pass=False
6. couch [1] --(in front of)--> laptop [2] score=0.3470 thr=0.2748 pass=True
7. potted plant [3] --(above)--> couch [1] score=0.3455 thr=0.3064 pass=True
8. clock [4] --(above)--> couch [1] score=0.2937 thr=0.3064 pass=False
9. couch [1] --(below)--> clock [4] score=0.2864 thr=0.3004 pass=False
10. clock [4] --(to the left of)--> potted plant [3] score=0.2728 thr=0.2703 pass=True

Frame: 482x640 | Detections: 4 (M2 boxes: 4) | M2 2x identical: True



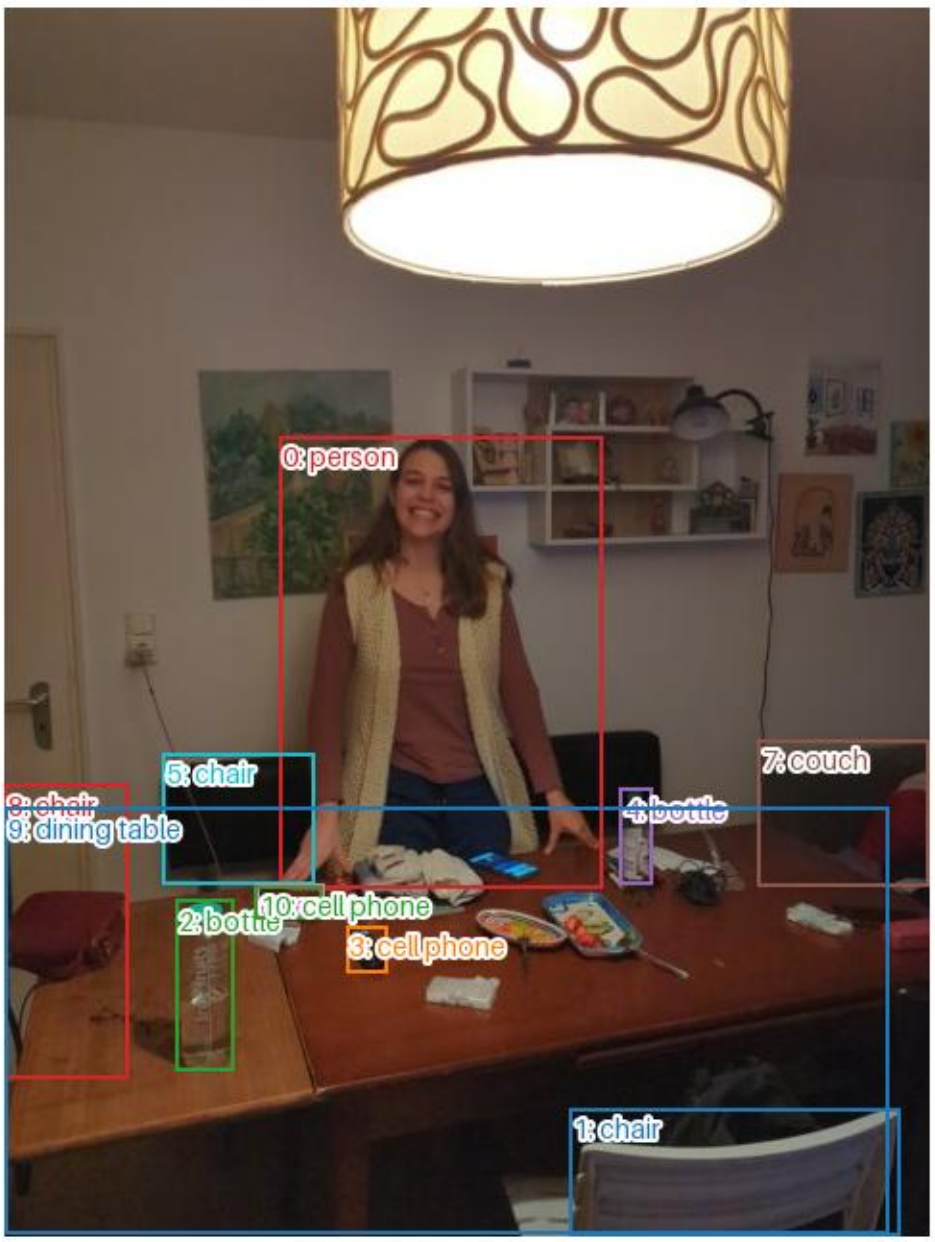

Shown M2 text (relate.context):

M2 relations:
- person [2] resting on bed [0].
- person [1] resting on bed [0].
- person [2] holding person [1].
- potted plant [3] above bed [0].

Raw top-10 M2 rows (score / threshold / pass):

1. person [2] --(resting on)--> bed [0] score=0.7460 thr=0.5001 pass=True
2. person [1] --(resting on)--> bed [0] score=0.7111 thr=0.5001 pass=True
3. person [2] --(holding)--> person [1] score=0.6441 thr=0.5001 pass=True
4. potted plant [3] --(above)--> bed [0] score=0.3917 thr=0.3064 pass=True
5. person [1] --(covering)--> person [2] score=0.3674 thr=0.5697 pass=False
6. bed [0] --(below)--> potted plant [3] score=0.2557 thr=0.3004 pass=False
7. person [2] --(below)--> potted plant [3] score=0.2444 thr=0.3004 pass=False
8. person [1] --(below)--> potted plant [3] score=0.2178 thr=0.3004 pass=False
9. potted plant [3] --(above)--> person [1] score=0.2162 thr=0.3064 pass=False
10. potted plant [3] --(above)--> person [2] score=0.1613 thr=0.3064 pass=False

Frame: 482x640 | Detections: 11 (M2 boxes: 11) | M2 2x identical: True




Shown M2 text (relate.context):

M2 relations:
- bottle [2] resting on dining table [9].
- cell phone [3] resting on dining table [9].
- cell phone [10] resting on dining table [9].
- remote [6] resting on dining table [9].
- bottle [4] resting on dining table [9].

Raw top-10 M2 rows (score / threshold / pass):

1. bottle [2] --(resting on)--> dining table [9] score=0.7290 thr=0.5001 pass=True
2. cell phone [3] --(resting on)--> dining table [9] score=0.7057 thr=0.5001 pass=True
3. cell phone [10] --(resting on)--> dining table [9] score=0.7051 thr=0.5001 pass=True
4. remote [6] --(resting on)--> dining table [9] score=0.7050 thr=0.5001 pass=True
5. bottle [4] --(resting on)--> dining table [9] score=0.6685 thr=0.5001 pass=True
6. chair [8] --(beside)--> dining table [9] score=0.6551 thr=0.5001 pass=True
7. chair [1] --(beside)--> dining table [9] score=0.6475 thr=0.5001 pass=True
8. person [0] --(wearing)--> dining table [9] score=0.5920 thr=0.5697 pass=True
9. person [0] --(wearing)--> chair [5] score=0.5885 thr=0.5697 pass=True
10. chair [5] --(beside)--> dining table [9] score=0.4294 thr=0.5001 pass=False

# bench_photos/person_sideon_040cm.jpg -- Page 14 / 61

Frame: 482x640 | Detections: 3 (M2 boxes: 3) | M2 2x identical: True




Shown M2 text (relate.context):

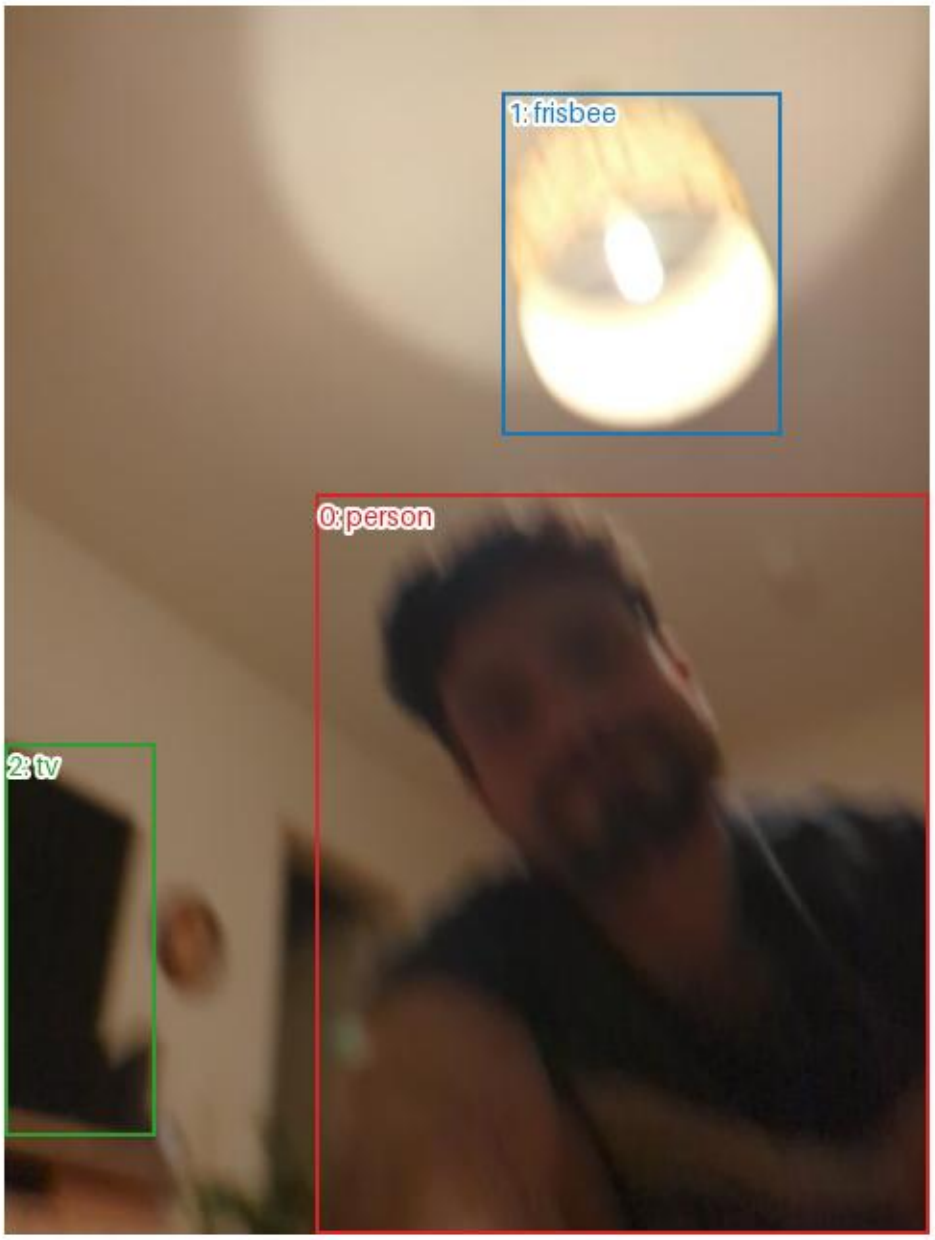
M2 relations:
- person [0] resting on couch [2].
- person [1] resting on couch [2].
- person [0] leaning against person [1].

Raw top-10 M2 rows (score / threshold / pass):

1. person [0] --(resting on)--> couch [2] score=0.6635 thr=0.5001 pass=True
2. person [1] --(resting on)--> couch [2] score=0.5990 thr=0.5001 pass=True
3. person [0] --(leaning against)--> person [1] score=0.5621 thr=0.5308 pass=True
4. person [1] --(beside)--> person [0] score=0.3865 thr=0.5001 pass=False
5. couch [2] --(below)--> person [1] score=0.1970 thr=0.3004 pass=False
6. couch [2] --(behind)--> person [0] score=0.0994 thr=0.3004 pass=False

# bench_photos/person_sideon_060cm.jpg -- Page 15 / 61

Frame: 482x640 | Detections: 3 (M2 boxes: 3) | M2 2x identical: True



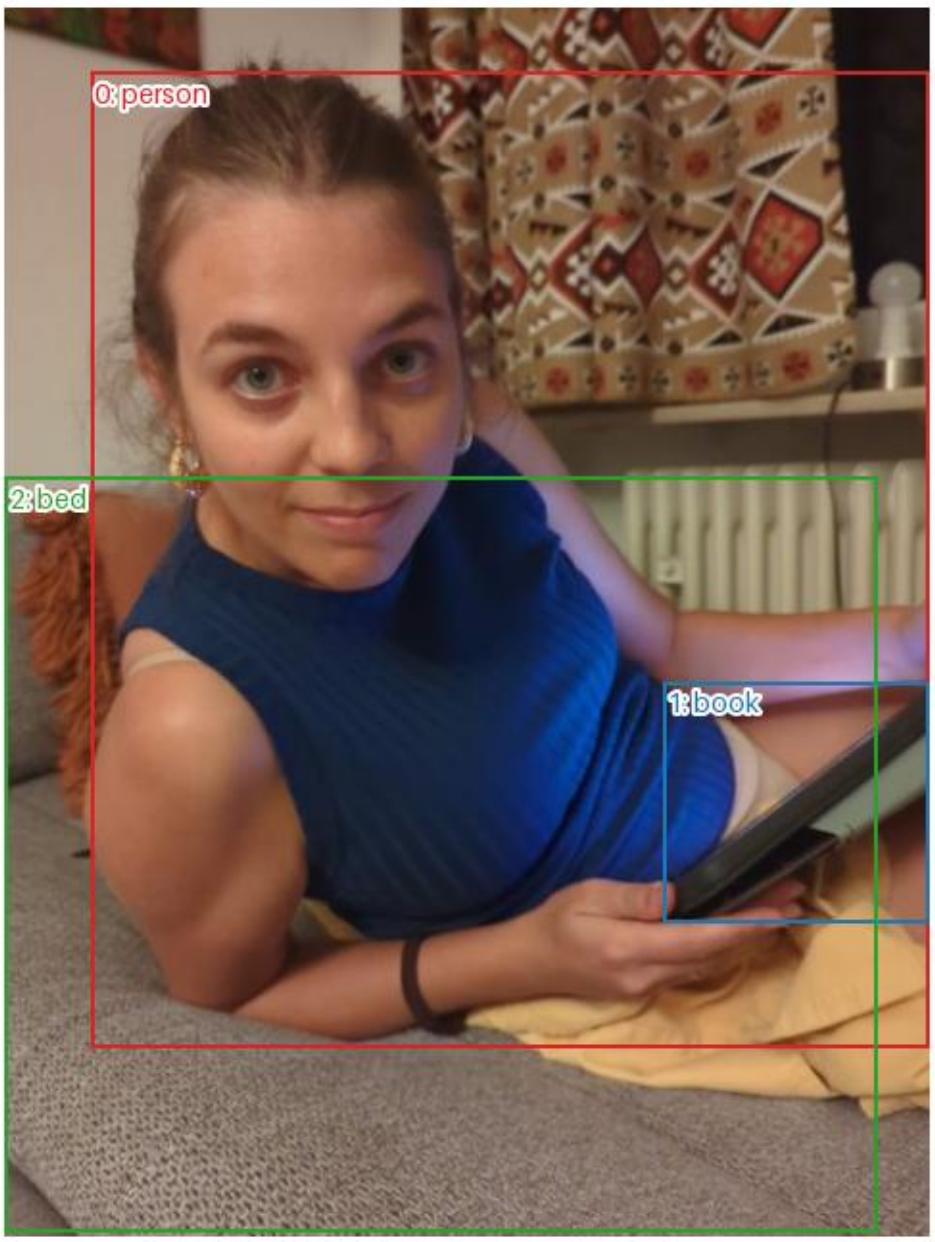

Shown M2 text (relate.context):

M2 relations:
- person [0] below frisbee [1].
- person [0] looking at tv [2].
- frisbee [1] above person [0].
- tv [2] behind person [0].

Raw top-10 M2 rows (score / threshold / pass):

1. person [0] --(below)--> frisbee [1] score=0.6010 thr=0.3004 pass=True
2. person [0] --(looking at)--> tv [2] score=0.5989 thr=0.4536 pass=True
3. frisbee [1] --(above)--> person [0] score=0.5342 thr=0.3064 pass=True
4. tv [2] --(behind)--> person [0] score=0.3888 thr=0.3004 pass=True
5. tv [2] --(below)--> frisbee [1] score=0.2816 thr=0.3004 pass=False
6. frisbee [1] --(to the right of)--> tv [2] score=0.2072 thr=0.2703 pass=False

Frame: 482x640 | Detections: 3 (M2 boxes: 3) | M2 2x identical: True




Shown M2 text (relate.context):

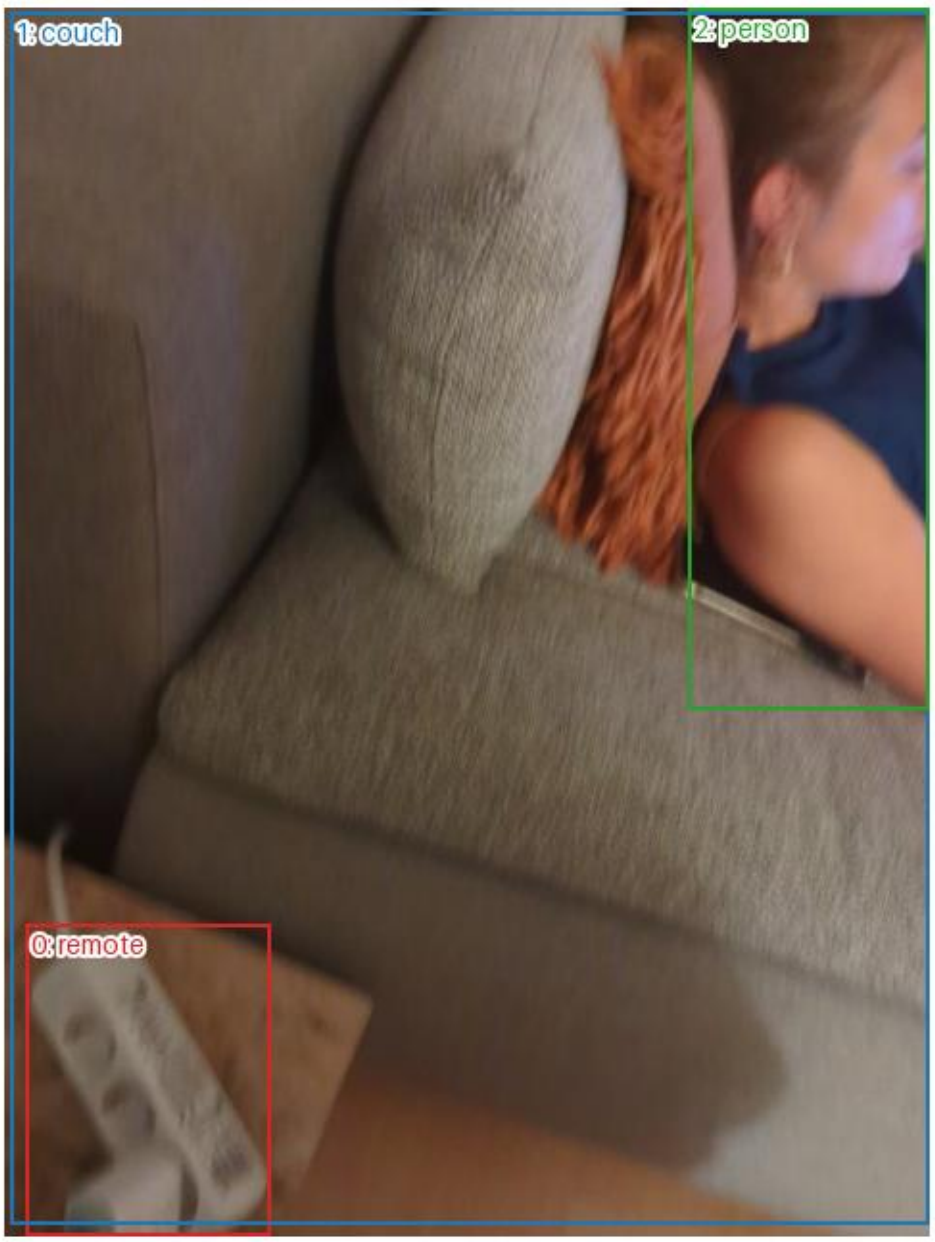
M2 relations:
- person [0] using book [1].
- person [0] sitting on bed [2].
- book [1] resting on person [0].

Raw top-10 M2 rows (score / threshold / pass):

1. person [0] --(using)--> book [1] score=0.7658 thr=0.5697 pass=True
2. person [0] --(sitting on)--> bed [2] score=0.6881 thr=0.5001 pass=True
3. book [1] --(resting on)--> person [0] score=0.5079 thr=0.5001 pass=True
4. book [1] --(resting on)--> bed [2] score=0.4983 thr=0.5001 pass=False
5. bed [2] --(to the left of)--> book [1] score=0.1748 thr=0.2703 pass=False
6. bed [2] --(behind)--> person [0] score=0.1294 thr=0.3004 pass=False

# bench_photos/person_sideon_100cm.jpg -- Page 17 / 61

Frame: 482x640 | Detections: 3 (M2 boxes: 3) | M2 2x identical: True



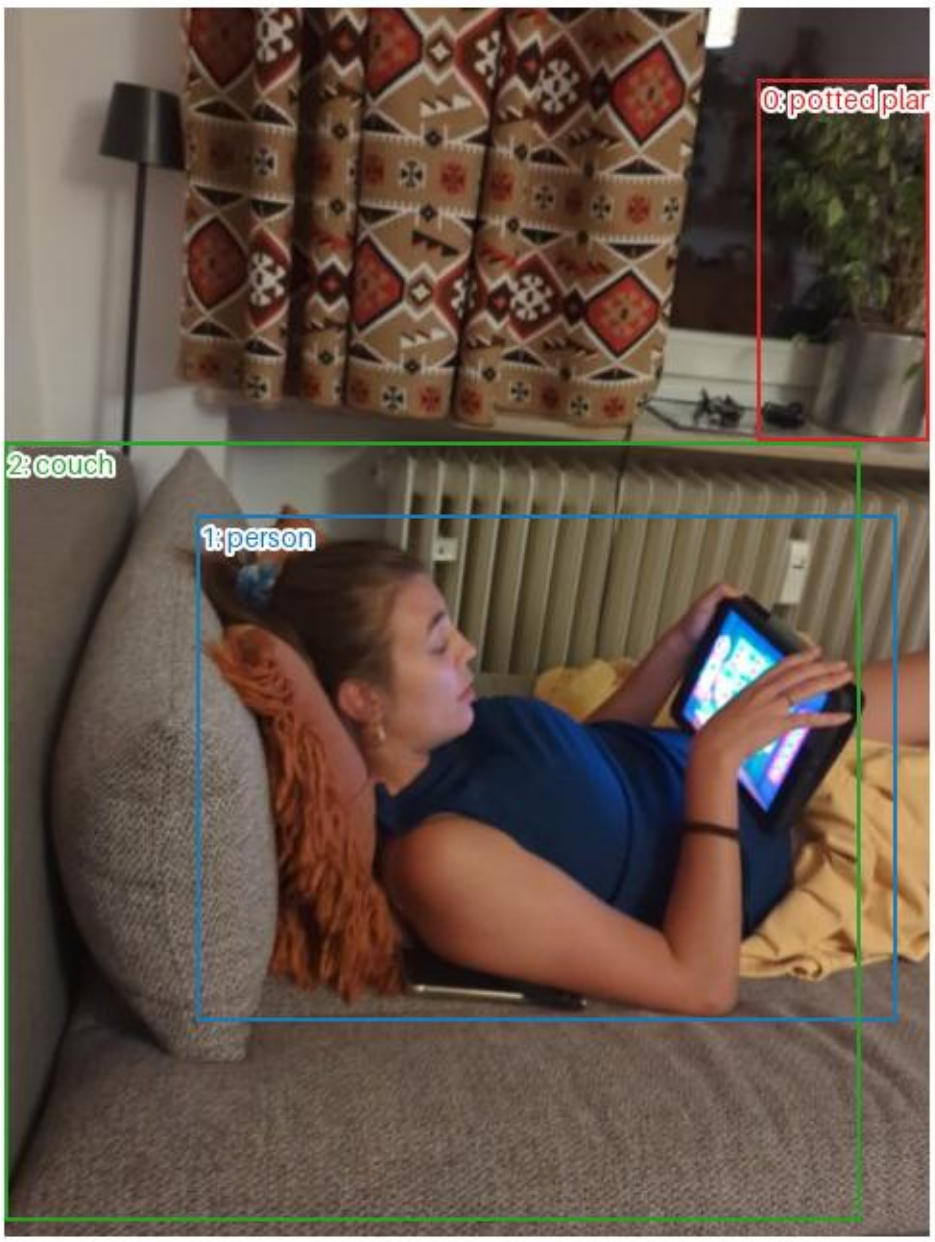

Shown M2 text (relate.context):

M2 relations:
- remote [0] resting on couch [1].
- person [2] leaning against couch [1].
- remote [0] to the left of person [2].
- person [2] to the right of remote [0].
- couch [1] behind remote [0].

Raw top-10 M2 rows (score / threshold / pass):

1. remote [0] --(resting on)--> couch [1] score=0.7369 thr=0.5001 pass=True
2. person [2] --(leaning against)--> couch [1] score=0.7080 thr=0.5308 pass=True
3. remote [0] --(to the left of)--> person [2] score=0.4002 thr=0.2703 pass=True
4. person [2] --(to the right of)--> remote [0] score=0.3819 thr=0.2703 pass=True
5. couch [1] --(behind)--> remote [0] score=0.3486 thr=0.3004 pass=True
6. couch [1] --(to the left of)--> person [2] score=0.1245 thr=0.2703 pass=False

Frame: 482x640 | Detections: 3 (M2 boxes: 3) | M2 2x identical: True




Shown M2 text (relate.context):

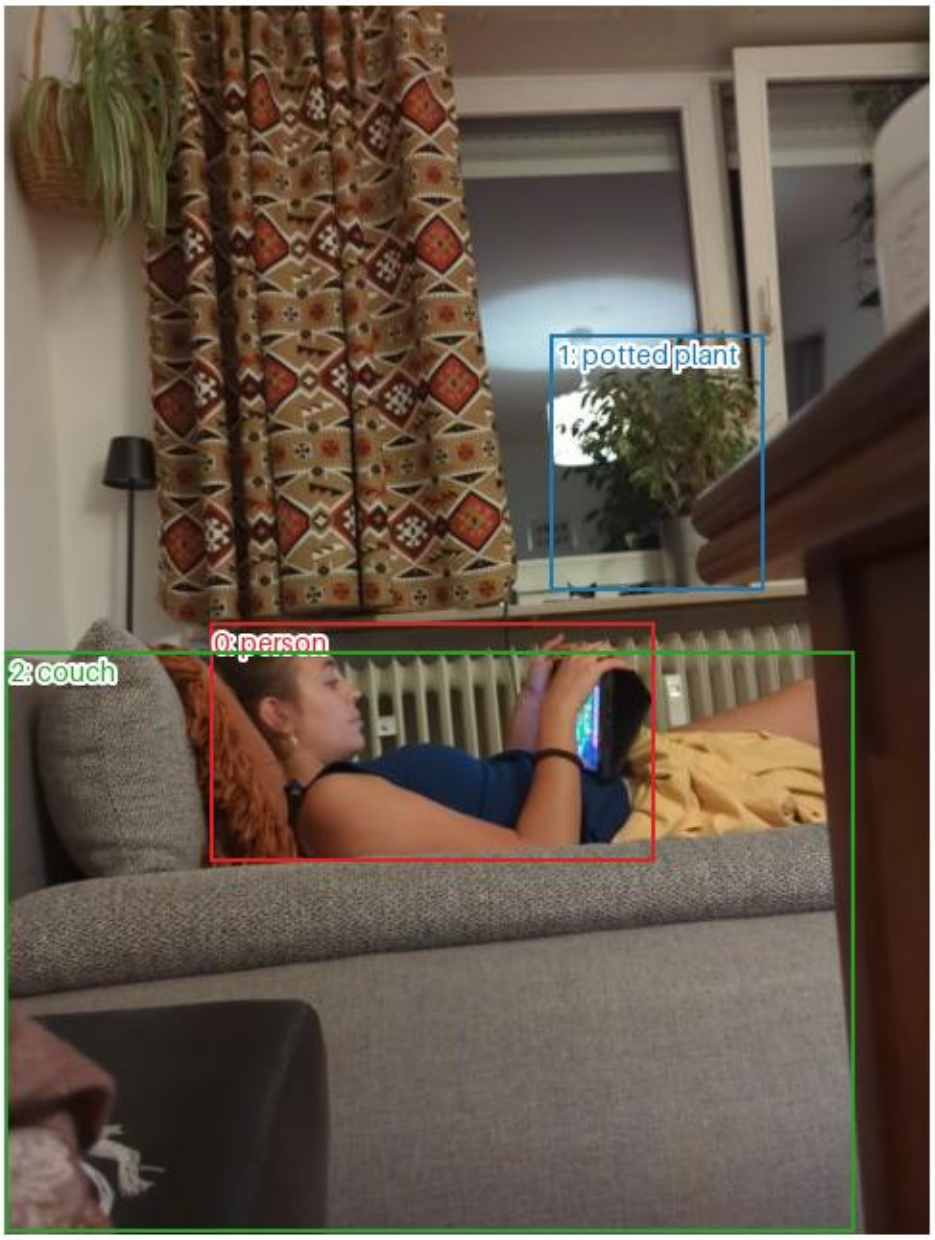
M2 relations:
- person [1] resting on couch [2].
- potted plant [0] behind person [1].

Raw top-10 M2 rows (score / threshold / pass):

1. person [1] --(resting on)--> couch [2] score=0.7145 thr=0.5001 pass=True
2. person [1] --(beside)--> potted plant [0] score=0.3647 thr=0.5001 pass=False
3. potted plant [0] --(behind)--> person [1] score=0.3588 thr=0.3004 pass=True
4. potted plant [0] --(sitting in)--> couch [2] score=0.3013 thr=0.5001 pass=False
5. couch [2] --(below)--> potted plant [0] score=0.1916 thr=0.3004 pass=False
6. couch [2] --(holding)--> person [1] score=0.1108 thr=0.5001 pass=False

# bench_photos/person_sideon_200cm.jpg -- Page 19 / 61

Frame: 482x640 | Detections: 3 (M2 boxes: 3) | M2 2x identical: True




Shown M2 text (relate.context):

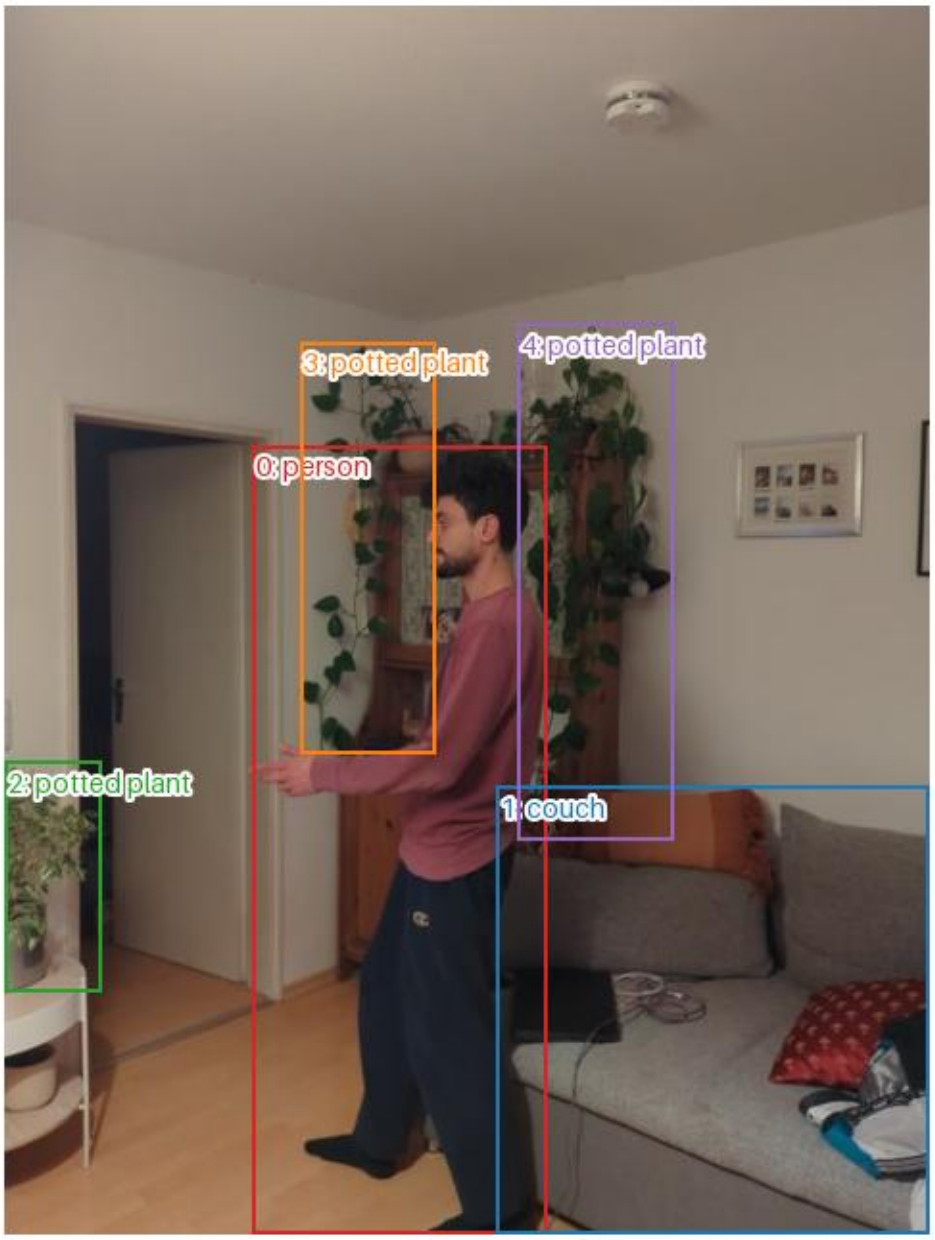
M2 relations:
- person [0] resting on couch [2].
- potted plant [1] behind person [0].

Raw top-10 M2 rows (score / threshold / pass):

1. person [0] --(resting on)--> couch [2] score=0.6688 thr=0.5001 pass=True
2. person [0] --(looking at)--> potted plant [1] score=0.4088 thr=0.4536 pass=False
3. potted plant [1] --(behind)--> person [0] score=0.3178 thr=0.3004 pass=True
4. potted plant [1] --(behind)--> couch [2] score=0.2967 thr=0.3004 pass=False
5. couch [2] --(below)--> potted plant [1] score=0.2364 thr=0.3004 pass=False
6. couch [2] --(holding)--> person [0] score=0.2075 thr=0.5001 pass=False

# bench_photos/person_sideon_300cm.jpg -- Page 20 / 61

Frame: 482x640 | Detections: 5 (M2 boxes: 5) | M2 2x identical: True



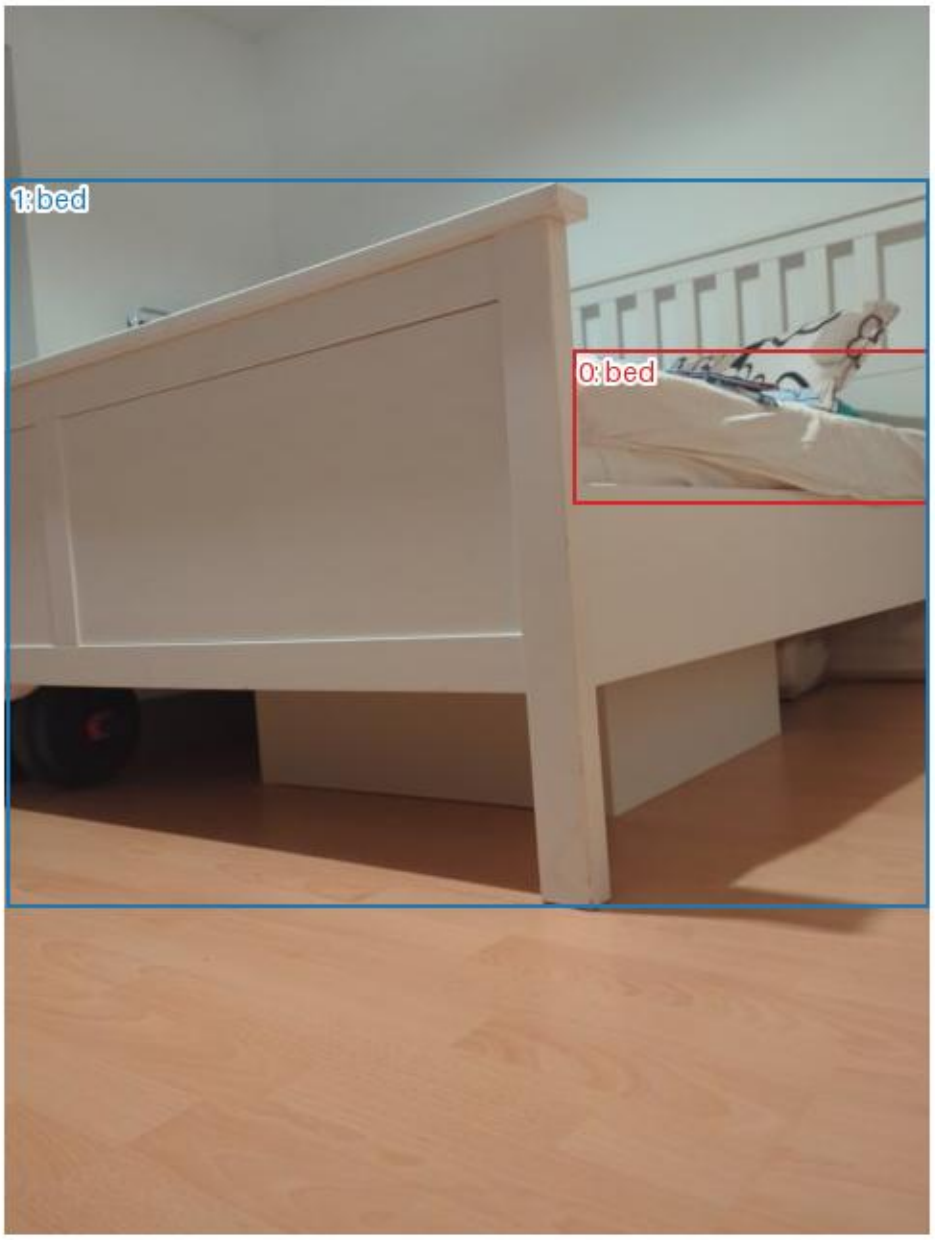

Shown M2 text (relate.context):

M2 relations:
- person [0] wearing couch [1].
- person [0] looking at potted plant [4].
- person [0] looking at potted plant [3].
- potted plant [4] behind person [0].
- potted plant [2] to the left of person [0].

Raw top-10 M2 rows (score / threshold / pass):

1. person [0] --(wearing)--> couch [1] score=0.7430 thr=0.5697 pass=True
2. person [0] --(looking at)--> potted plant [4] score=0.5178 thr=0.4536 pass=True
3. person [0] --(looking at)--> potted plant [3] score=0.5112 thr=0.4536 pass=True
4. potted plant [3] --(hanging from)--> person [0] score=0.4738 thr=0.5697 pass=False
5. person [0] --(beside)--> potted plant [2] score=0.4095 thr=0.5001 pass=False
6. potted plant [4] --(behind)--> person [0] score=0.3556 thr=0.3004 pass=True
7. potted plant [4] --(beside)--> couch [1] score=0.3350 thr=0.5001 pass=False
8. potted plant [3] --(beside)--> potted plant [4] score=0.3200 thr=0.5001 pass=False
9. potted plant [2] --(to the left of)--> person [0] score=0.3128 thr=0.2703 pass=True
10. couch [1] --(to the right of)--> person [0] score=0.3013 thr=0.2703 pass=True

Frame: 482x640 | Detections: 2 (M2 boxes: 2) | M2 2x identical: True




Shown M2 text (relate.context):

M2 relations:
- bed [0] resting on bed [1].

Raw top-10 M2 rows (score / threshold / pass):

1. bed [0] --(resting on)--> bed [1] score=0.7459 thr=0.5001 pass=True
2. bed [1] --(beside)--> bed [0] score=0.3167 thr=0.5001 pass=False

# bench_photos/roomsig_bed_3.jpg -- Page 22 / 61

Frame: 482x640 | Detections: 2 (M2 boxes: 2) | M2 2x identical: True



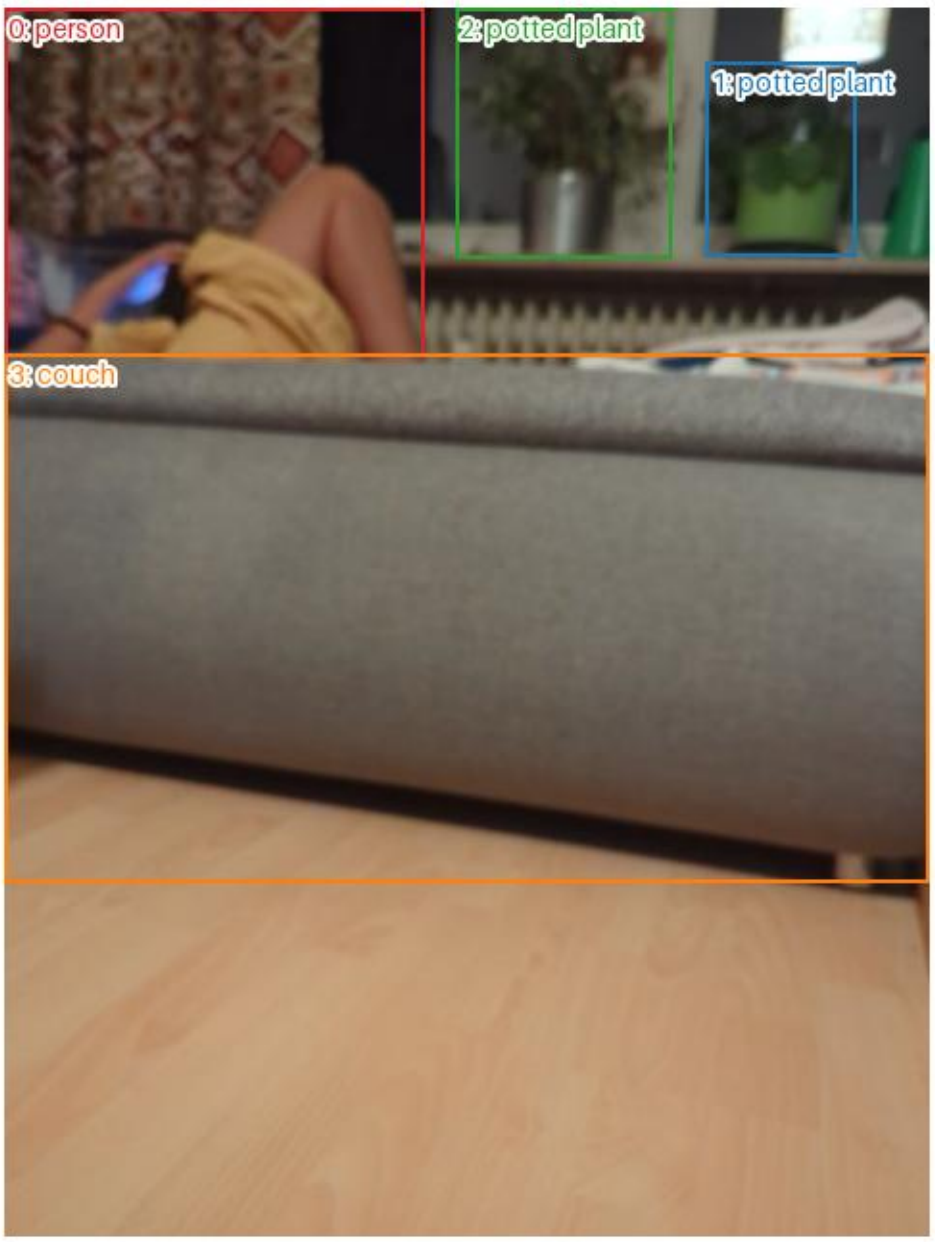

Shown M2 text (relate.context):

M2 relations:
- bed [0] behind chair [1].

Raw top-10 M2 rows (score / threshold / pass):

1. chair [1] --(covering)--> bed [0] score=0.4778 thr=0.5697 pass=False
2. bed [0] --(behind)--> chair [1] score=0.4111 thr=0.3004 pass=True

Frame: 482x640 | Detections: 4 (M2 boxes: 4) | M2 2x identical: True



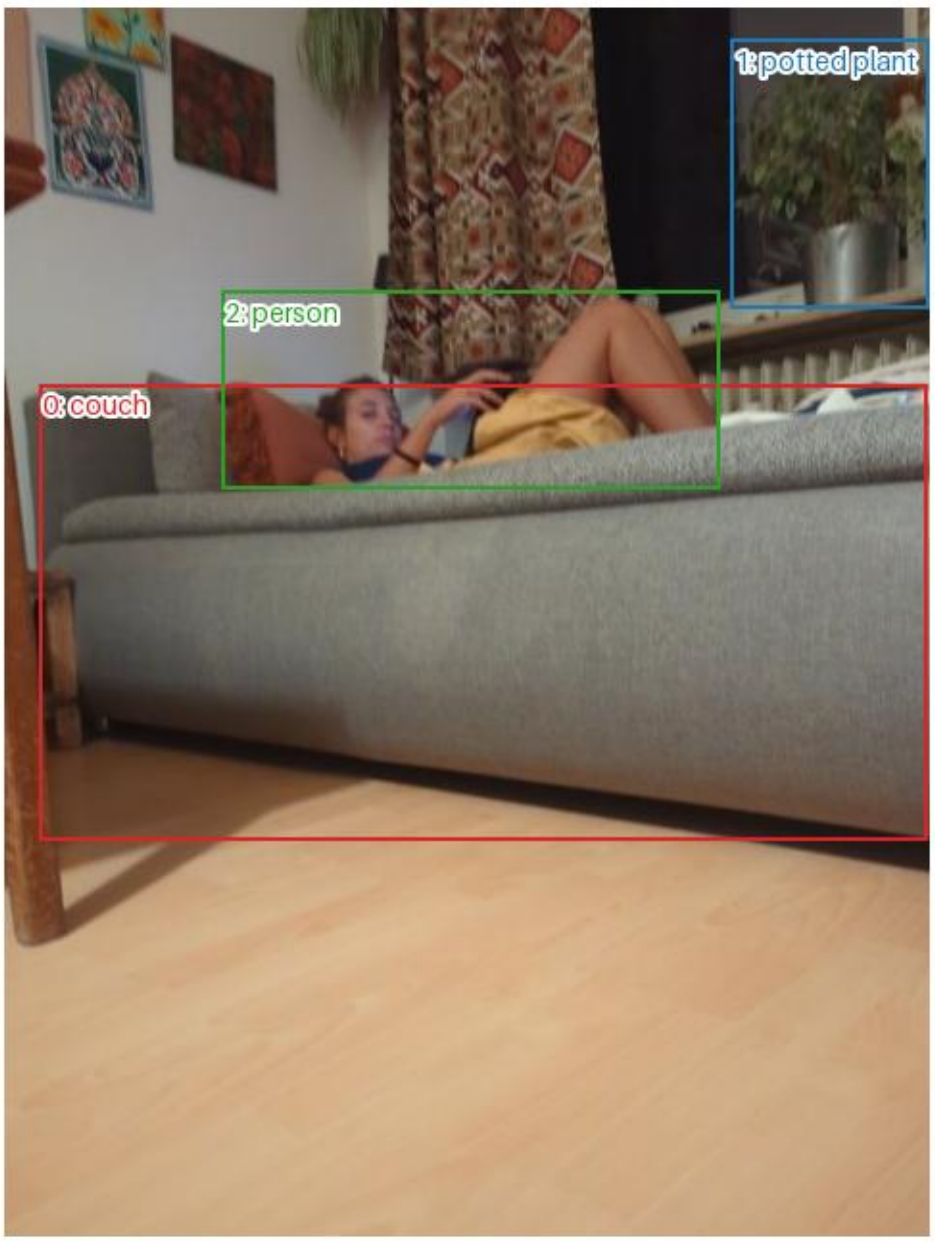

Shown M2 text (relate.context):

M2 relations:
- person [0] resting on couch [3].
- potted plant [2] beside potted plant [1].
- potted plant [1] to the right of potted plant [2].
- person [0] to the left of potted plant [2].
- potted plant [2] to the right of person [0].

Raw top-10 M2 rows (score / threshold / pass):

1. person [0] --(resting on)--> couch [3] score=0.5073 thr=0.5001 pass=True
2. potted plant [2] --(beside)--> potted plant [1] score=0.5012 thr=0.5001 pass=True
3. potted plant [1] --(to the right of)--> potted plant [2] score=0.4322 thr=0.2703 pass=True
4. person [0] --(to the left of)--> potted plant [2] score=0.3765 thr=0.2703 pass=True
5. potted plant [2] --(to the right of)--> person [0] score=0.3655 thr=0.2703 pass=True
6. potted plant [2] --(behind)--> couch [3] score=0.3525 thr=0.3004 pass=True
7. potted plant [1] --(behind)--> couch [3] score=0.3331 thr=0.3004 pass=True
8. couch [3] --(in front of)--> person [0] score=0.3185 thr=0.2748 pass=True
9. person [0] --(to the left of)--> potted plant [1] score=0.2329 thr=0.2703 pass=False
10. couch [3] --(in front of)--> potted plant [1] score=0.1824 thr=0.2748 pass=False

Frame: 482x640 | Detections: 3 (M2 boxes: 3) | M2 2x identical: True



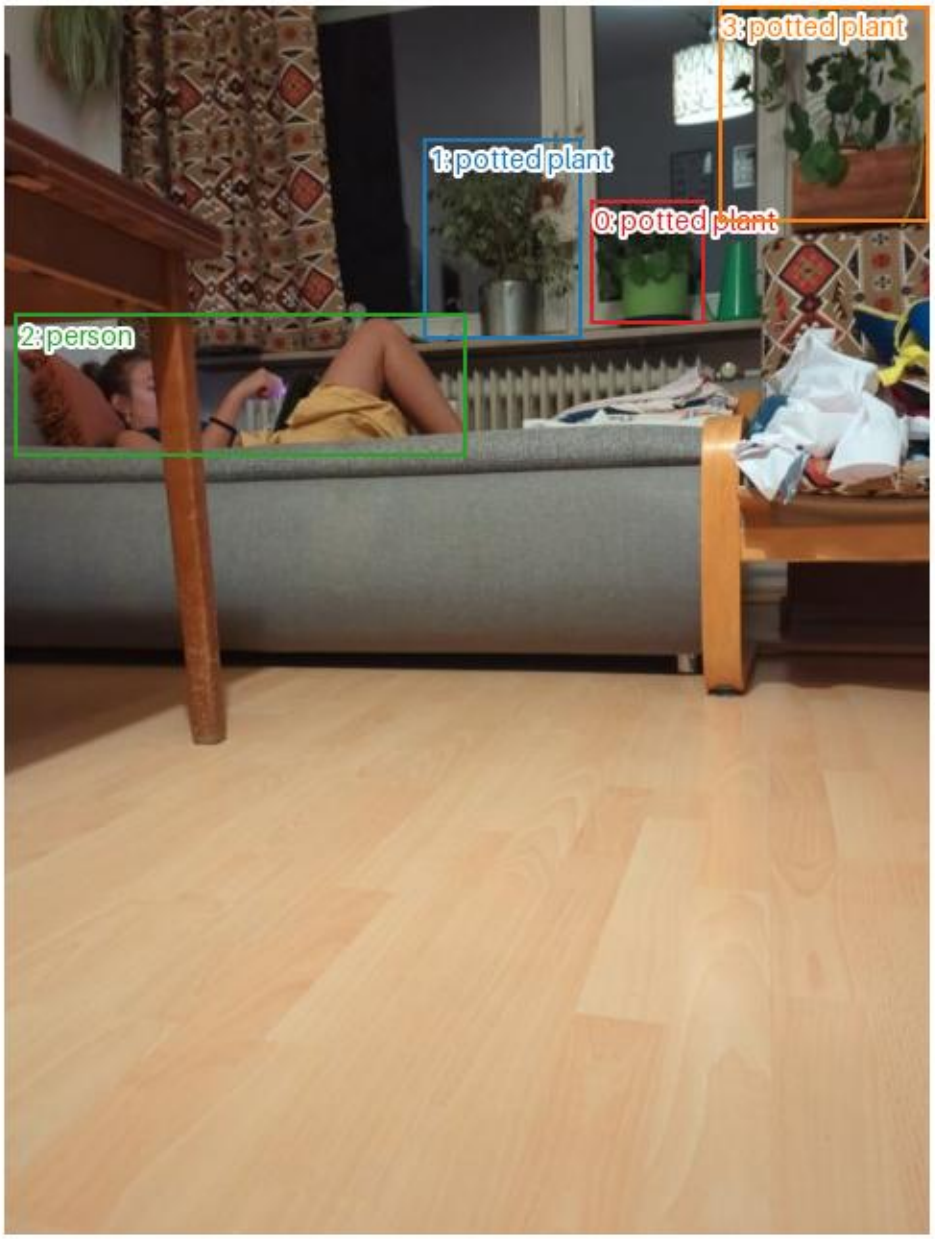

Shown M2 text (relate.context):

M2 relations:
- person [2] resting on couch [0].
- potted plant [1] to the right of person [2].

Raw top-10 M2 rows (score / threshold / pass):

1. person [2] --(resting on)--> couch [0] score=0.6369 thr=0.5001 pass=True
2. potted plant [1] --(beside)--> couch [0] score=0.3024 thr=0.5001 pass=False
3. potted plant [1] --(to the right of)--> person [2] score=0.2889 thr=0.2703 pass=True
4. person [2] --(beside)--> potted plant [1] score=0.2839 thr=0.5001 pass=False
5. couch [0] --(in front of)--> potted plant [1] score=0.2246 thr=0.2748 pass=False
6. couch [0] --(covering)--> person [2] score=0.1051 thr=0.5697 pass=False

Frame: 482x640 | Detections: 4 (M2 boxes: 4) | M2 2x identical: True



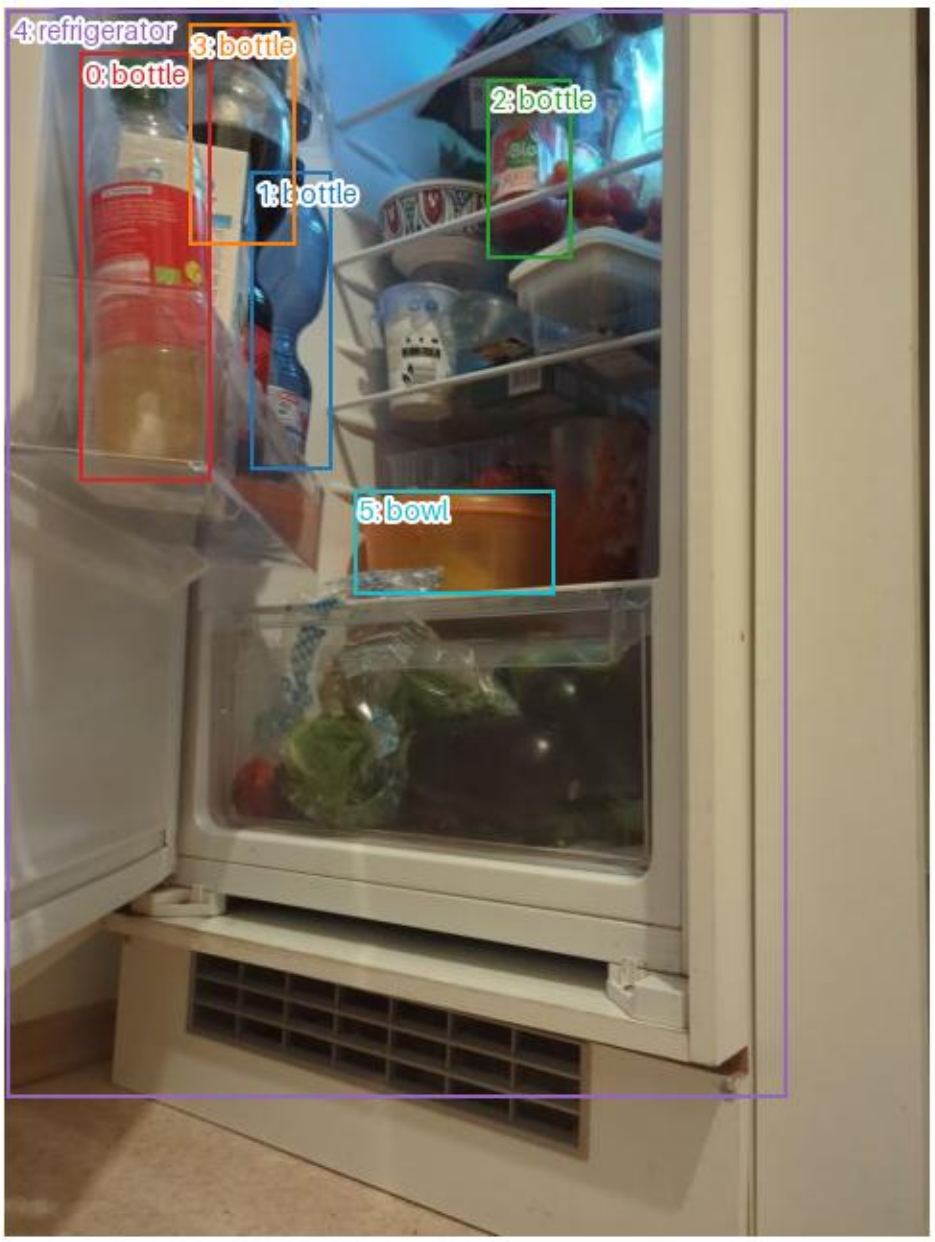

Shown M2 text (relate.context):

M2 relations:
- none above threshold.

Raw top-10 M2 rows (score / threshold / pass):

1. potted plant [1] --(beside)--> potted plant [0] score=0.3751 thr=0.5001 pass=False
2. person [2] --(resting on)--> potted plant [1] score=0.3452 thr=0.5001 pass=False
3. potted plant [0] --(beside)--> potted plant [1] score=0.3326 thr=0.5001 pass=False
4. potted plant [0] --(beside)--> potted plant [3] score=0.2929 thr=0.5001 pass=False
5. potted plant [3] --(above)--> potted plant [0] score=0.2788 thr=0.3064 pass=False
6. person [2] --(resting on)--> potted plant [0] score=0.2449 thr=0.5001 pass=False
7. potted plant [1] --(to the left of)--> potted plant [3] score=0.2435 thr=0.2703 pass=False
8. potted plant [3] --(to the right of)--> potted plant [1] score=0.2398 thr=0.2703 pass=False
9. potted plant [1] --(sitting on)--> person [2] score=0.1848 thr=0.5001 pass=False
10. potted plant [0] --(to the right of)--> person [2] score=0.1812 thr=0.2703 pass=False

Frame: 482x640 | Detections: 6 (M2 boxes: 6) | M2 2x identical: True




Shown M2 text (relate.context):

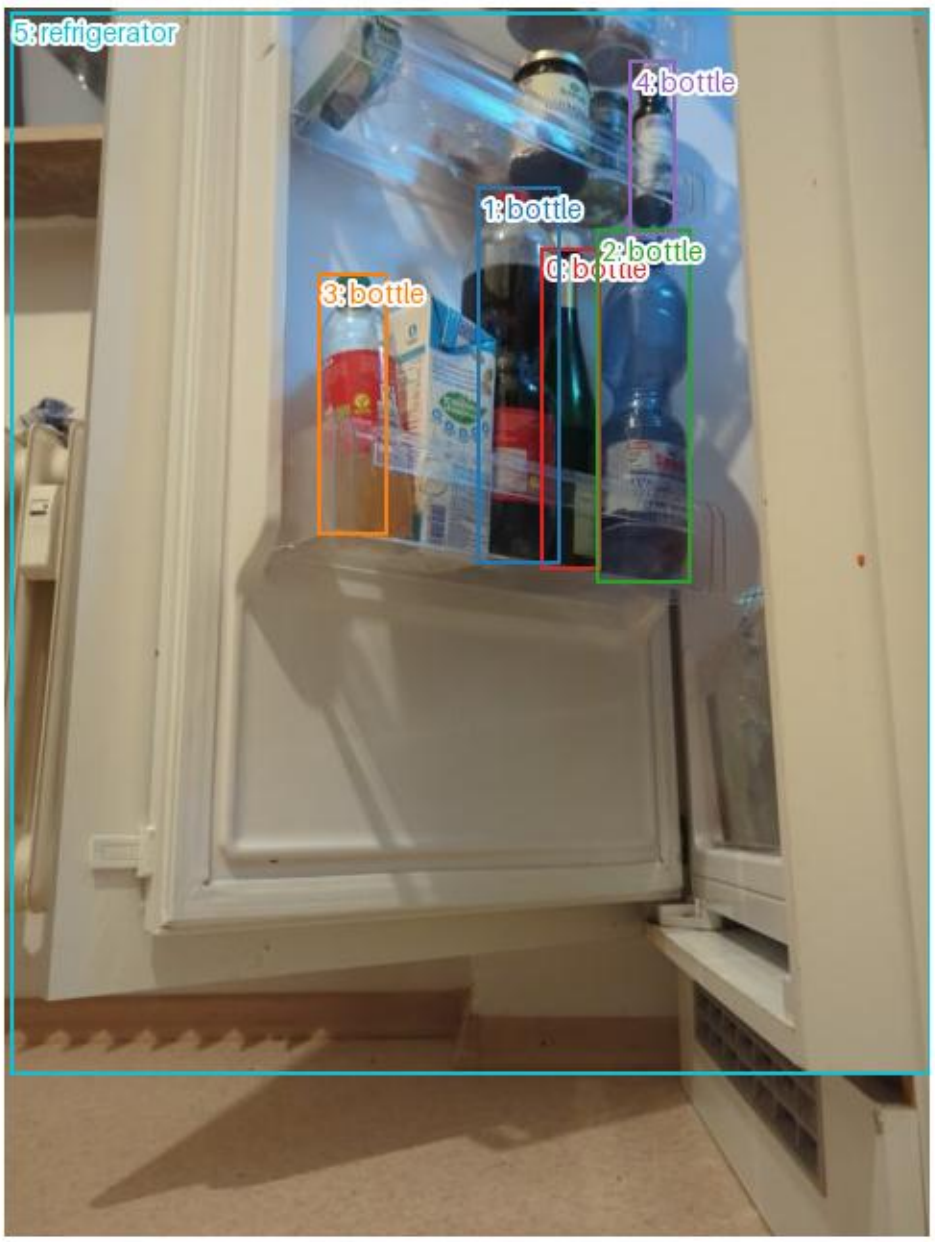
M2 relations:
- bottle [1] contained in refrigerator [4].
- bottle [0] contained in refrigerator [4].
- bottle [2] contained in refrigerator [4].
- bottle [3] contained in refrigerator [4].
- bowl [5] contained in refrigerator [4].

Raw top-10 M2 rows (score / threshold / pass):

1. bottle [1] --(contained in)--> refrigerator [4] score=0.7649 thr=0.5697 pass=True
2. bottle [0] --(contained in)--> refrigerator [4] score=0.7466 thr=0.5697 pass=True
3. bottle [2] --(contained in)--> refrigerator [4] score=0.7035 thr=0.5697 pass=True
4. bottle [3] --(contained in)--> refrigerator [4] score=0.6971 thr=0.5697 pass=True
5. bowl [5] --(contained in)--> refrigerator [4] score=0.6895 thr=0.5697 pass=True
6. bottle [1] --(below)--> bottle [3] score=0.3587 thr=0.3004 pass=True
7. bottle [0] --(resting on)--> bottle [3] score=0.3581 thr=0.5001 pass=False
8. refrigerator [4] --(holding)--> bowl [5] score=0.3528 thr=0.5001 pass=False
9. bottle [1] --(to the right of)--> bottle [0] score=0.3374 thr=0.2703 pass=True
10. bottle [3] --(above)--> bottle [1] score=0.3209 thr=0.3064 pass=True

# bench_photos/roomsig_refrigerator_2.jpg -- Page 27 / 61

Frame: 482x640 | Detections: 6 (M2 boxes: 6) | M2 2x identical: True




Shown M2 text (relate.context):

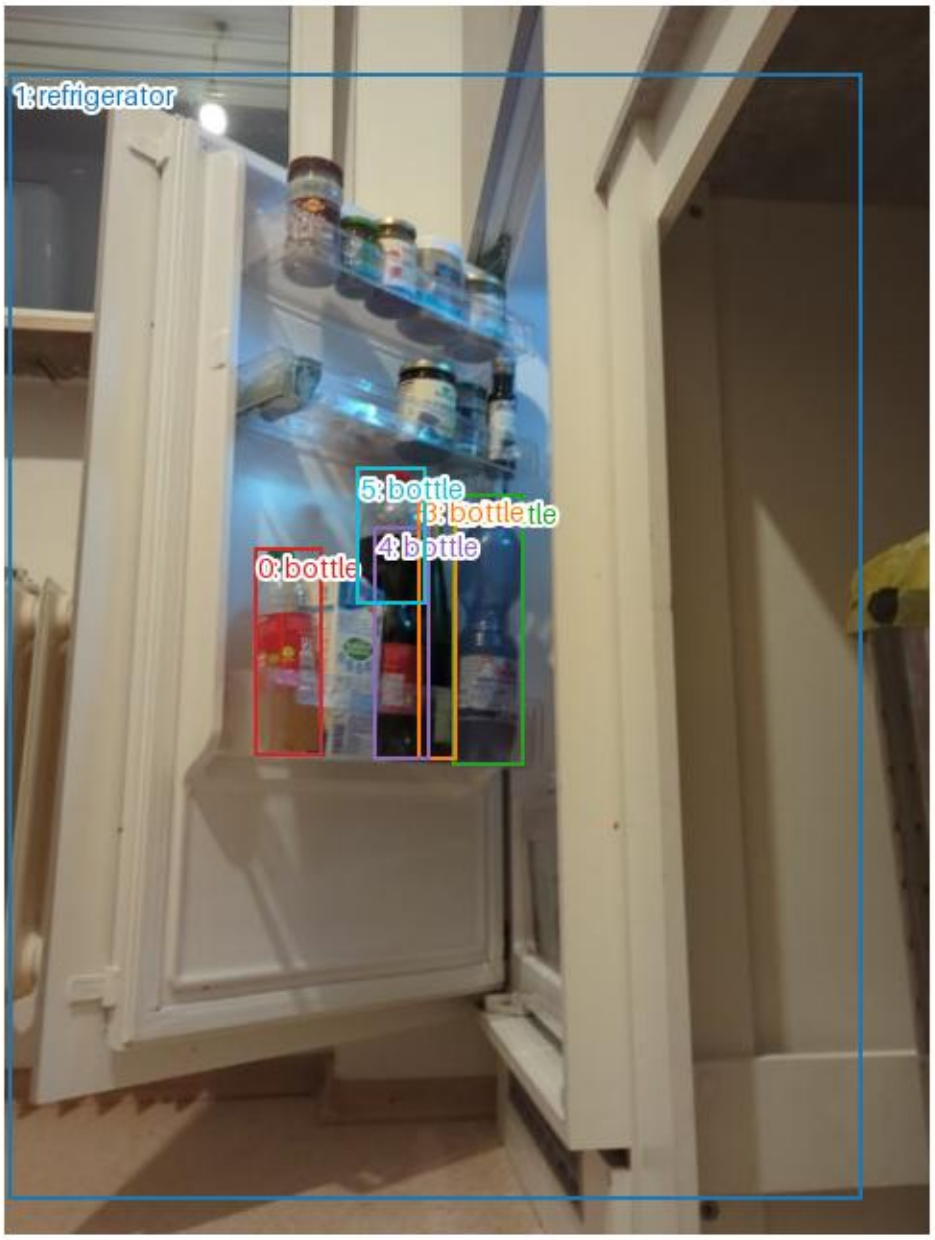
M2 relations:
- bottle [3] contained in refrigerator [5].
- bottle [0] contained in refrigerator [5].
- bottle [2] contained in refrigerator [5].
- bottle [1] contained in refrigerator [5].
- bottle [4] contained in refrigerator [5].

Raw top-10 M2 rows (score / threshold / pass):

1. bottle [3] --(contained in)--> refrigerator [5] score=0.8020 thr=0.5697 pass=True
2. bottle [0] --(contained in)--> refrigerator [5] score=0.7805 thr=0.5697 pass=True
3. bottle [2] --(contained in)--> refrigerator [5] score=0.7781 thr=0.5697 pass=True
4. bottle [1] --(contained in)--> refrigerator [5] score=0.7776 thr=0.5697 pass=True
5. bottle [4] --(contained in)--> refrigerator [5] score=0.7202 thr=0.5697 pass=True
6. bottle [0] --(to the right of)--> bottle [1] score=0.5060 thr=0.2703 pass=True
7. bottle [1] --(resting on)--> bottle [0] score=0.4452 thr=0.5001 pass=False
8. bottle [2] --(to the right of)--> bottle [0] score=0.3914 thr=0.2703 pass=True
9. bottle [0] --(to the left of)--> bottle [2] score=0.3866 thr=0.2703 pass=True
10. bottle [1] --(to the left of)--> bottle [2] score=0.3614 thr=0.2703 pass=True

# bench_photos/roomsig_refrigerator_3.jpg -- Page 28 / 61

Frame: 482x640 | Detections: 6 (M2 boxes: 6) | M2 2x identical: True




Shown M2 text (relate.context):

M2 relations:
- bottle [4] contained in refrigerator [1].
- bottle [0] contained in refrigerator [1].
- bottle [3] contained in refrigerator [1].
- bottle [2] contained in refrigerator [1].
- bottle [5] contained in refrigerator [1].

Raw top-10 M2 rows (score / threshold / pass):

1. bottle [4] --(contained in)--> refrigerator [1] score=0.7922 thr=0.5697 pass=True
2. bottle [0] --(contained in)--> refrigerator [1] score=0.7903 thr=0.5697 pass=True
3. bottle [3] --(contained in)--> refrigerator [1] score=0.7831 thr=0.5697 pass=True
4. bottle [2] --(contained in)--> refrigerator [1] score=0.7723 thr=0.5697 pass=True
5. bottle [5] --(contained in)--> refrigerator [1] score=0.7423 thr=0.5697 pass=True
6. bottle [3] --(to the right of)--> bottle [4] score=0.3994 thr=0.2703 pass=True
7. bottle [4] --(to the left of)--> bottle [3] score=0.3913 thr=0.2703 pass=True
8. bottle [4] --(below)--> bottle [5] score=0.3664 thr=0.3004 pass=True
9. bottle [3] --(to the left of)--> bottle [2] score=0.3496 thr=0.2703 pass=True
10. bottle [2] --(to the right of)--> bottle [3] score=0.3452 thr=0.2703 pass=True

# bench_photos/roomsig_toilet_1.jpg -- Page 29 / 61

Frame: 482x640 | Detections: 3 (M2 boxes: 3) | M2 2x identical: True




Shown M2 text (relate.context):

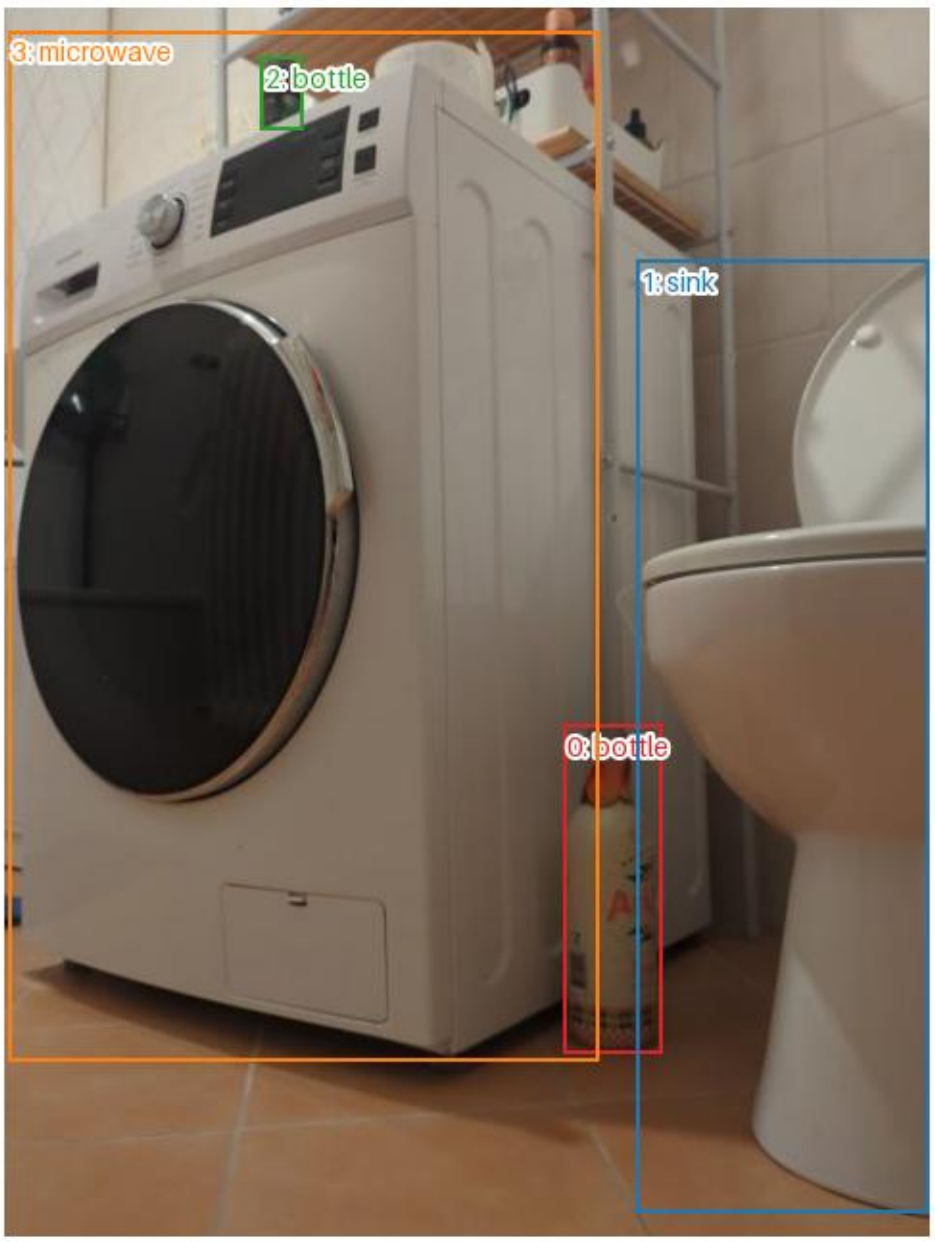
M2 relations:
- bottle [0] resting on sink [2].
- bottle [0] resting on toilet [1].
- toilet [1] in front of bottle [0].
- sink [2] in front of bottle [0].

Raw top-10 M2 rows (score / threshold / pass):

1. bottle [0] --(resting on)--> sink [2] score=0.6284 thr=0.5001 pass=True
2. bottle [0] --(resting on)--> toilet [1] score=0.6279 thr=0.5001 pass=True
3. toilet [1] --(beside)--> sink [2] score=0.4415 thr=0.5001 pass=False
4. sink [2] --(beside)--> toilet [1] score=0.4402 thr=0.5001 pass=False
5. toilet [1] --(in front of)--> bottle [0] score=0.2822 thr=0.2748 pass=True
6. sink [2] --(in front of)--> bottle [0] score=0.2816 thr=0.2748 pass=True

# bench_photos/roomsig_toilet_2.jpg -- Page 30 / 61

Frame: 482x640 | Detections: 4 (M2 boxes: 4) | M2 2x identical: True



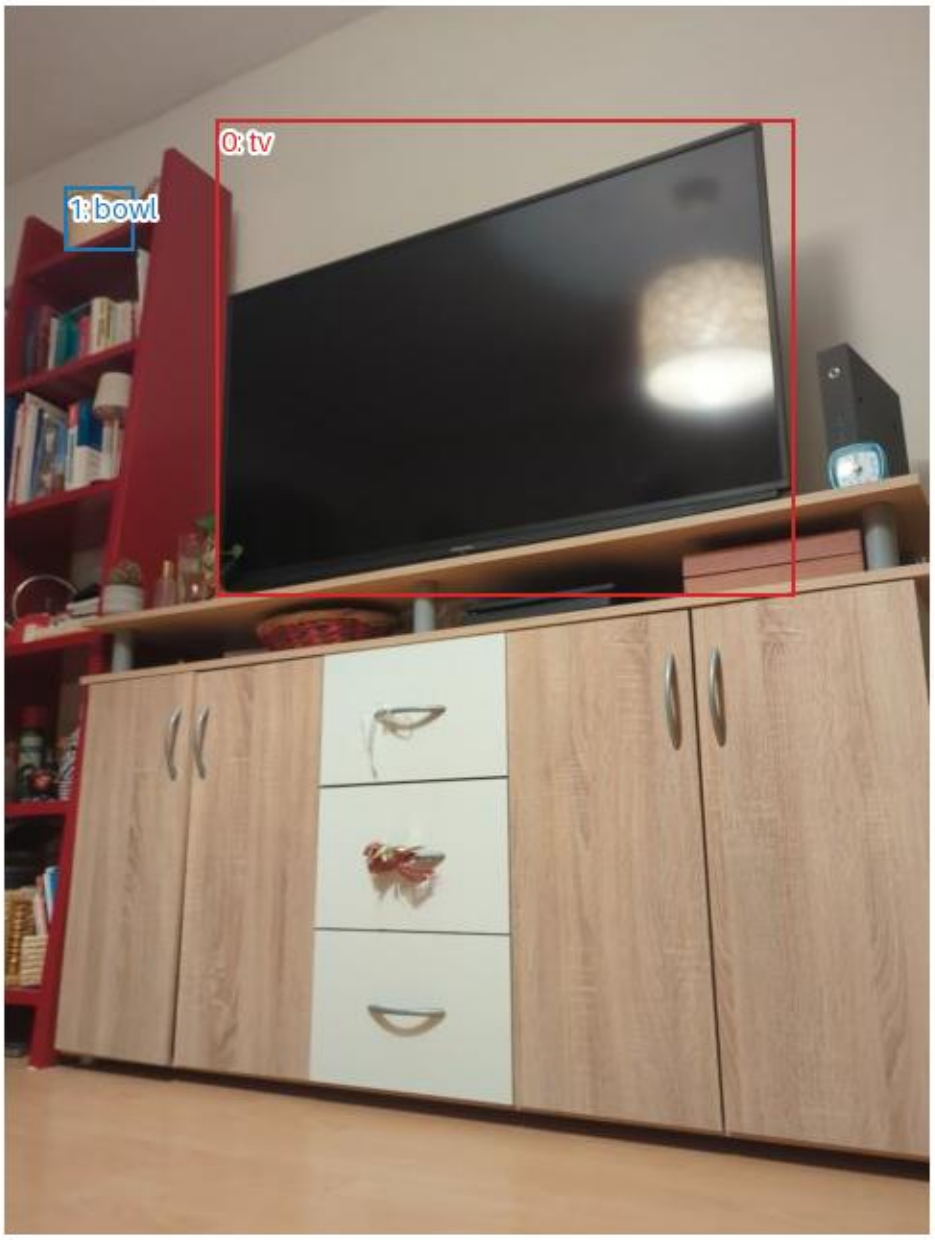

Shown M2 text (relate.context):

M2 relations:
- bottle [0] resting on microwave [3].
- bottle [0] resting on sink [1].
- bottle [2] resting on microwave [3].
- bottle [2] behind bottle [0].
- bottle [0] in front of bottle [2].

Raw top-10 M2 rows (score / threshold / pass):

1. bottle [0] --(resting on)--> microwave [3] score=0.7286 thr=0.5001 pass=True
2. bottle [0] --(resting on)--> sink [1] score=0.6361 thr=0.5001 pass=True
3. bottle [2] --(resting on)--> microwave [3] score=0.5578 thr=0.5001 pass=True
4. sink [1] --(beside)--> microwave [3] score=0.4741 thr=0.5001 pass=False
5. microwave [3] --(beside)--> sink [1] score=0.4622 thr=0.5001 pass=False
6. bottle [2] --(behind)--> bottle [0] score=0.4555 thr=0.3004 pass=True
7. bottle [0] --(in front of)--> bottle [2] score=0.4476 thr=0.2748 pass=True
8. microwave [3] --(below)--> bottle [2] score=0.4287 thr=0.3004 pass=True
9. sink [1] --(to the right of)--> bottle [0] score=0.3758 thr=0.2703 pass=True
10. microwave [3] --(beside)--> bottle [0] score=0.2915 thr=0.5001 pass=False

Frame: 482x640 | Detections: 2 (M2 boxes: 2) | M2 2x identical: True



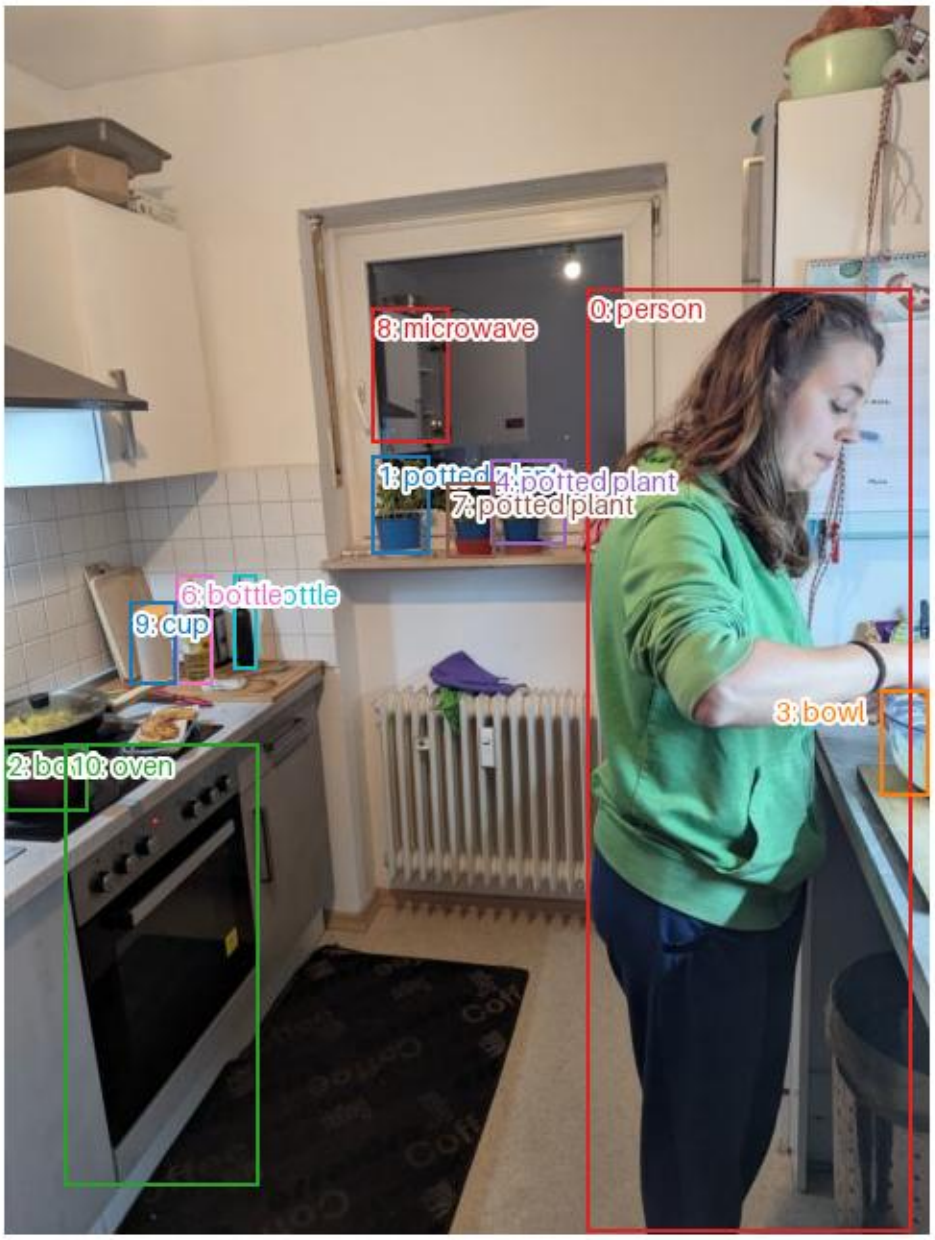

Shown M2 text (relate.context):

M2 relations:
- none above threshold.

Raw top-10 M2 rows (score / threshold / pass):

1. bowl [1] --(beside)--> tv [0] score=0.3366 thr=0.5001 pass=False
2. tv [0] --(to the right of)--> bowl [1] score=0.1721 thr=0.2703 pass=False

Frame: 482x640 | Detections: 11 (M2 boxes: 11) | M2 2x identical: True



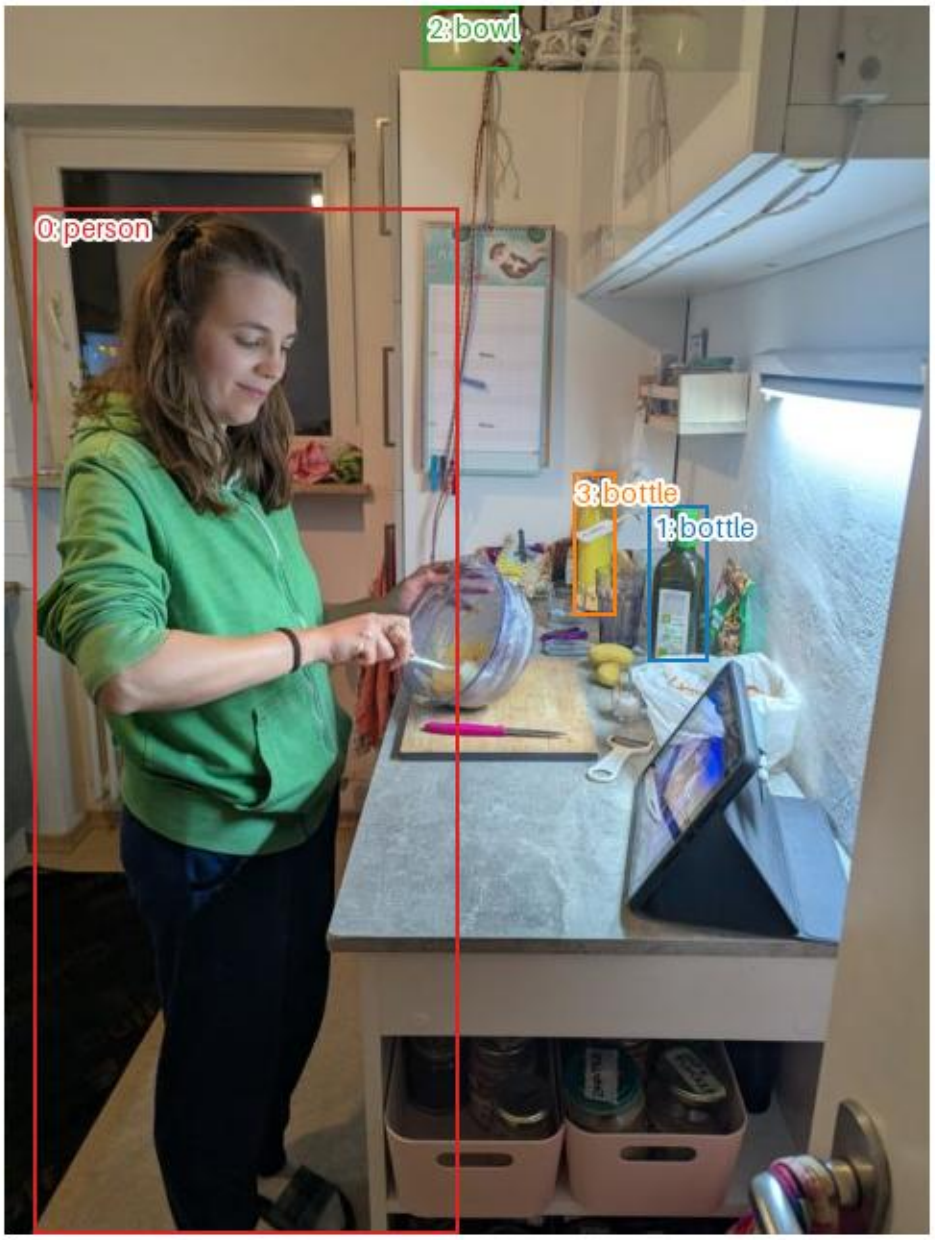

Shown M2 text (relate.context):

M2 relations:
- person [0] holding bowl [3].

Raw top-10 M2 rows (score / threshold / pass):

1. person [0] --(holding)--> bowl [3] score=0.5622 thr=0.5001 pass=True
2. bowl [2] --(resting on)--> oven [10] score=0.4692 thr=0.5001 pass=False
3. person [0] --(using)--> oven [10] score=0.3908 thr=0.5697 pass=False
4. potted plant [4] --(resting on)--> potted plant [7] score=0.3785 thr=0.5001 pass=False
5. potted plant [7] --(resting on)--> potted plant [4] score=0.3737 thr=0.5001 pass=False
6. bowl [3] --(on)--> person [0] score=0.2690 thr=0.3692 pass=False
7. potted plant [1] --(beside)--> potted plant [7] score=0.2470 thr=0.5001 pass=False
8. potted plant [1] --(below)--> microwave [8] score=0.2434 thr=0.3004 pass=False
9. person [0] --(in front of)--> microwave [8] score=0.2122 thr=0.2748 pass=False
10. microwave [8] --(behind)--> person [0] score=0.2093 thr=0.3004 pass=False

Frame: 482x640 | Detections: 4 (M2 boxes: 4) | M2 2x identical: True



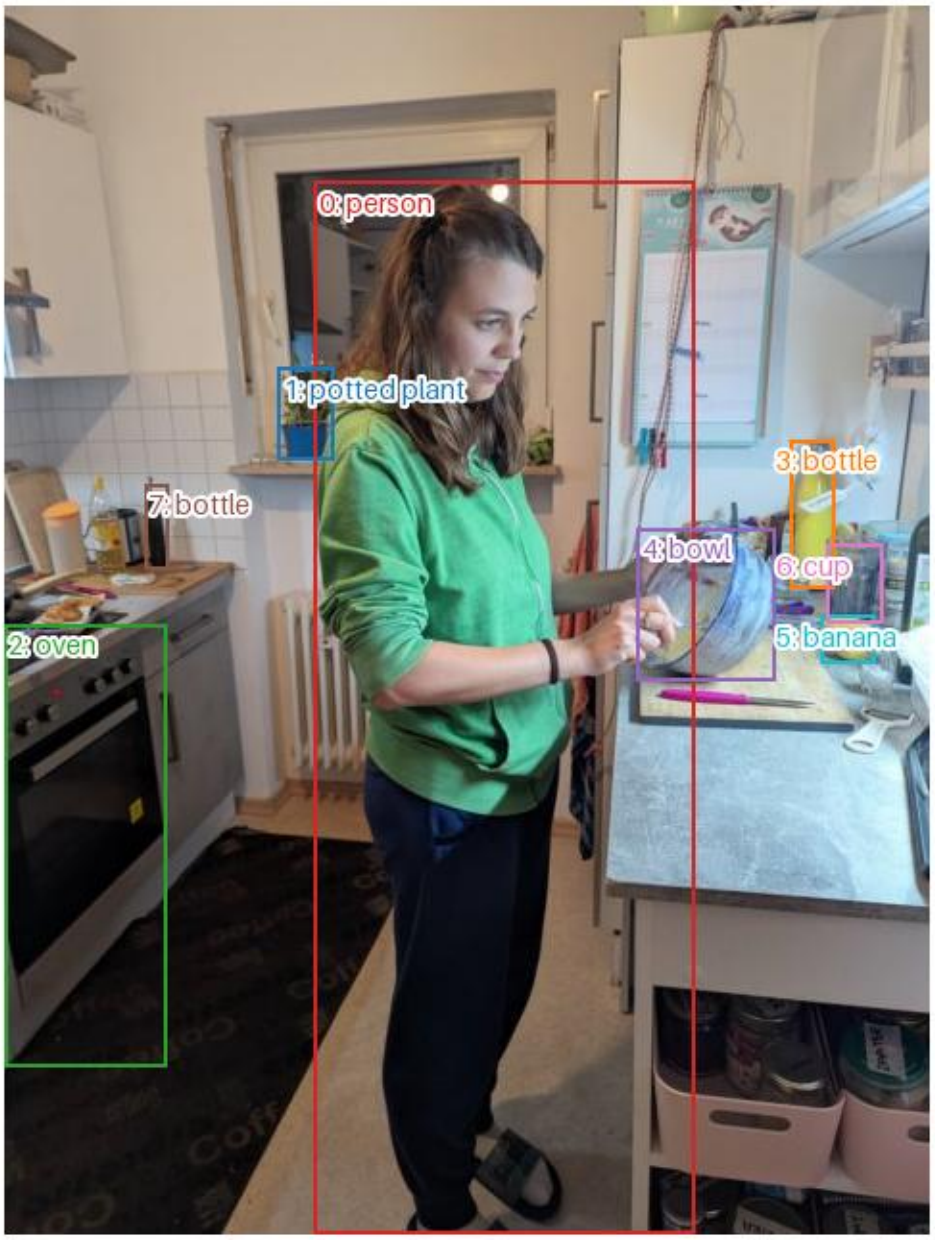

Shown M2 text (relate.context):

M2 relations:
- person [0] below bowl [2].
- bowl [2] above person [0].

Raw top-10 M2 rows (score / threshold / pass):

1. person [0] --(below)--> bowl [2] score=0.4221 thr=0.3004 pass=True
2. person [0] --(using)--> bottle [1] score=0.3950 thr=0.5697 pass=False
3. person [0] --(using)--> bottle [3] score=0.3296 thr=0.5697 pass=False
4. bowl [2] --(above)--> person [0] score=0.3181 thr=0.3064 pass=True
5. bottle [3] --(below)--> bowl [2] score=0.2500 thr=0.3004 pass=False
6. bottle [3] --(beside)--> bottle [1] score=0.2181 thr=0.5001 pass=False
7. bottle [1] --(beside)--> bottle [3] score=0.1878 thr=0.5001 pass=False
8. bottle [1] --(below)--> bowl [2] score=0.1779 thr=0.3004 pass=False
9. bowl [2] --(above)--> bottle [3] score=0.1410 thr=0.3064 pass=False
10. bowl [2] --(above)--> bottle [1] score=0.1166 thr=0.3064 pass=False

Frame: 482x640 | Detections: 8 (M2 boxes: 8) | M2 2x identical: True




Shown M2 text (relate.context):

M2 relations:
- person [0] holding bowl [4].
- person [0] wearing potted plant [1].
- person [0] beside oven [2].

Raw top-10 M2 rows (score / threshold / pass):

1. person [0] --(holding)--> bowl [4] score=0.7991 thr=0.5001 pass=True
2. person [0] --(wearing)--> potted plant [1] score=0.6250 thr=0.5697 pass=True
3. person [0] --(beside)--> oven [2] score=0.5095 thr=0.5001 pass=True
4. bottle [3] --(resting on)--> cup [6] score=0.3221 thr=0.5001 pass=False
5. potted plant [1] --(sitting on)--> person [0] score=0.2892 thr=0.5001 pass=False
6. banana [5] --(below)--> cup [6] score=0.2856 thr=0.3004 pass=False
7. cup [6] --(below)--> bottle [3] score=0.2795 thr=0.3004 pass=False
8. bowl [4] --(on)--> person [0] score=0.2676 thr=0.3692 pass=False
9. cup [6] --(above)--> banana [5] score=0.2623 thr=0.3064 pass=False
10. bottle [3] --(above)--> banana [5] score=0.2406 thr=0.3064 pass=False

Frame: 482x640 | Detections: 2 (M2 boxes: 2) | M2 2x identical: True




Shown M2 text (relate.context):

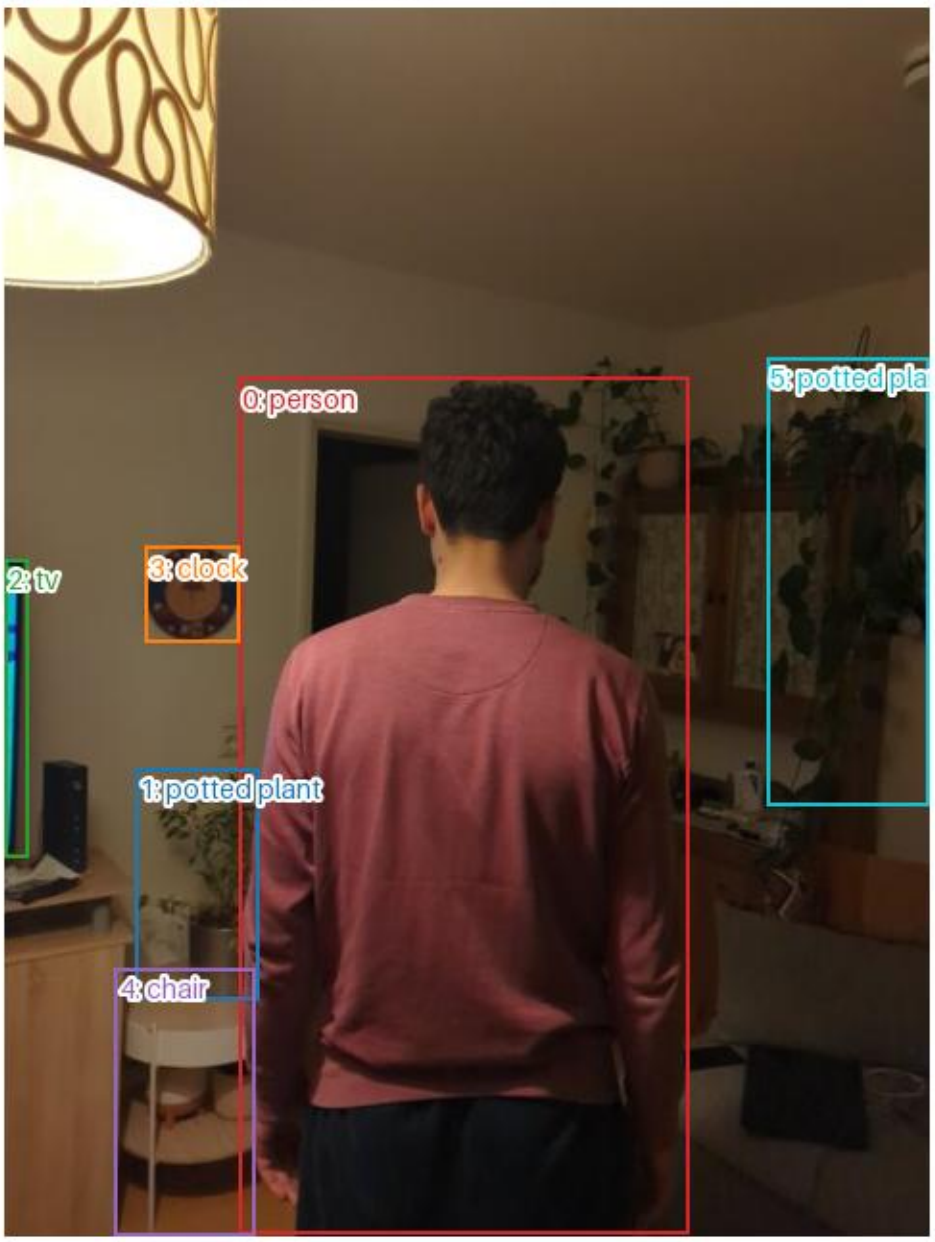
M2 relations:
- person [0] beside person [1].
- person [1] to the left of person [0].

Raw top-10 M2 rows (score / threshold / pass):

1. person [0] --(beside)--> person [1] score=0.5385 thr=0.5001 pass=True
2. person [1] --(to the left of)--> person [0] score=0.3118 thr=0.2703 pass=True

# test_photos/abdel_close_back.jpg -- Page 36 / 61

Frame: 482x640 | Detections: 6 (M2 boxes: 6) | M2 2x identical: True



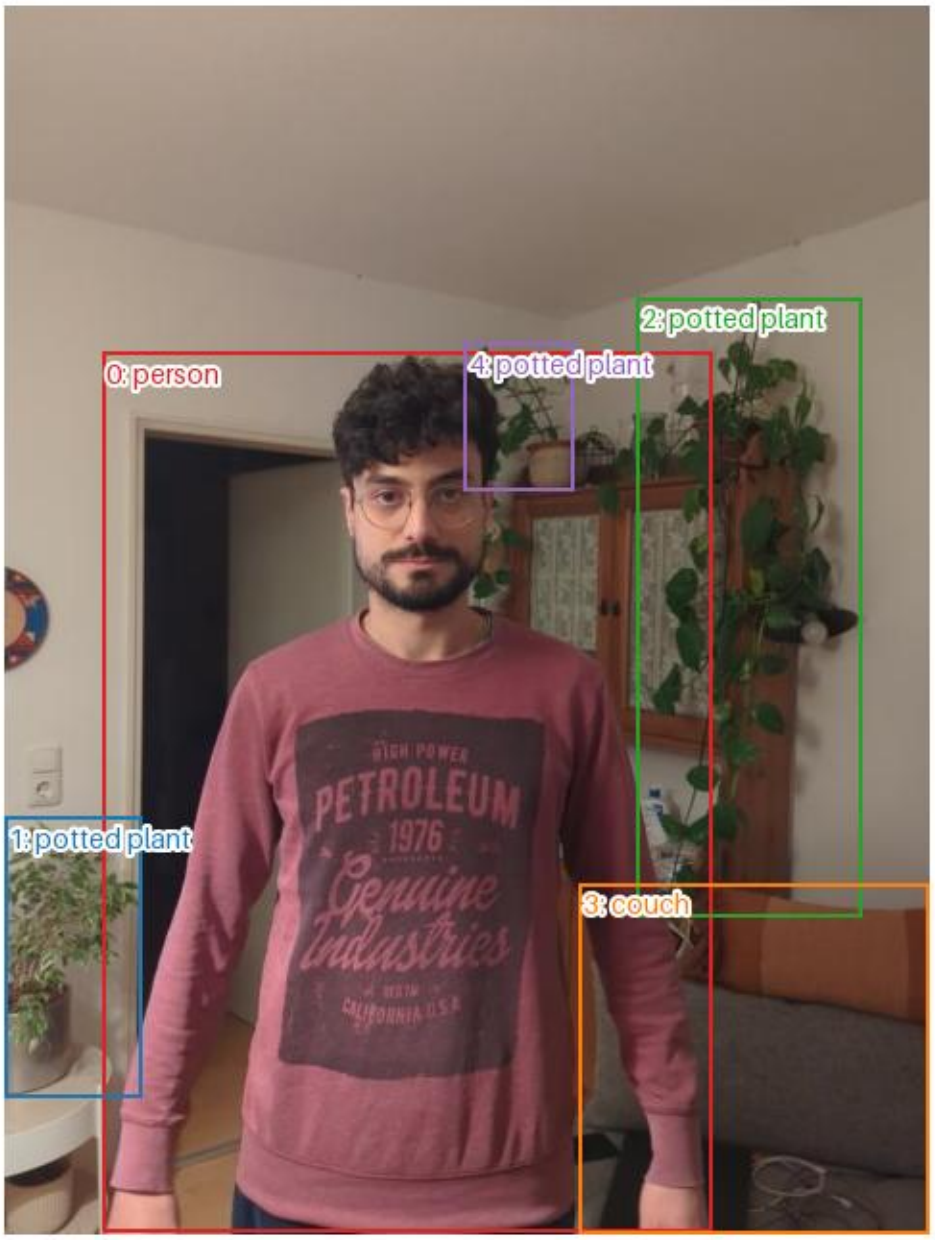

Shown M2 text (relate.context):

M2 relations:
- person [0] watching tv [2].
- person [0] wearing chair [4].
- potted plant [1] resting on chair [4].

Raw top-10 M2 rows (score / threshold / pass):

1. person [0] --(watching)--> tv [2] score=0.7080 thr=0.5308 pass=True
2. person [0] --(wearing)--> chair [4] score=0.6742 thr=0.5697 pass=True
3. potted plant [1] --(resting on)--> chair [4] score=0.5663 thr=0.5001 pass=True
4. person [0] --(wearing)--> potted plant [1] score=0.5569 thr=0.5697 pass=False
5. potted plant [1] --(beside)--> person [0] score=0.4456 thr=0.5001 pass=False
6. person [0] --(beside)--> potted plant [5] score=0.4152 thr=0.5001 pass=False
7. potted plant [5] --(beside)--> person [0] score=0.4101 thr=0.5001 pass=False
8. person [0] --(beside)--> clock [3] score=0.3915 thr=0.5001 pass=False
9. clock [3] --(hanging from)--> person [0] score=0.3135 thr=0.5697 pass=False
10. chair [4] --(behind)--> person [0] score=0.2862 thr=0.3004 pass=False

Frame: 482x640 | Detections: 5 (M2 boxes: 5) | M2 2x identical: True




Shown M2 text (relate.context):

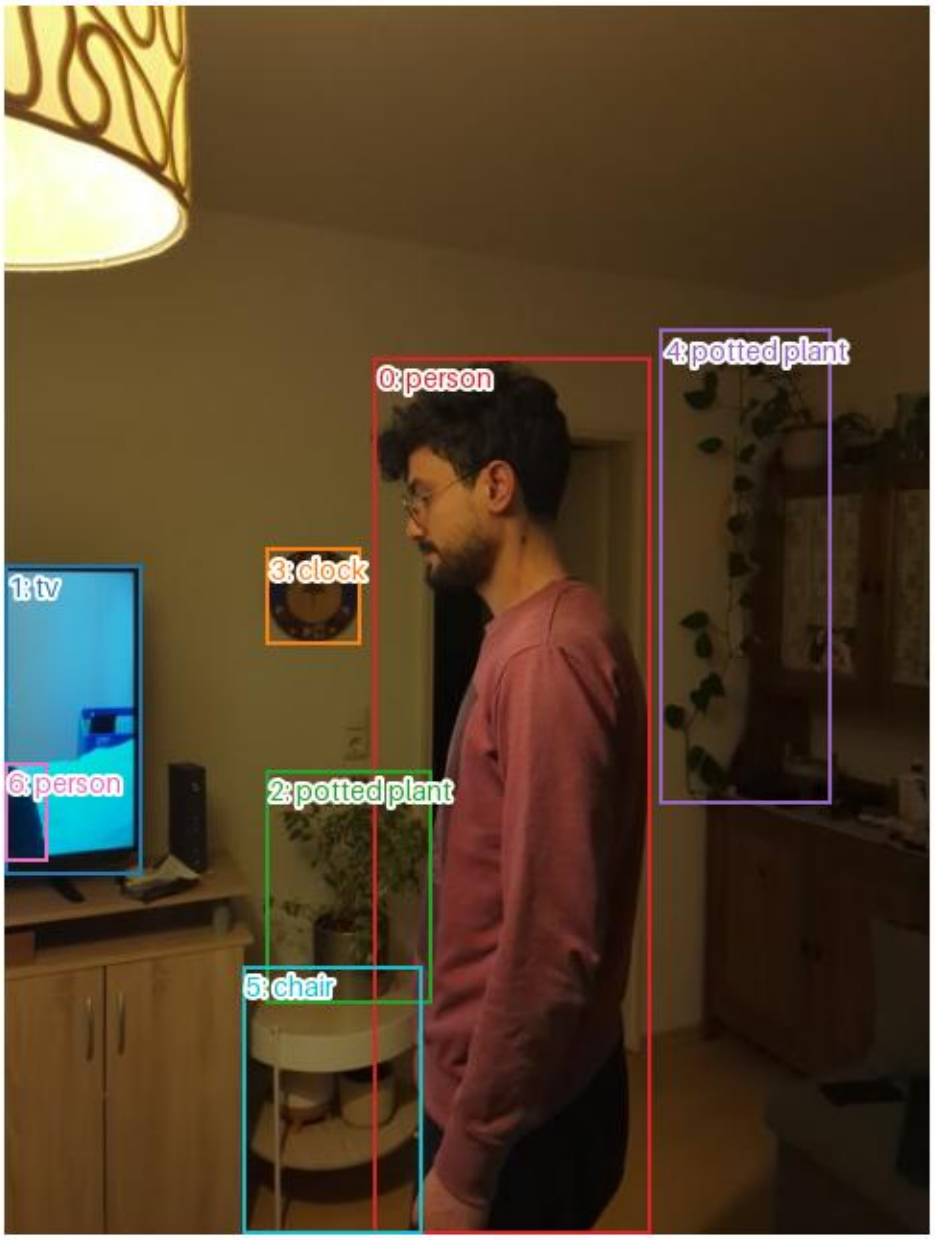
M2 relations:
- person [0] wearing couch [3].
- potted plant [1] to the left of person [0].

Raw top-10 M2 rows (score / threshold / pass):

1. person [0] --(wearing)--> couch [3] score=0.7487 thr=0.5697 pass=True
2. person [0] --(wearing)--> potted plant [2] score=0.5578 thr=0.5697 pass=False
3. person [0] --(wearing)--> potted plant [1] score=0.5336 thr=0.5697 pass=False
4. person [0] --(wearing)--> potted plant [4] score=0.5034 thr=0.5697 pass=False
5. potted plant [2] --(beside)--> person [0] score=0.4758 thr=0.5001 pass=False
6. potted plant [2] --(beside)--> couch [3] score=0.4393 thr=0.5001 pass=False
7. potted plant [4] --(beside)--> potted plant [2] score=0.4297 thr=0.5001 pass=False
8. potted plant [1] --(to the left of)--> person [0] score=0.3954 thr=0.2703 pass=True
9. potted plant [4] --(hanging from)--> person [0] score=0.3705 thr=0.5697 pass=False
10. potted plant [2] --(beside)--> potted plant [4] score=0.3486 thr=0.5001 pass=False

test_photos/abdel_close_left.jpg -- Page 38 / 61

Frame: 482x640 | Detections: 7 (M2 boxes: 7) | M2 2x identical: True



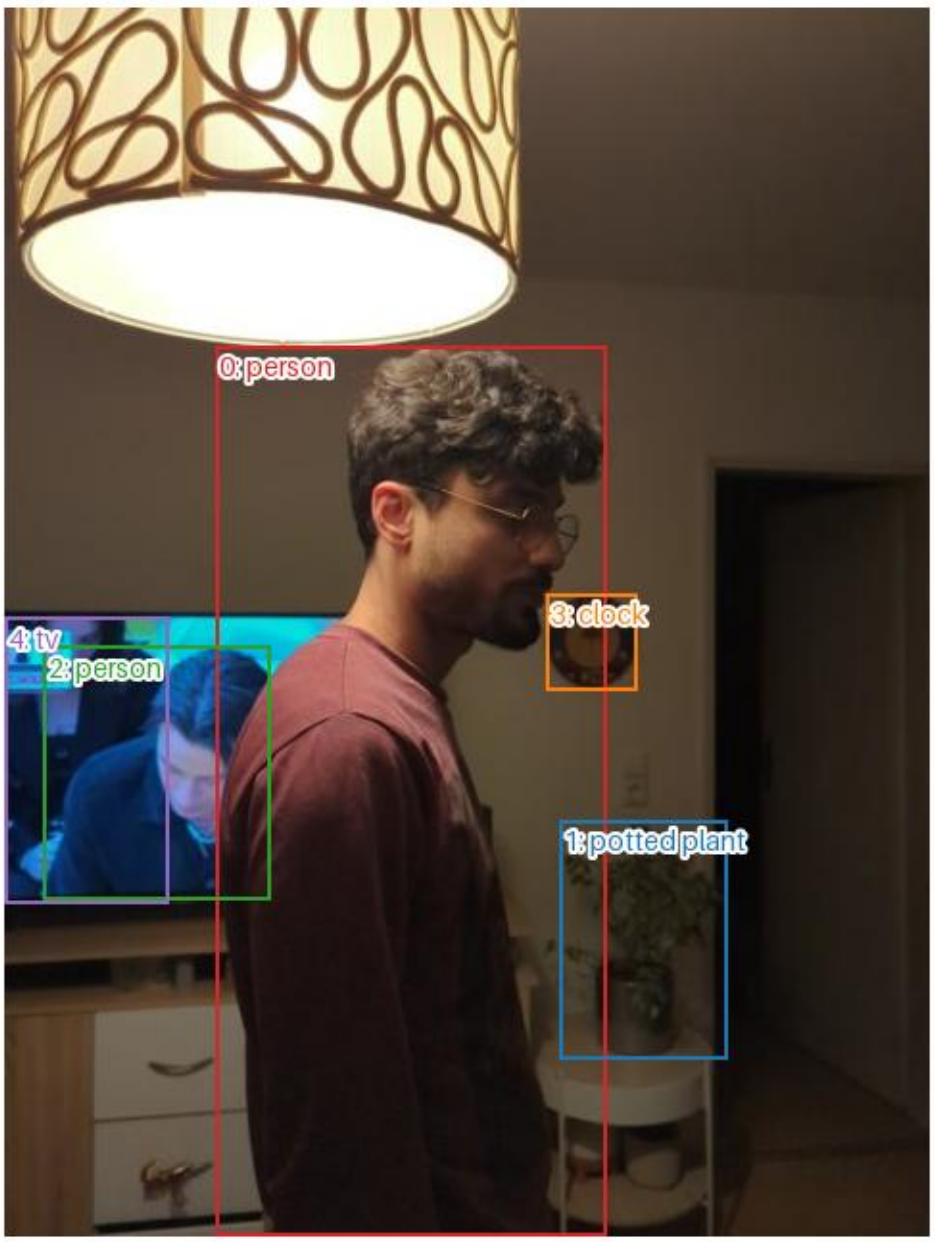

Shown M2 text (relate.context):

M2 relations:
- person [0] watching tv [1].
- person [6] watching tv [1].
- potted plant [2] resting on chair [5].
- person [0] standing beside chair [5].

Raw top-10 M2 rows (score / threshold / pass):

1. person [0] --(watching)--> tv [1] score=0.7728 thr=0.5308 pass=True
2. person [6] --(watching)--> tv [1] score=0.7598 thr=0.5308 pass=True
3. potted plant [2] --(resting on)--> chair [5] score=0.6198 thr=0.5001 pass=True
4. person [0] --(standing beside)--> chair [5] score=0.5754 thr=0.5308 pass=True
5. person [0] --(beside)--> potted plant [2] score=0.4744 thr=0.5001 pass=False
6. potted plant [4] --(beside)--> person [0] score=0.4733 thr=0.5001 pass=False
7. person [0] --(beside)--> potted plant [4] score=0.4608 thr=0.5001 pass=False
8. potted plant [2] --(beside)--> person [0] score=0.3556 thr=0.5001 pass=False
9. person [0] --(beside)--> clock [3] score=0.3546 thr=0.5001 pass=False
10. clock [3] --(hanging from)--> person [0] score=0.3150 thr=0.5697 pass=False

Frame: 482x640 | Detections: 5 (M2 boxes: 5) | M2 2x identical: True



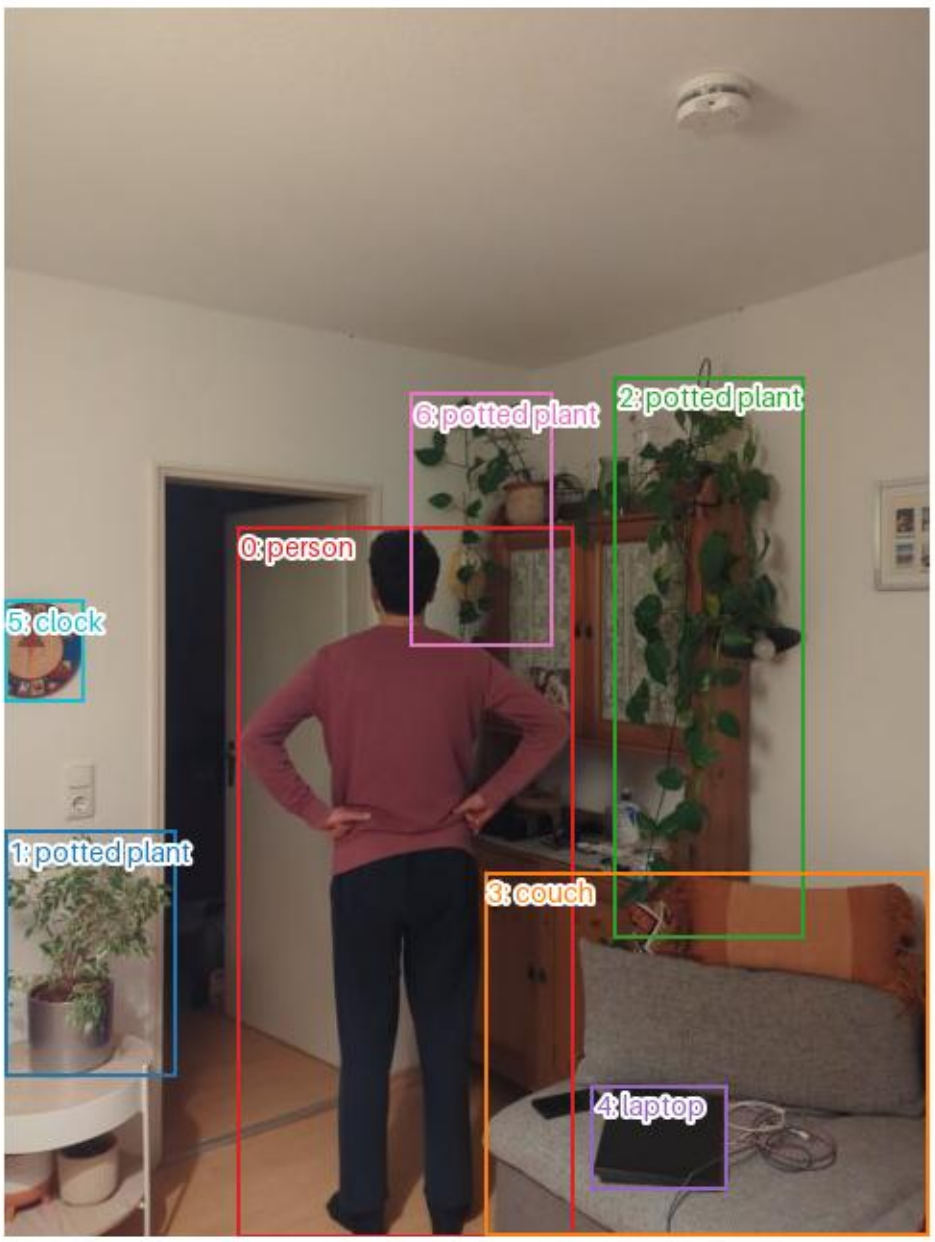

Shown M2 text (relate.context):

M2 relations:
- person [0] beside potted plant [1].
- person [0] beside person [2].
- person [0] wearing clock [3].
- person [2] looking at person [0].
- person [0] watching tv [4].

Raw top-10 M2 rows (score / threshold / pass):

1. person [0] --(beside)--> potted plant [1] score=0.6090 thr=0.5001 pass=True
2. person [0] --(beside)--> person [2] score=0.6019 thr=0.5001 pass=True
3. person [0] --(wearing)--> clock [3] score=0.5978 thr=0.5697 pass=True
4. person [2] --(looking at)--> person [0] score=0.5868 thr=0.4536 pass=True
5. person [0] --(watching)--> tv [4] score=0.5664 thr=0.5308 pass=True
6. tv [4] --(watching)--> person [0] score=0.5564 thr=0.5308 pass=True
7. person [2] --(watching)--> tv [4] score=0.5421 thr=0.5308 pass=True
8. potted plant [1] --(beside)--> person [0] score=0.5217 thr=0.5001 pass=True
9. tv [4] --(watching)--> person [2] score=0.4950 thr=0.5308 pass=False
10. clock [3] --(hanging from)--> person [0] score=0.3553 thr=0.5697 pass=False

Frame: 482x640 | Detections: 7 (M2 boxes: 7) | M2 2x identical: True




Shown M2 text (relate.context):

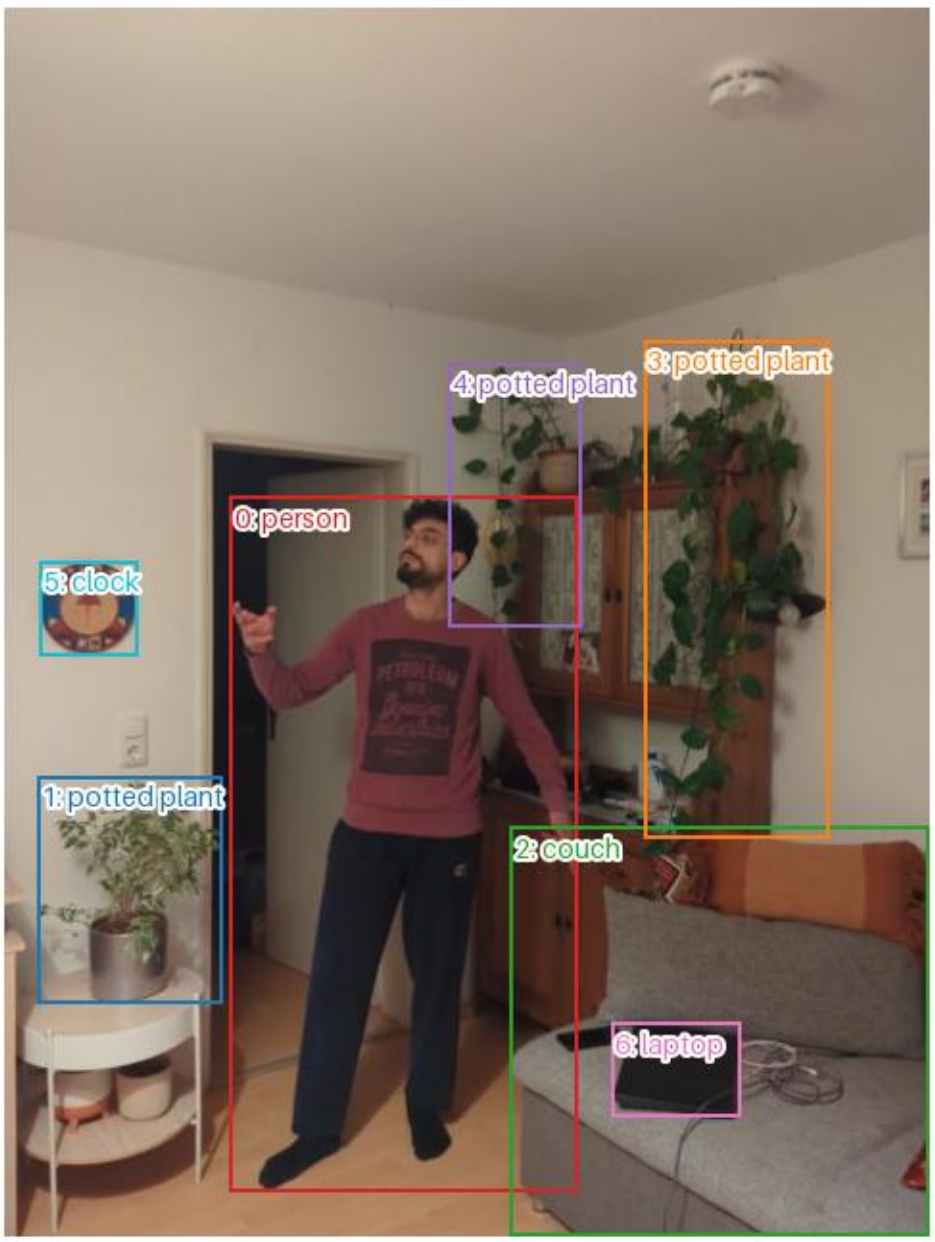
M2 relations:
- person [0] wearing couch [3].
- laptop [4] resting on couch [3].
- potted plant [1] to the left of person [0].

Raw top-10 M2 rows (score / threshold / pass):

1. person [0] --(wearing)--> couch [3] score=0.6791 thr=0.5697 pass=True
2. laptop [4] --(resting on)--> couch [3] score=0.6329 thr=0.5001 pass=True
3. person [0] --(beside)--> potted plant [1] score=0.4473 thr=0.5001 pass=False
4. person [0] --(looking at)--> clock [5] score=0.4375 thr=0.4536 pass=False
5. couch [3] --(holding)--> laptop [4] score=0.4272 thr=0.5001 pass=False
6. potted plant [6] --(beside)--> potted plant [2] score=0.4135 thr=0.5001 pass=False
7. person [0] --(looking at)--> potted plant [2] score=0.3706 thr=0.4536 pass=False
8. potted plant [2] --(beside)--> couch [3] score=0.3470 thr=0.5001 pass=False
9. person [0] --(looking at)--> potted plant [6] score=0.3467 thr=0.4536 pass=False
10. potted plant [1] --(to the left of)--> person [0] score=0.3375 thr=0.2703 pass=True

test_photos/abdel_far_front.jpg -- Page 41 / 61

Frame: 482x640 | Detections: 7 (M2 boxes: 7) | M2 2x identical: True




Shown M2 text (relate.context):

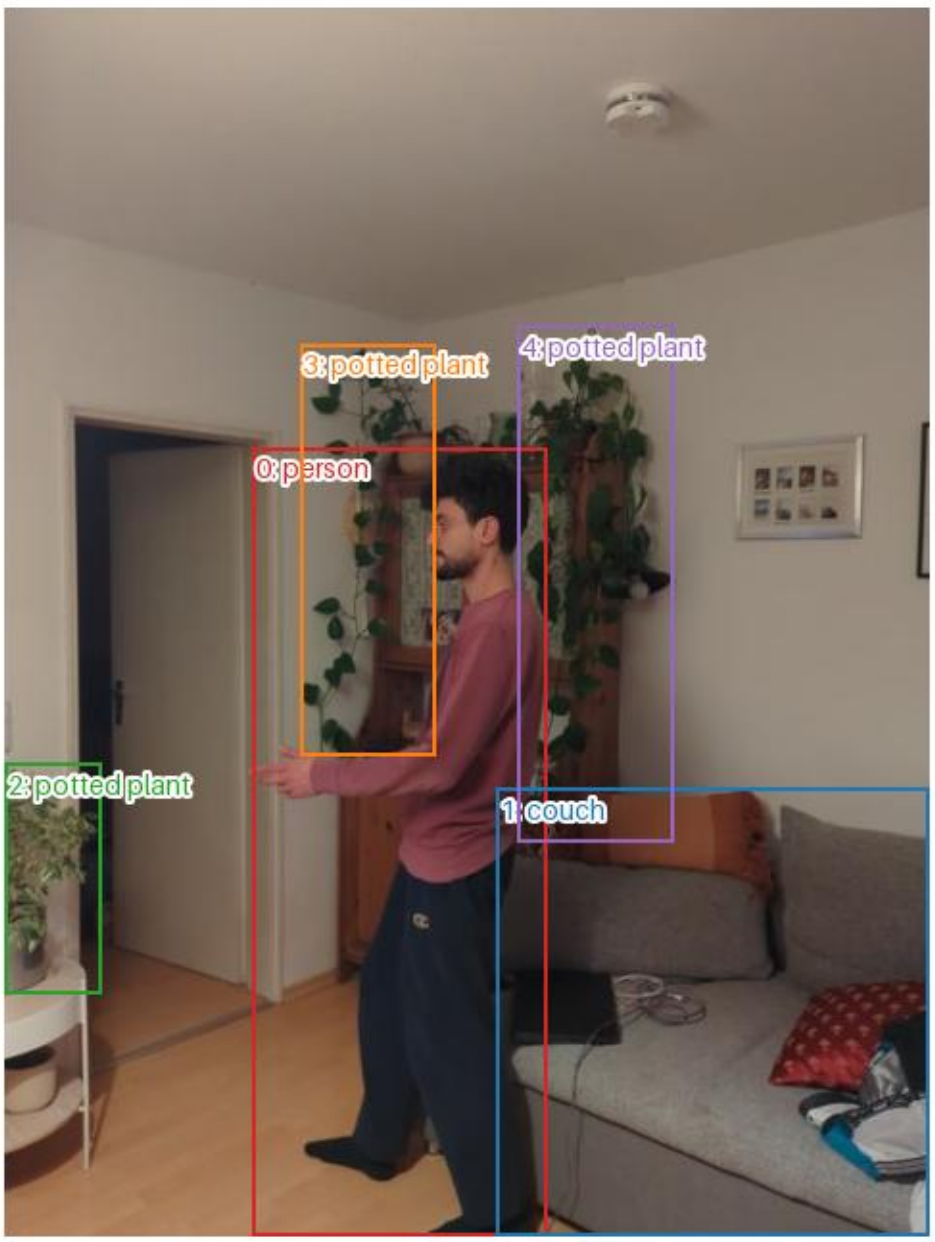
M2 relations:
- person [0] wearing couch [2].
- laptop [6] resting on couch [2].
- person [0] beside potted plant [1].
- couch [2] to the right of person [0].

Raw top-10 M2 rows (score / threshold / pass):

1. person [0] --(wearing)--> couch [2] score=0.6790 thr=0.5697 pass=True
2. laptop [6] --(resting on)--> couch [2] score=0.6311 thr=0.5001 pass=True
3. person [0] --(beside)--> potted plant [1] score=0.5064 thr=0.5001 pass=True
4. person [0] --(wearing)--> potted plant [4] score=0.4352 thr=0.5697 pass=False
5. person [0] --(using)--> laptop [6] score=0.4338 thr=0.5697 pass=False
6. potted plant [1] --(beside)--> person [0] score=0.4009 thr=0.5001 pass=False
7. potted plant [3] --(beside)--> couch [2] score=0.3471 thr=0.5001 pass=False
8. potted plant [4] --(hanging from)--> person [0] score=0.3286 thr=0.5697 pass=False
9. couch [2] --(holding)--> laptop [6] score=0.3148 thr=0.5001 pass=False
10. couch [2] --(to the right of)--> person [0] score=0.3070 thr=0.2703 pass=True

test_photos/abdel_far_left.jpg -- Page 42 / 61

Frame: 482x640 | Detections: 5 (M2 boxes: 5) | M2 2x identical: True



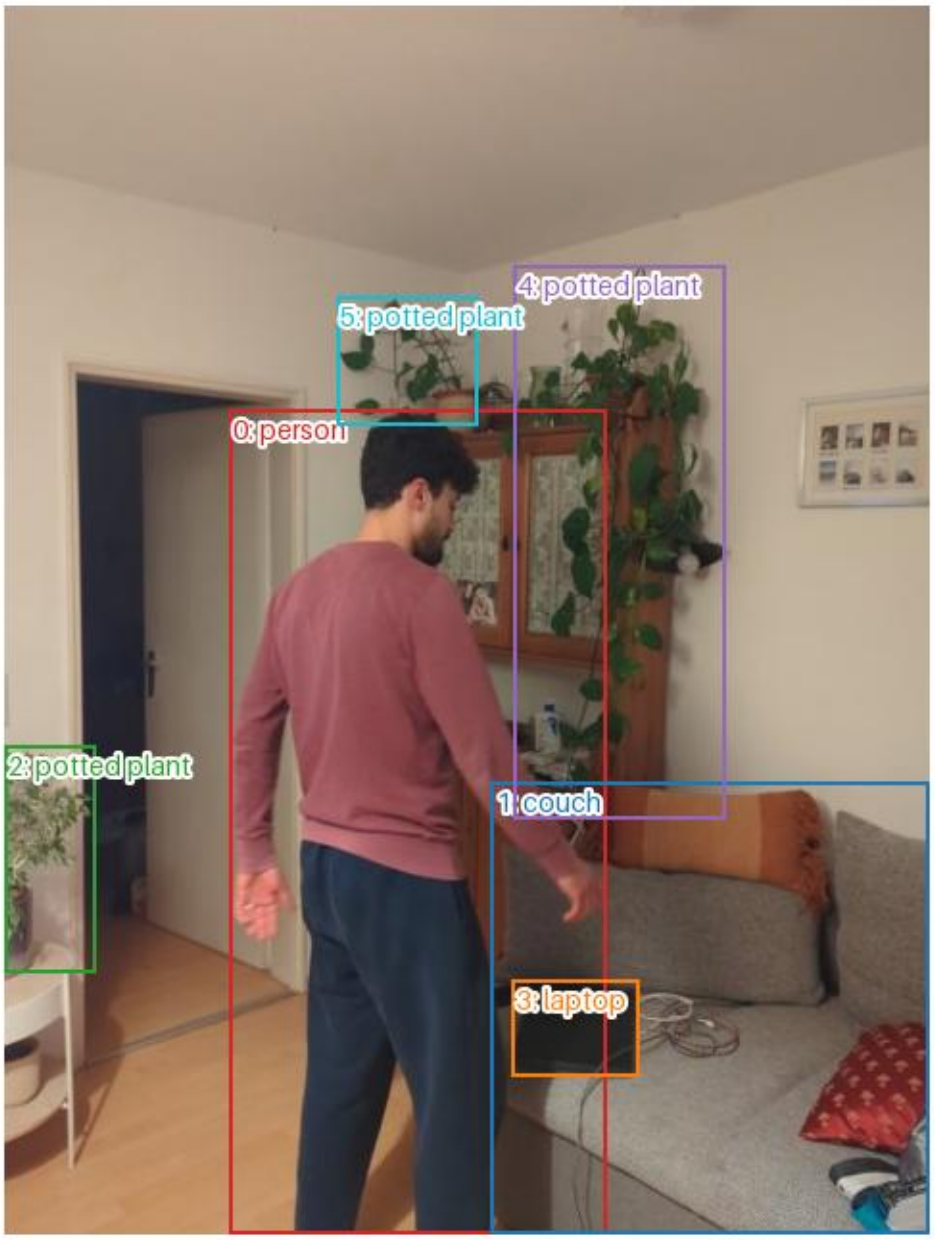

Shown M2 text (relate.context):

M2 relations:
- person [0] wearing couch [1].
- person [0] looking at potted plant [4].
- person [0] looking at potted plant [3].
- potted plant [4] behind person [0].
- potted plant [2] to the left of person [0].

Raw top-10 M2 rows (score / threshold / pass):

1. person [0] --(wearing)--> couch [1] score=0.7430 thr=0.5697 pass=True
2. person [0] --(looking at)--> potted plant [4] score=0.5178 thr=0.4536 pass=True
3. person [0] --(looking at)--> potted plant [3] score=0.5112 thr=0.4536 pass=True
4. potted plant [3] --(hanging from)--> person [0] score=0.4738 thr=0.5697 pass=False
5. person [0] --(beside)--> potted plant [2] score=0.4095 thr=0.5001 pass=False
6. potted plant [4] --(behind)--> person [0] score=0.3556 thr=0.3004 pass=True
7. potted plant [4] --(beside)--> couch [1] score=0.3350 thr=0.5001 pass=False
8. potted plant [3] --(beside)--> potted plant [4] score=0.3200 thr=0.5001 pass=False
9. potted plant [2] --(to the left of)--> person [0] score=0.3128 thr=0.2703 pass=True
10. couch [1] --(to the right of)--> person [0] score=0.3013 thr=0.2703 pass=True

Frame: 482x640 | Detections: 6 (M2 boxes: 6) | M2 2x identical: True



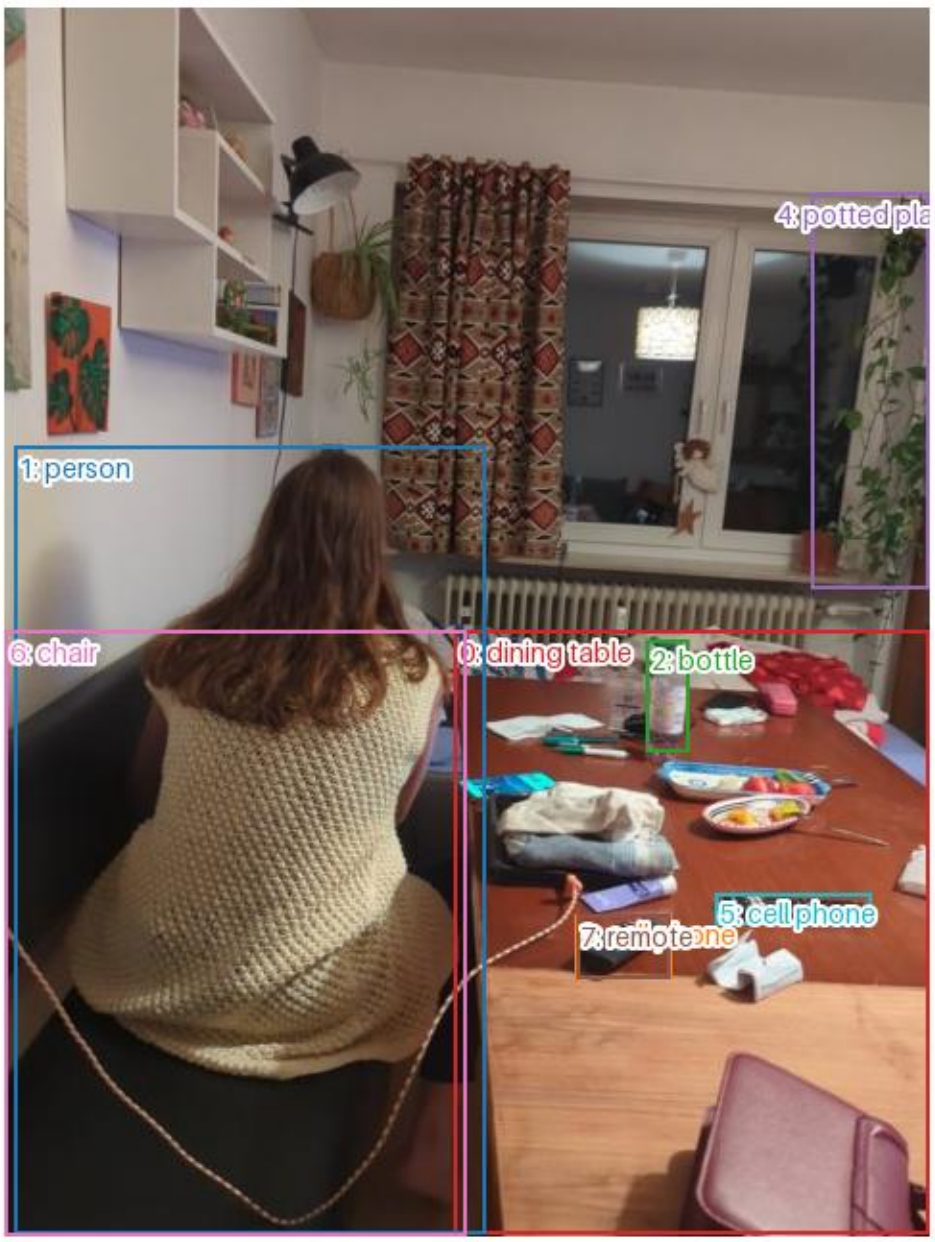

Shown M2 text (relate.context):

M2 relations:
- person [0] wearing couch [1].
- laptop [3] resting on couch [1].

Raw top-10 M2 rows (score / threshold / pass):

1. person [0] --(wearing)--> couch [1] score=0.7183 thr=0.5697 pass=True
2. laptop [3] --(resting on)--> couch [1] score=0.5901 thr=0.5001 pass=True
3. person [0] --(using)--> laptop [3] score=0.5635 thr=0.5697 pass=False
4. person [0] --(beside)--> potted plant [2] score=0.4608 thr=0.5001 pass=False
5. potted plant [4] --(beside)--> couch [1] score=0.4050 thr=0.5001 pass=False
6. person [0] --(wearing)--> potted plant [4] score=0.3449 thr=0.5697 pass=False
7. couch [1] --(holding)--> laptop [3] score=0.3356 thr=0.5001 pass=False
8. person [0] --(wearing)--> potted plant [5] score=0.3215 thr=0.5697 pass=False
9. potted plant [4] --(beside)--> potted plant [5] score=0.3172 thr=0.5001 pass=False
10. potted plant [4] --(behind)--> person [0] score=0.2937 thr=0.3004 pass=False

Frame: 482x640 | Detections: 8 (M2 boxes: 8) | M2 2x identical: True



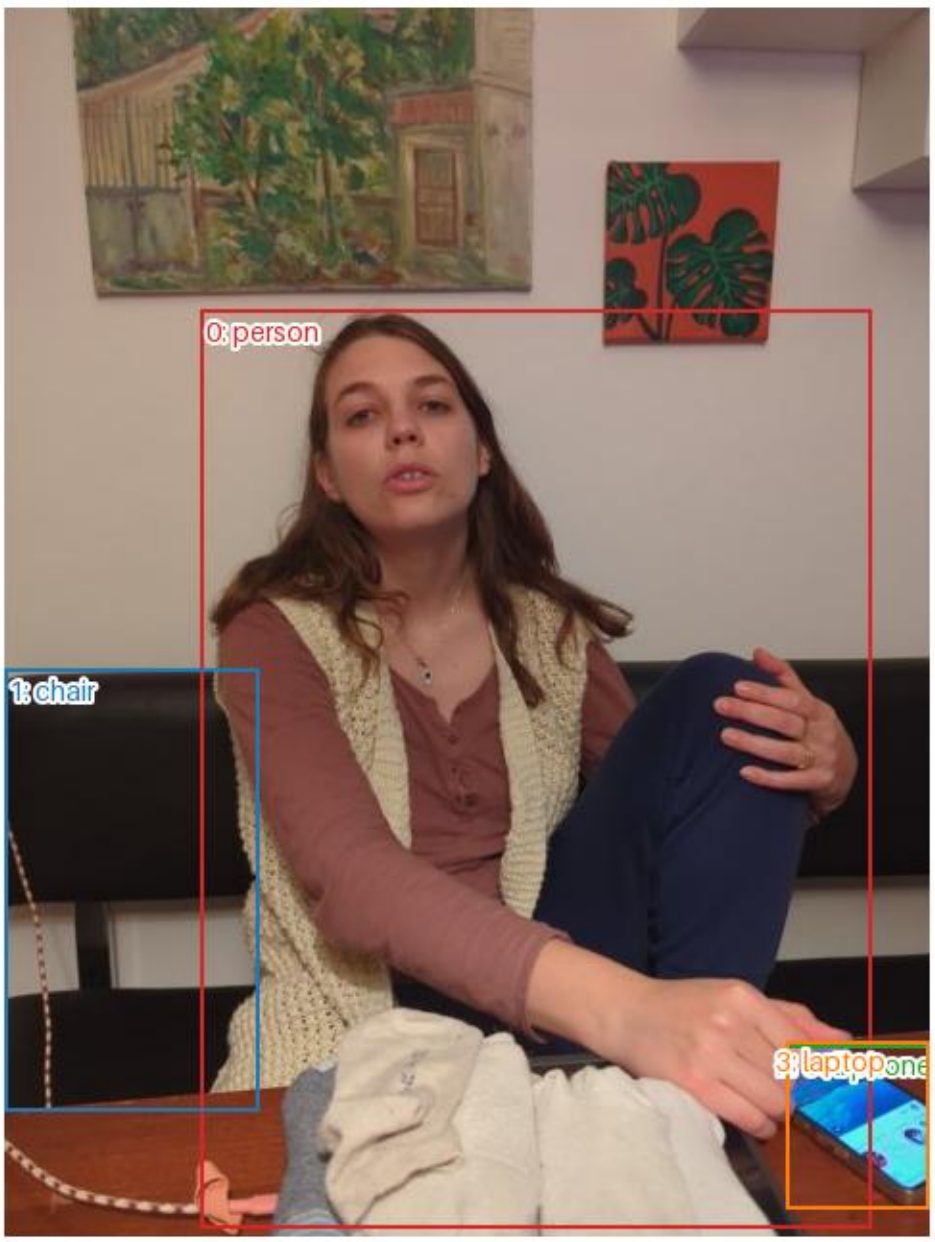

Shown M2 text (relate.context):

M2 relations:
- person [1] sitting in chair [6].
- cell phone [3] resting on dining table [0].
- remote [7] resting on dining table [0].
- cell phone [5] resting on dining table [0].
- bottle [2] resting on dining table [0].

Raw top-10 M2 rows (score / threshold / pass):

1. person [1] --(sitting in)--> chair [6] score=0.7691 thr=0.5001 pass=True
2. cell phone [3] --(resting on)--> dining table [0] score=0.7416 thr=0.5001 pass=True
3. remote [7] --(resting on)--> dining table [0] score=0.7416 thr=0.5001 pass=True
4. cell phone [5] --(resting on)--> dining table [0] score=0.7144 thr=0.5001 pass=True
5. bottle [2] --(resting on)--> dining table [0] score=0.6913 thr=0.5001 pass=True
6. person [1] --(beside)--> dining table [0] score=0.6518 thr=0.5001 pass=True
7. dining table [0] --(beside)--> chair [6] score=0.4537 thr=0.5001 pass=False
8. potted plant [4] --(beside)--> dining table [0] score=0.3599 thr=0.5001 pass=False
9. chair [6] --(beside)--> dining table [0] score=0.3337 thr=0.5001 pass=False
10. dining table [0] --(beside)--> potted plant [4] score=0.2773 thr=0.5001 pass=False

Frame: 482x640 | Detections: 4 (M2 boxes: 4) | M2 2x identical: True




Shown M2 text (relate.context):

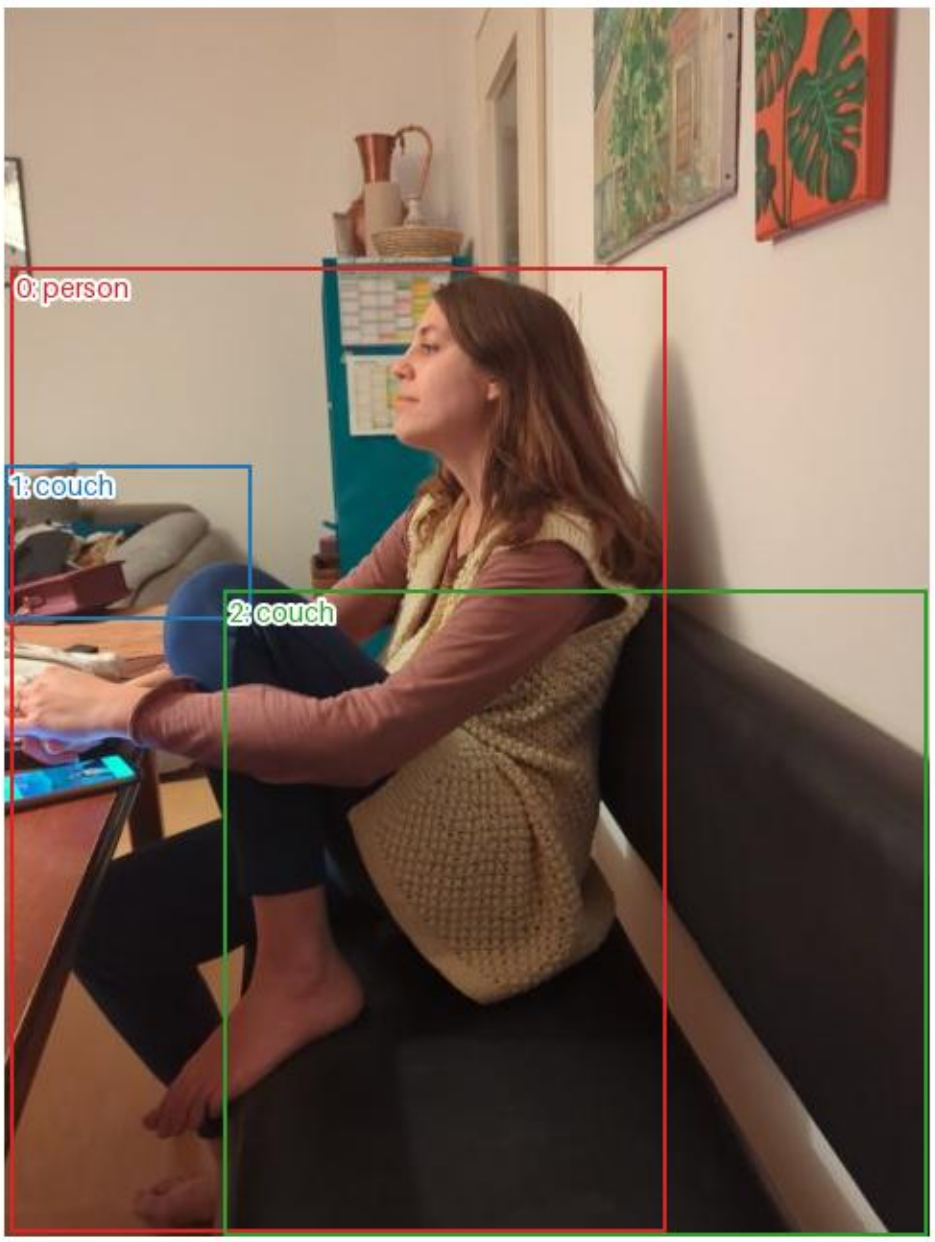
M2 relations:
- person [0] sitting on chair [1].
- person [0] holding laptop [3].
- person [0] holding cell phone [2].
- laptop [3] on cell phone [2].
- cell phone [2] behind laptop [3].

Raw top-10 M2 rows (score / threshold / pass):

1. person [0] --(sitting on)--> chair [1] score=0.6304 thr=0.5001 pass=True
2. person [0] --(holding)--> laptop [3] score=0.5811 thr=0.5001 pass=True
3. person [0] --(holding)--> cell phone [2] score=0.5766 thr=0.5001 pass=True
4. laptop [3] --(on)--> cell phone [2] score=0.4601 thr=0.3692 pass=True
5. cell phone [2] --(behind)--> laptop [3] score=0.4062 thr=0.3004 pass=True
6. laptop [3] --(on)--> person [0] score=0.4044 thr=0.3692 pass=True
7. cell phone [2] --(on)--> person [0] score=0.4032 thr=0.3692 pass=True
8. chair [1] --(behind)--> person [0] score=0.2739 thr=0.3004 pass=False
9. cell phone [2] --(to the right of)--> chair [1] score=0.1792 thr=0.2703 pass=False
10. laptop [3] --(to the right of)--> chair [1] score=0.1778 thr=0.2703 pass=False

test_photos/chiara_close_left.jpg -- Page 46 / 61

Frame: 482x640 | Detections: 3 (M2 boxes: 3) | M2 2x identical: True



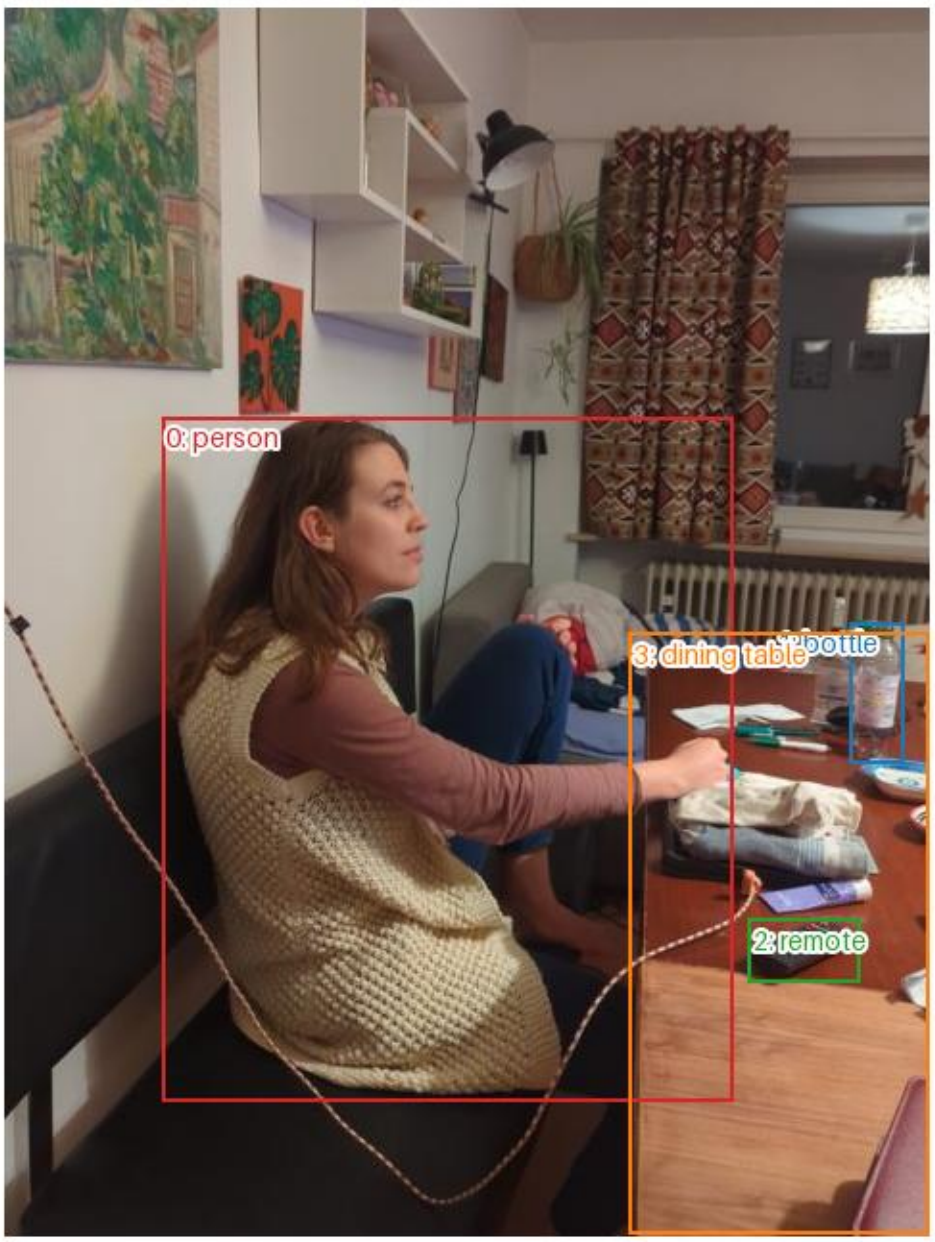

Shown M2 text (relate.context):

M2 relations:
- person [0] sitting on couch [2].
- person [0] leaning against couch [1].
- couch [1] to the left of couch [2].

Raw top-10 M2 rows (score / threshold / pass):

1. person [0] --(sitting on)--> couch [2] score=0.7932 thr=0.5001 pass=True
2. person [0] --(leaning against)--> couch [1] score=0.6054 thr=0.5308 pass=True
3. couch [1] --(to the left of)--> couch [2] score=0.3684 thr=0.2703 pass=True
4. couch [1] --(behind)--> person [0] score=0.2628 thr=0.3004 pass=False
5. couch [2] --(to the right of)--> couch [1] score=0.1924 thr=0.2703 pass=False
6. couch [2] --(behind)--> person [0] score=0.0750 thr=0.3004 pass=False

Frame: 482x640 | Detections: 4 (M2 boxes: 4) | M2 2x identical: True




Shown M2 text (relate.context):

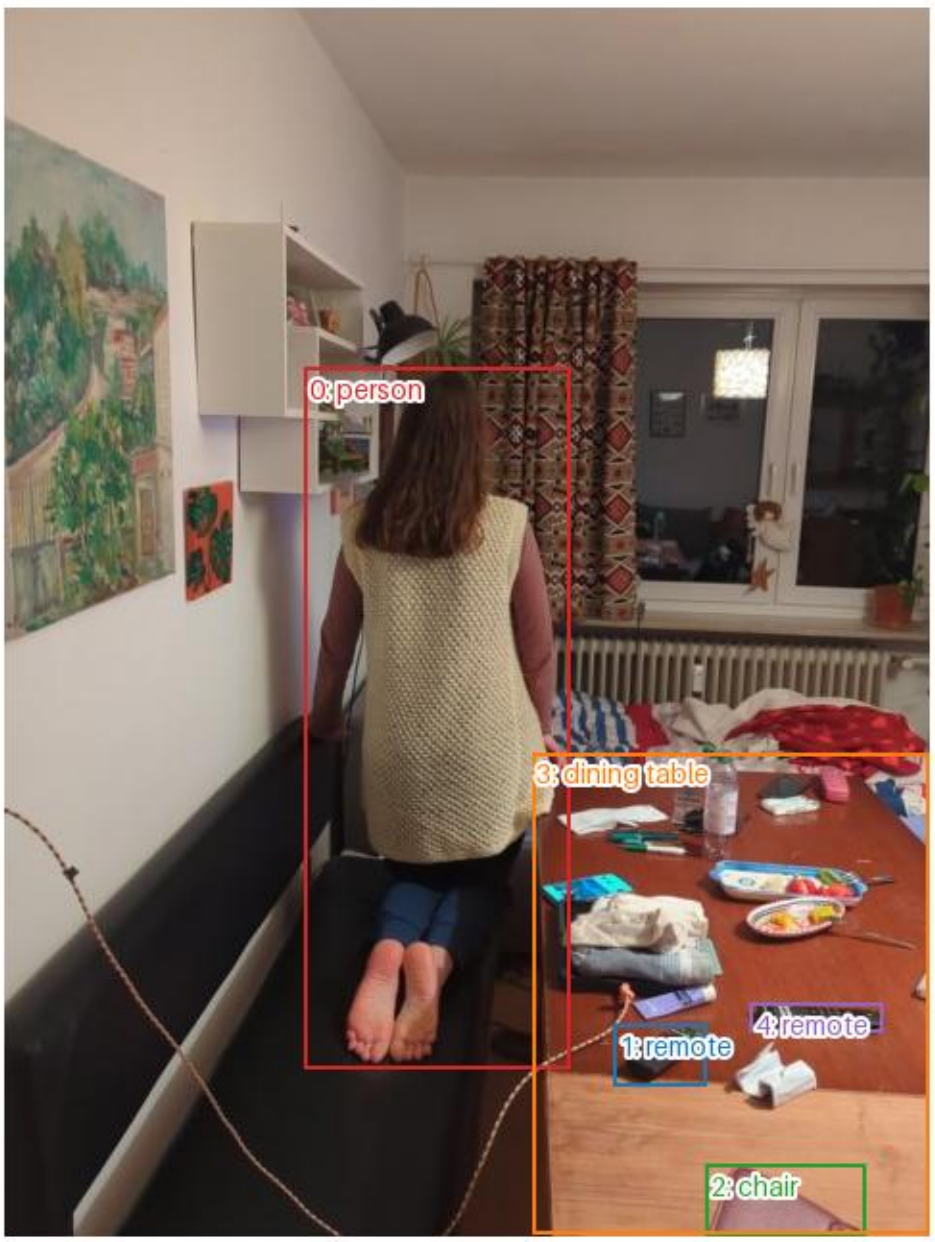
M2 relations:
- remote [2] resting on dining table [3].
- bottle [1] resting on dining table [3].
- person [0] beside dining table [3].
- remote [2] below bottle [1].

Raw top-10 M2 rows (score / threshold / pass):

1. remote [2] --(resting on)--> dining table [3] score=0.7321 thr=0.5001 pass=True
2. bottle [1] --(resting on)--> dining table [3] score=0.7254 thr=0.5001 pass=True
3. person [0] --(using)--> remote [2] score=0.5397 thr=0.5697 pass=False
4. person [0] --(beside)--> dining table [3] score=0.5194 thr=0.5001 pass=True
5. remote [2] --(below)--> bottle [1] score=0.3160 thr=0.3004 pass=True
6. bottle [1] --(above)--> remote [2] score=0.2786 thr=0.3064 pass=False
7. dining table [3] --(holding)--> bottle [1] score=0.2521 thr=0.5001 pass=False
8. dining table [3] --(holding)--> remote [2] score=0.2017 thr=0.5001 pass=False
9. dining table [3] --(behind)--> person [0] score=0.1608 thr=0.3004 pass=False
10. bottle [1] --(to the right of)--> person [0] score=0.1080 thr=0.2703 pass=False

# test_photos/chiara_far_back.jpg -- Page 48 / 61

Frame: 482x640 | Detections: 5 (M2 boxes: 5) | M2 2x identical: True




Shown M2 text (relate.context):

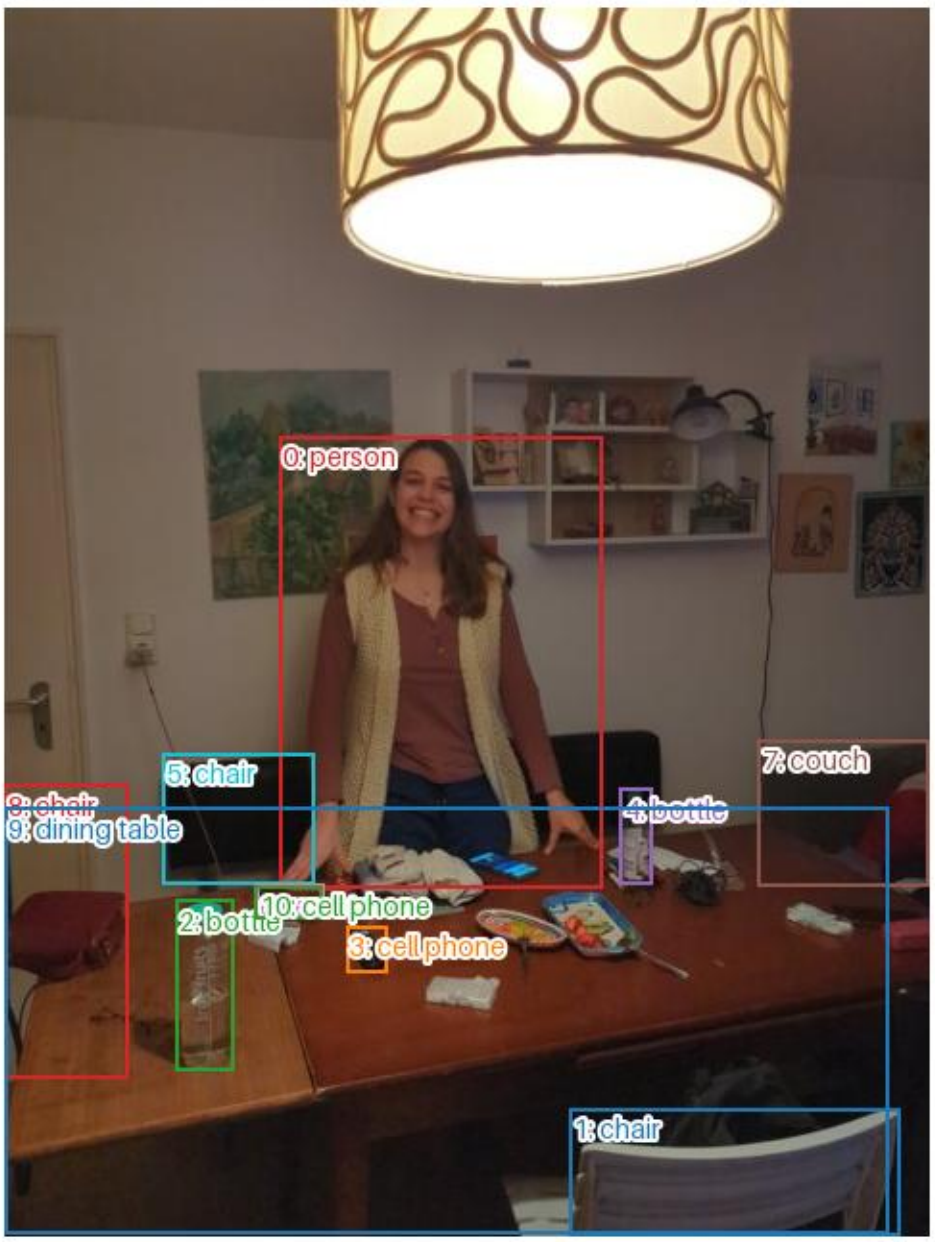
M2 relations:
- remote [1] resting on dining table [3].
- remote [4] resting on dining table [3].
- chair [2] resting on dining table [3].

Raw top-10 M2 rows (score / threshold / pass):

1. remote [1] --(resting on)--> dining table [3] score=0.7495 thr=0.5001 pass=True
2. remote [4] --(resting on)--> dining table [3] score=0.7296 thr=0.5001 pass=True
3. chair [2] --(resting on)--> dining table [3] score=0.6832 thr=0.5001 pass=True
4. person [0] --(wearing)--> dining table [3] score=0.5534 thr=0.5697 pass=False
5. dining table [3] --(holding)--> remote [4] score=0.3239 thr=0.5001 pass=False
6. dining table [3] --(holding)--> chair [2] score=0.3198 thr=0.5001 pass=False
7. dining table [3] --(holding)--> remote [1] score=0.3114 thr=0.5001 pass=False
8. remote [4] --(to the right of)--> remote [1] score=0.2441 thr=0.2703 pass=False
9. remote [1] --(to the left of)--> remote [4] score=0.2376 thr=0.2703 pass=False
10. person [0] --(using)--> remote [1] score=0.2074 thr=0.5697 pass=False

# test_photos/chiara_far_front.jpg -- Page 49 / 61

Frame: 482x640 | Detections: 11 (M2 boxes: 11) | M2 2x identical: True




Shown M2 text (relate.context):

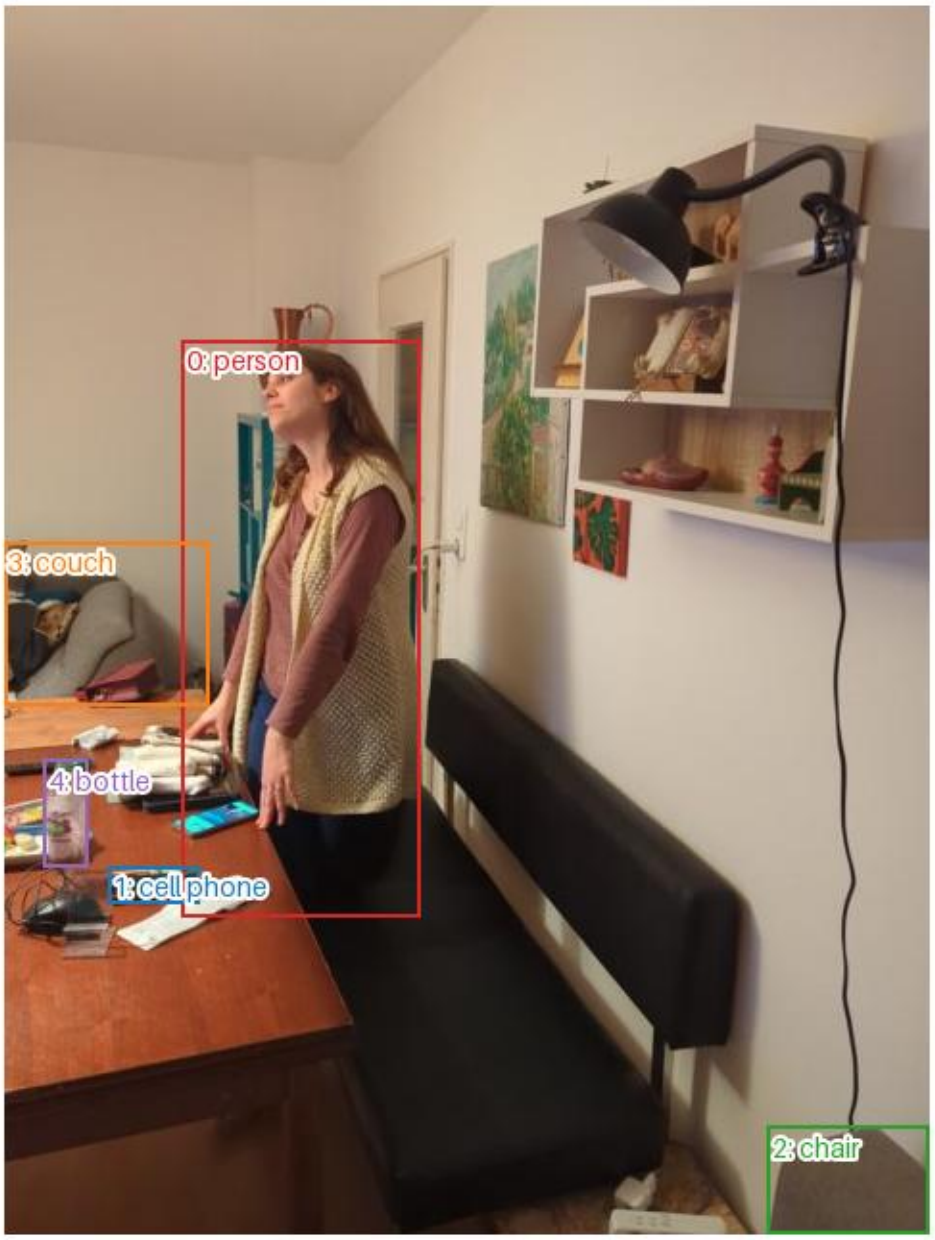
M2 relations:
- bottle [2] resting on dining table [9].
- cell phone [3] resting on dining table [9].
- cell phone [10] resting on dining table [9].
- remote [6] resting on dining table [9].
- bottle [4] resting on dining table [9].

Raw top-10 M2 rows (score / threshold / pass):

1. bottle [2] --(resting on)--> dining table [9] score=0.7290 thr=0.5001 pass=True
2. cell phone [3] --(resting on)--> dining table [9] score=0.7057 thr=0.5001 pass=True
3. cell phone [10] --(resting on)--> dining table [9] score=0.7051 thr=0.5001 pass=True
4. remote [6] --(resting on)--> dining table [9] score=0.7050 thr=0.5001 pass=True
5. bottle [4] --(resting on)--> dining table [9] score=0.6685 thr=0.5001 pass=True
6. chair [8] --(beside)--> dining table [9] score=0.6551 thr=0.5001 pass=True
7. chair [1] --(beside)--> dining table [9] score=0.6475 thr=0.5001 pass=True
8. person [0] --(wearing)--> dining table [9] score=0.5920 thr=0.5697 pass=True
9. person [0] --(wearing)--> chair [5] score=0.5885 thr=0.5697 pass=True
10. chair [5] --(beside)--> dining table [9] score=0.4294 thr=0.5001 pass=False

# test_photos/chiara_far_left.jpg -- Page 50 / 61

Frame: 482x640 | Detections: 5 (M2 boxes: 5) | M2 2x identical: True




Shown M2 text (relate.context):

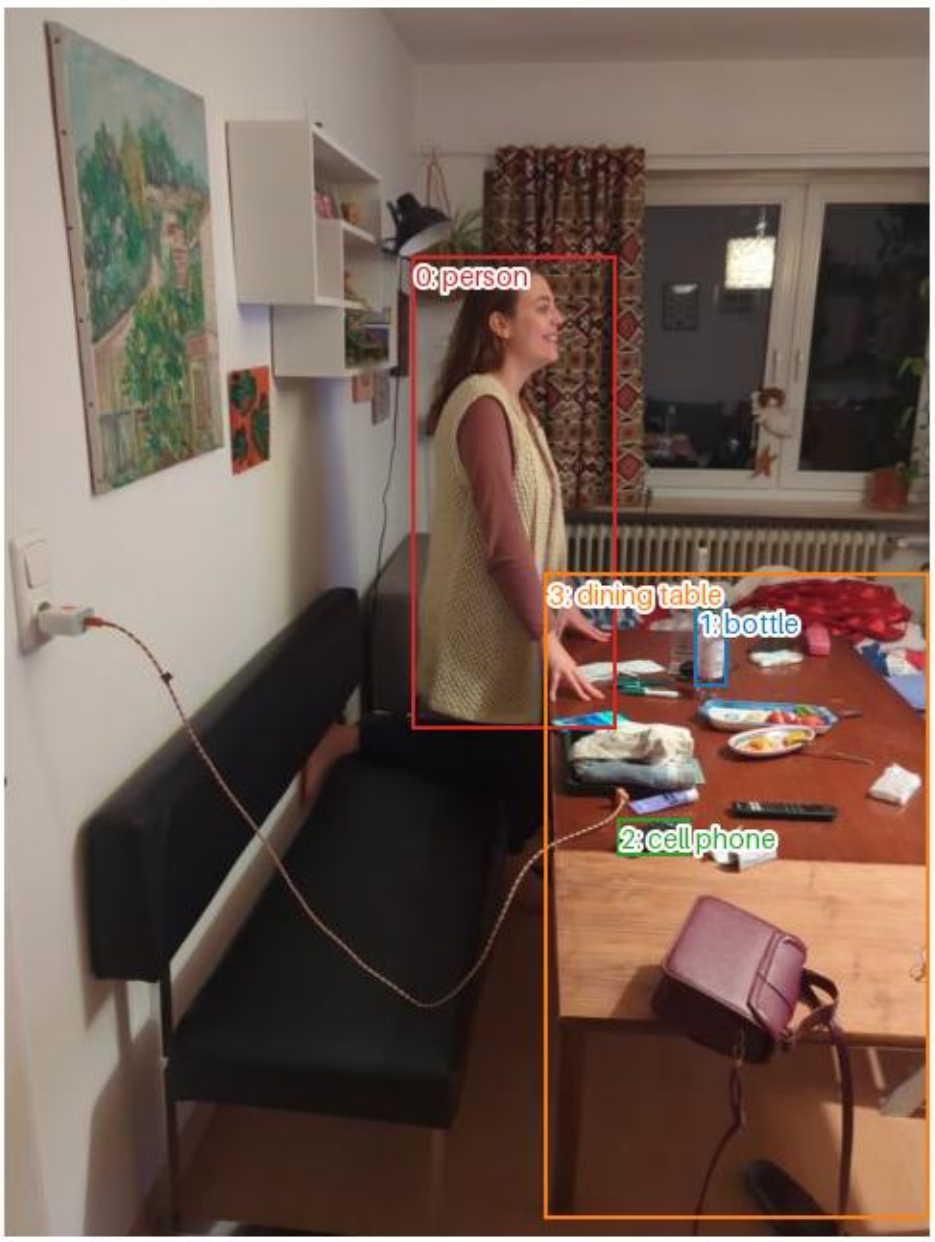
M2 relations:
- person [0] standing beside couch [3].

Raw top-10 M2 rows (score / threshold / pass):

1. person [0] --(standing beside)--> couch [3] score=0.5657 thr=0.5308 pass=True
2. person [0] --(wearing)--> cell phone [1] score=0.4829 thr=0.5697 pass=False
3. person [0] --(standing beside)--> chair [2] score=0.3493 thr=0.5308 pass=False
4. couch [3] --(behind)--> person [0] score=0.2627 thr=0.3004 pass=False
5. person [0] --(standing beside)--> bottle [4] score=0.2622 thr=0.5308 pass=False
6. chair [2] --(beside)--> person [0] score=0.2148 thr=0.5001 pass=False
7. bottle [4] --(to the left of)--> cell phone [1] score=0.1625 thr=0.2703 pass=False
8. chair [2] --(to the right of)--> couch [3] score=0.1437 thr=0.2703 pass=False
9. bottle [4] --(below)--> couch [3] score=0.1355 thr=0.3004 pass=False
10. couch [3] --(to the left of)--> chair [2] score=0.1308 thr=0.2703 pass=False

test_photos/chiara_far_right.jpg -- Page 51 / 61

Frame: 482x640 | Detections: 4 (M2 boxes: 4) | M2 2x identical: True




Shown M2 text (relate.context):

M2 relations:
- bottle [1] resting on dining table [3].
- cell phone [2] resting on dining table [3].
- person [0] leaning against dining table [3].
- dining table [3] in front of person [0].

Raw top-10 M2 rows (score / threshold / pass):

1. bottle [1] --(resting on)--> dining table [3] score=0.7213 thr=0.5001 pass=True
2. cell phone [2] --(resting on)--> dining table [3] score=0.6995 thr=0.5001 pass=True
3. person [0] --(leaning against)--> dining table [3] score=0.6531 thr=0.5308 pass=True
4. dining table [3] --(in front of)--> person [0] score=0.2918 thr=0.2748 pass=True
5. cell phone [2] --(below)--> bottle [1] score=0.2659 thr=0.3004 pass=False
6. dining table [3] --(holding)--> bottle [1] score=0.2614 thr=0.5001 pass=False
7. dining table [3] --(holding)--> cell phone [2] score=0.2278 thr=0.5001 pass=False
8. person [0] --(to the left of)--> bottle [1] score=0.2191 thr=0.2703 pass=False
9. bottle [1] --(above)--> cell phone [2] score=0.1963 thr=0.3064 pass=False
10. bottle [1] --(to the right of)--> person [0] score=0.1893 thr=0.2703 pass=False

# test_photos/empty_1.jpg -- Page 52 / 61

Frame: 482x640 | Detections: 3 (M2 boxes: 3) | M2 2x identical: True




Shown M2 text (relate.context):

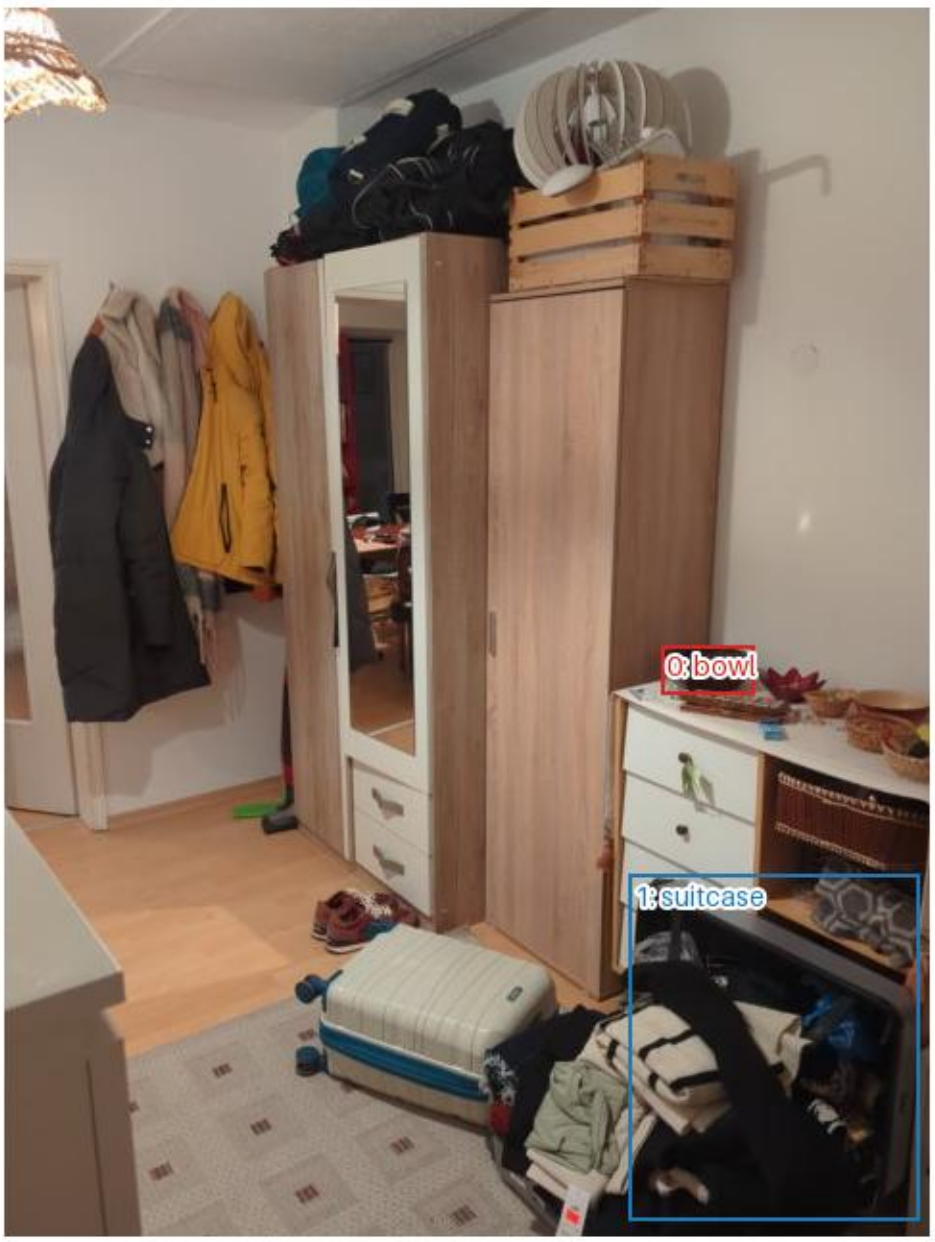
M2 relations:
- bottle [1] resting on bottle [2].
- bottle [1] to the left of toilet [0].
- bottle [2] to the left of toilet [0].
- toilet [0] to the right of bottle [2].
- toilet [0] to the right of bottle [1].

Raw top-10 M2 rows (score / threshold / pass):

1. bottle [1] --(resting on)--> bottle [2] score=0.6340 thr=0.5001 pass=True
2. bottle [2] --(contained in)--> bottle [1] score=0.4028 thr=0.5697 pass=False
3. bottle [1] --(to the left of)--> toilet [0] score=0.3909 thr=0.2703 pass=True
4. bottle [2] --(to the left of)--> toilet [0] score=0.3870 thr=0.2703 pass=True
5. toilet [0] --(to the right of)--> bottle [2] score=0.3749 thr=0.2703 pass=True
6. toilet [0] --(to the right of)--> bottle [1] score=0.3325 thr=0.2703 pass=True

test_photos/empty_2.jpg -- Page 53 / 61

Frame: 482x640 | Detections: 2 (M2 boxes: 2) | M2 2x identical: True




Shown M2 text (relate.context):

M2 relations:

- none above threshold.

Raw top-10 M2 rows (score / threshold / pass):

1. bowl [0] --(contained in)--> suitcase [1] score=0.2727 thr=0.5697 pass=False
2. suitcase [1] --(contained in)--> bowl [0] score=0.2683 thr=0.5697 pass=False

# test_photos/empty_3.jpg -- Page 54 / 61

Frame: 482x640 | Detections: 5 (M2 boxes: 5) | M2 2x identical: True



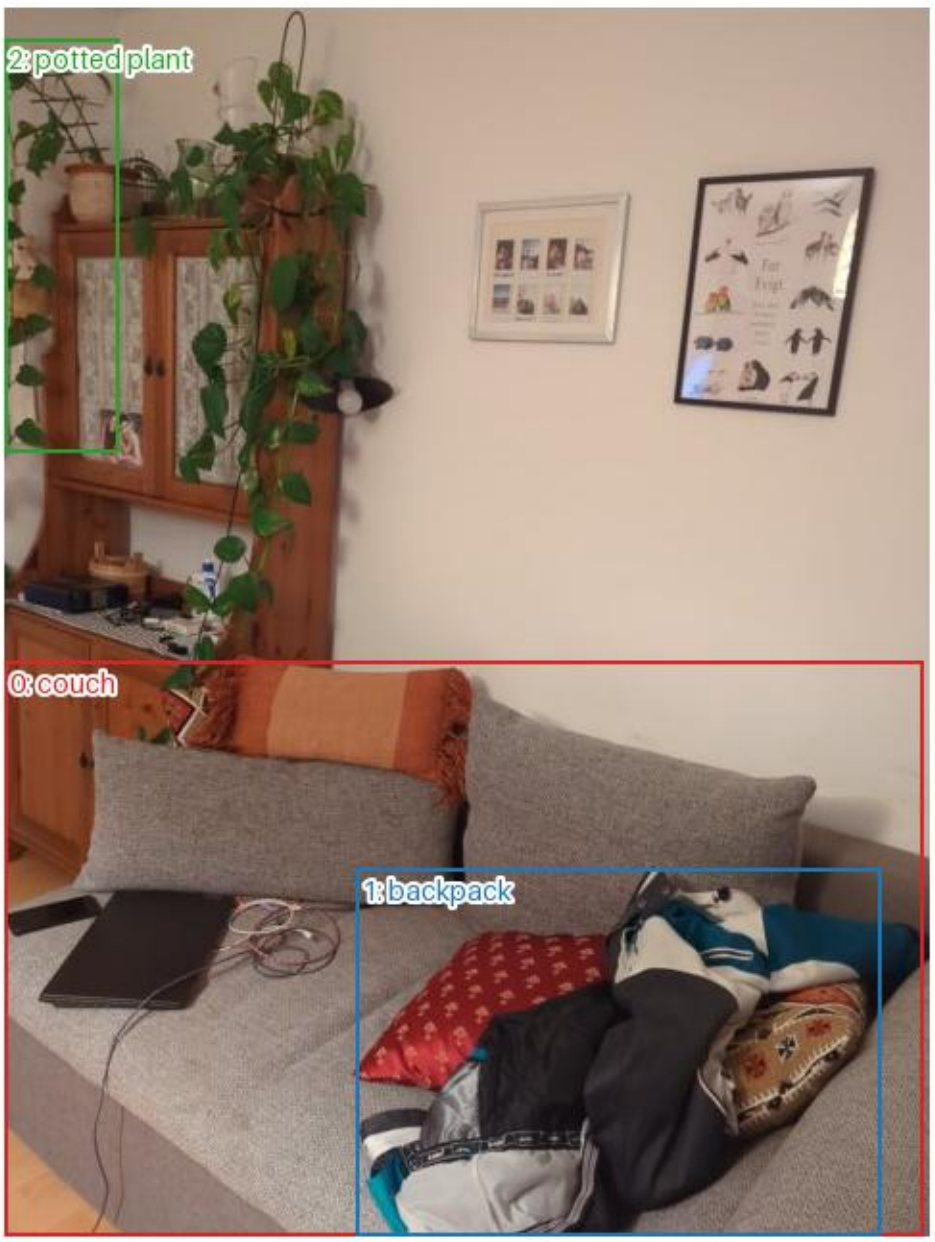

Shown M2 text (relate.context):

M2 relations:
- potted plant [1] behind oven [4].

Raw top-10 M2 rows (score / threshold / pass):

1. potted plant [2] --(resting on)--> potted plant [1] score=0.3794 thr=0.5001 pass=False
2. bottle [3] --(resting on)--> bottle [0] score=0.3249 thr=0.5001 pass=False
3. bottle [0] --(contained in)--> bottle [3] score=0.3230 thr=0.5697 pass=False
4. potted plant [1] --(beside)--> potted plant [2] score=0.3075 thr=0.5001 pass=False
5. potted plant [1] --(behind)--> oven [4] score=0.3008 thr=0.3004 pass=True
6. bottle [3] --(to the right of)--> potted plant [2] score=0.2112 thr=0.2703 pass=False
7. potted plant [2] --(to the left of)--> bottle [3] score=0.2107 thr=0.2703 pass=False
8. bottle [0] --(to the right of)--> potted plant [2] score=0.1747 thr=0.2703 pass=False
9. potted plant [2] --(behind)--> oven [4] score=0.1712 thr=0.3004 pass=False
10. potted plant [2] --(to the left of)--> bottle [0] score=0.1690 thr=0.2703 pass=False

Frame: 482x640 | Detections: 3 (M2 boxes: 3) | M2 2x identical: True



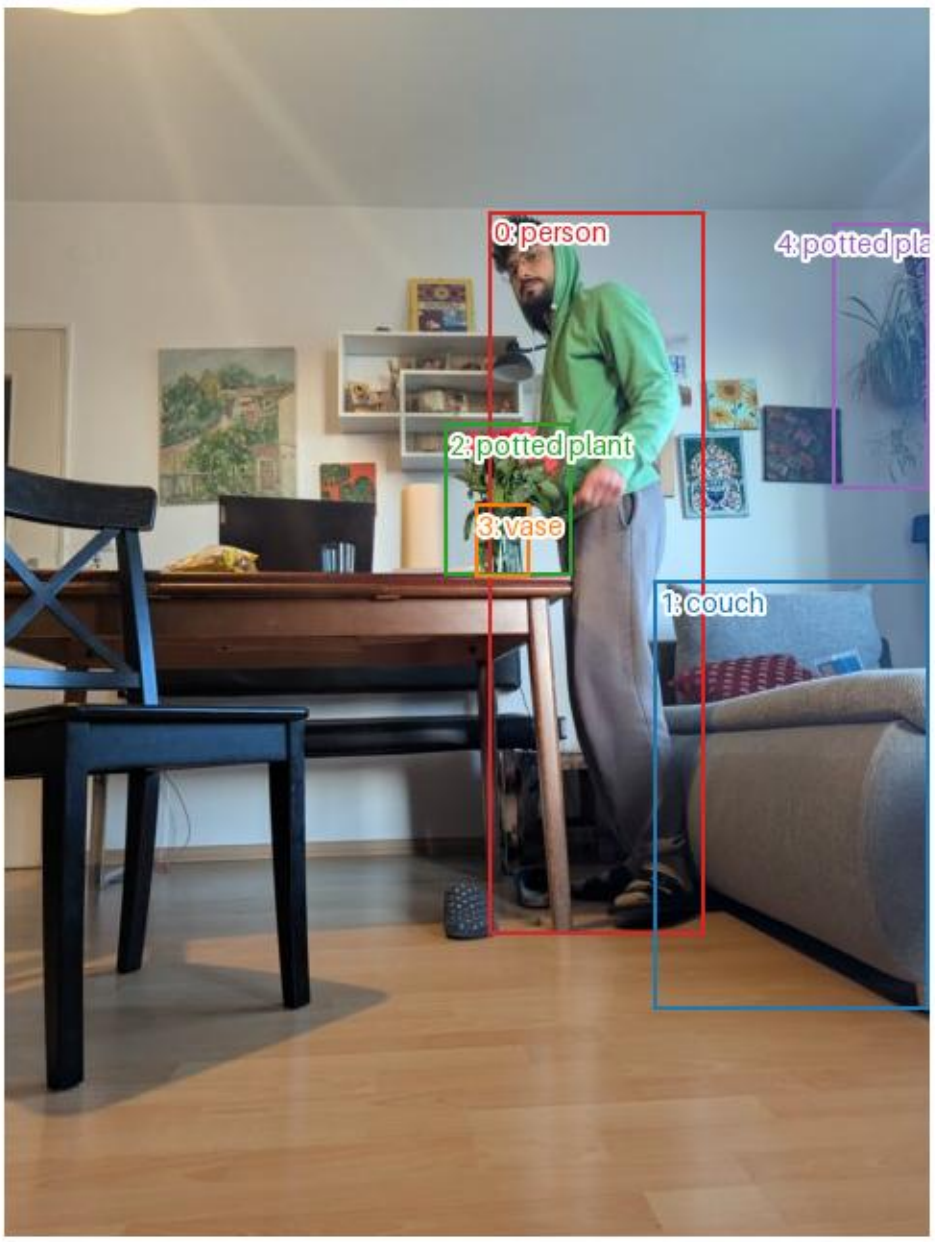

Shown M2 text (relate.context):

M2 relations:
- backpack [1] resting on couch [0].

Raw top-10 M2 rows (score / threshold / pass):

1. backpack [1] --(resting on)--> couch [0] score=0.7141 thr=0.5001 pass=True
2. couch [0] --(holding)--> backpack [1] score=0.4955 thr=0.5001 pass=False
3. potted plant [2] --(above)--> couch [0] score=0.2900 thr=0.3064 pass=False
4. couch [0] --(below)--> potted plant [2] score=0.2676 thr=0.3004 pass=False
5. potted plant [2] --(behind)--> backpack [1] score=0.2621 thr=0.3004 pass=False
6. backpack [1] --(in front of)--> potted plant [2] score=0.2229 thr=0.2748 pass=False

Frame: 482x640 | Detections: 5 (M2 boxes: 5) | M2 2x identical: True



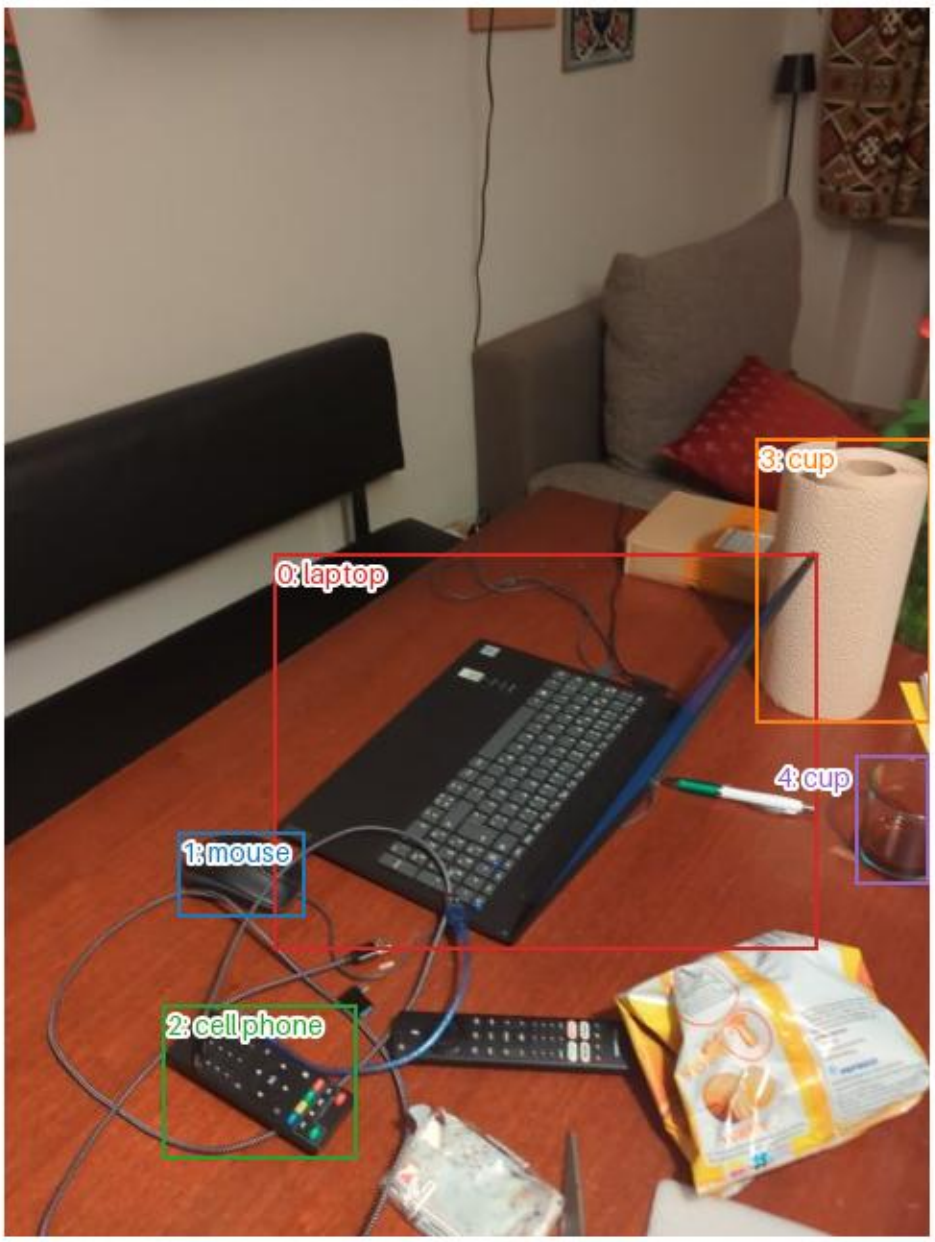

Shown M2 text (relate.context):

M2 relations:
- person [0] leaning against couch [1].
- person [0] holding potted plant [2].
- potted plant [2] contained in vase [3].
- person [0] holding vase [3].

Raw top-10 M2 rows (score / threshold / pass):

1. person [0] --(leaning against)--> couch [1] score=0.6921 thr=0.5308 pass=True
2. person [0] --(holding)--> potted plant [2] score=0.6783 thr=0.5001 pass=True
3. potted plant [2] --(contained in)--> vase [3] score=0.6143 thr=0.5697 pass=True
4. person [0] --(holding)--> vase [3] score=0.5218 thr=0.5001 pass=True
5. vase [3] --(holding)--> potted plant [2] score=0.4128 thr=0.5001 pass=False
6. potted plant [4] --(beside)--> couch [1] score=0.3409 thr=0.5001 pass=False
7. couch [1] --(beside)--> person [0] score=0.3175 thr=0.5001 pass=False
8. person [0] --(beside)--> potted plant [4] score=0.3156 thr=0.5001 pass=False
9. couch [1] --(below)--> potted plant [4] score=0.2496 thr=0.3004 pass=False
10. potted plant [4] --(to the right of)--> person [0] score=0.1991 thr=0.2703 pass=False

Frame: 482x640 | Detections: 5 (M2 boxes: 5) | M2 2x identical: True



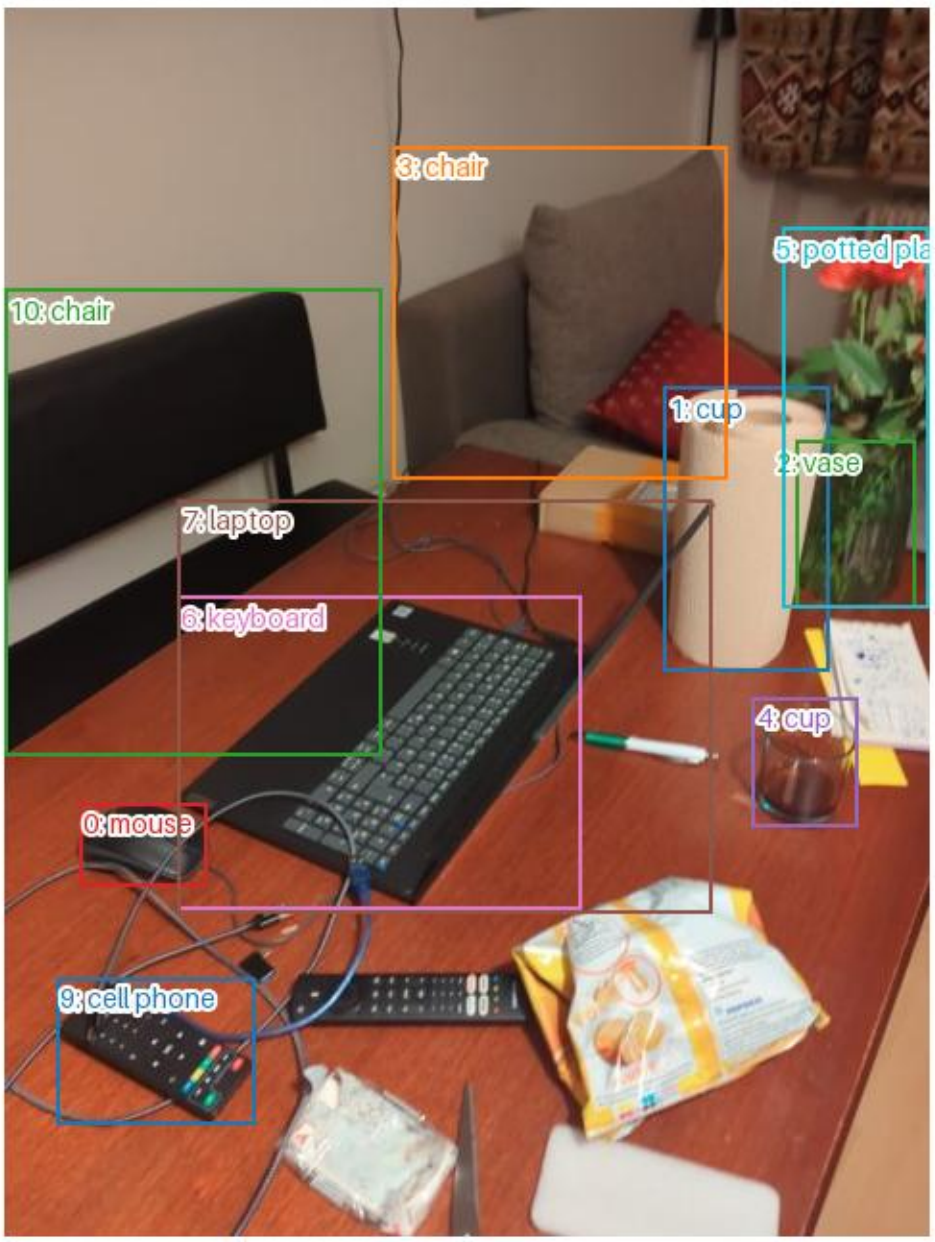

Shown M2 text (relate.context):

M2 relations:
- laptop [0] behind cell phone [2].
- cup [3] above cup [4].
- cell phone [2] in front of laptop [0].
- laptop [0] to the right of mouse [1].

Raw top-10 M2 rows (score / threshold / pass):

1. cup [3] --(beside)--> laptop [0] score=0.4261 thr=0.5001 pass=False
2. laptop [0] --(behind)--> cell phone [2] score=0.4144 thr=0.3004 pass=True
3. laptop [0] --(resting on)--> cup [3] score=0.3995 thr=0.5001 pass=False
4. cup [4] --(beside)--> cup [3] score=0.3889 thr=0.5001 pass=False
5. cup [3] --(above)--> cup [4] score=0.3792 thr=0.3064 pass=True
6. cup [4] --(beside)--> laptop [0] score=0.3678 thr=0.5001 pass=False
7. mouse [1] --(beside)--> laptop [0] score=0.3220 thr=0.5001 pass=False
8. cell phone [2] --(in front of)--> laptop [0] score=0.3196 thr=0.2748 pass=True
9. mouse [1] --(above)--> cell phone [2] score=0.2921 thr=0.3064 pass=False
10. laptop [0] --(to the right of)--> mouse [1] score=0.2878 thr=0.2703 pass=True

Frame: 482x640 | Detections: 11 (M2 boxes: 11) | M2 2x identical: True



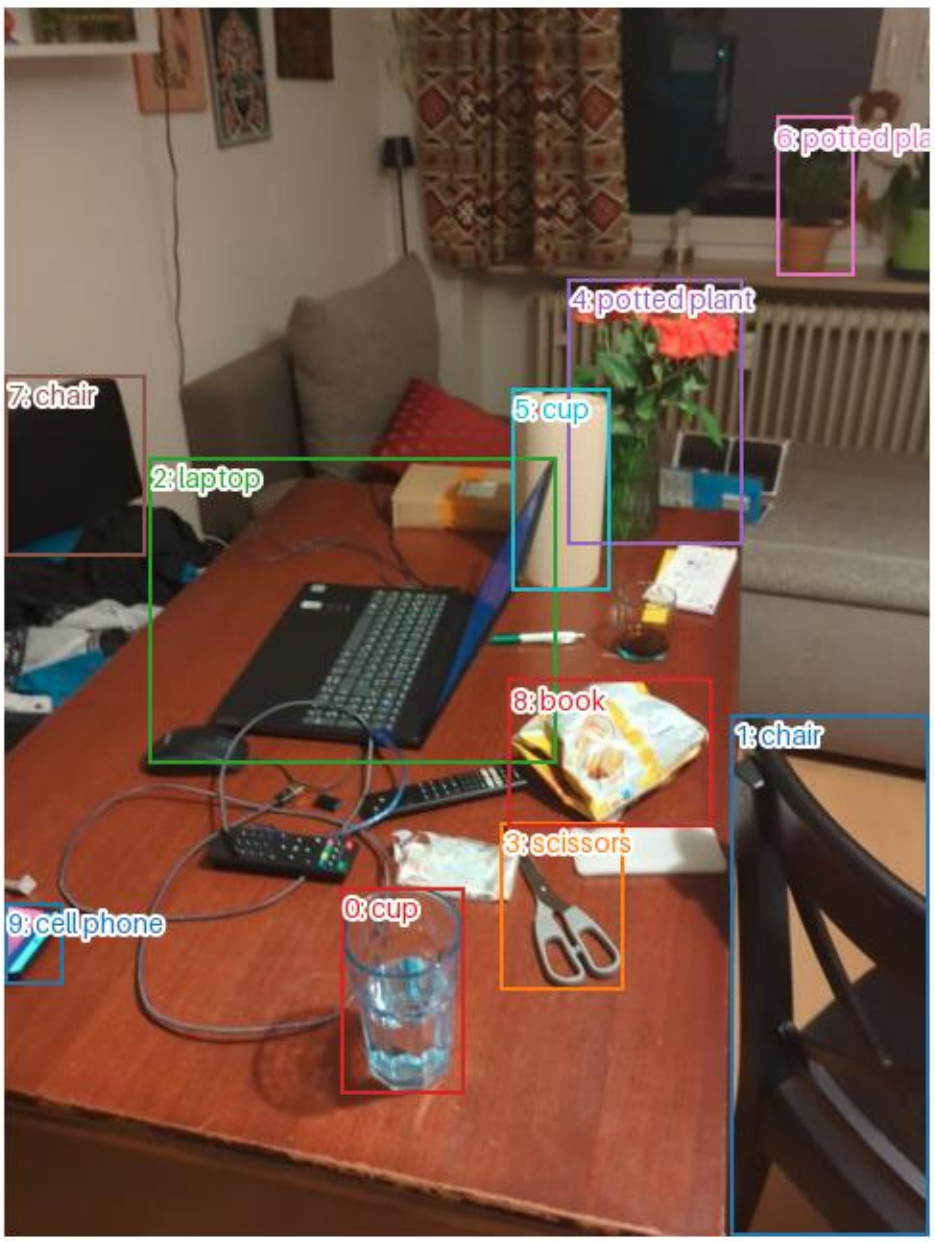

Shown M2 text (relate.context):

M2 relations:
- potted plant [5] contained in vase [2].
- vase [2] holding potted plant [5].
- keyboard [6] attached to laptop [7].
- laptop [7] resting on keyboard [6].

Raw top-10 M2 rows (score / threshold / pass):

1. potted plant [5] --(contained in)--> vase [2] score=0.7166 thr=0.5697 pass=True
2. vase [2] --(holding)--> potted plant [5] score=0.6109 thr=0.5001 pass=True
3. keyboard [6] --(attached to)--> laptop [7] score=0.5512 thr=0.4749 pass=True
4. laptop [7] --(resting on)--> keyboard [6] score=0.5080 thr=0.5001 pass=True
5. cup [1] --(resting on)--> vase [2] score=0.4418 thr=0.5001 pass=False
6. cup [1] --(holding)--> potted plant [5] score=0.3568 thr=0.5001 pass=False
7. cell phone [9] --(attached to)--> remote [8] score=0.3233 thr=0.4749 pass=False
8. cup [1] --(resting on)--> laptop [7] score=0.2989 thr=0.5001 pass=False
9. mouse [0] --(attached to)--> keyboard [6] score=0.2983 thr=0.4749 pass=False
10. remote [8] --(attached to)--> cell phone [9] score=0.2976 thr=0.4749 pass=False

Frame: 482x640 | Detections: 10 (M2 boxes: 10) | M2 2x identical: True



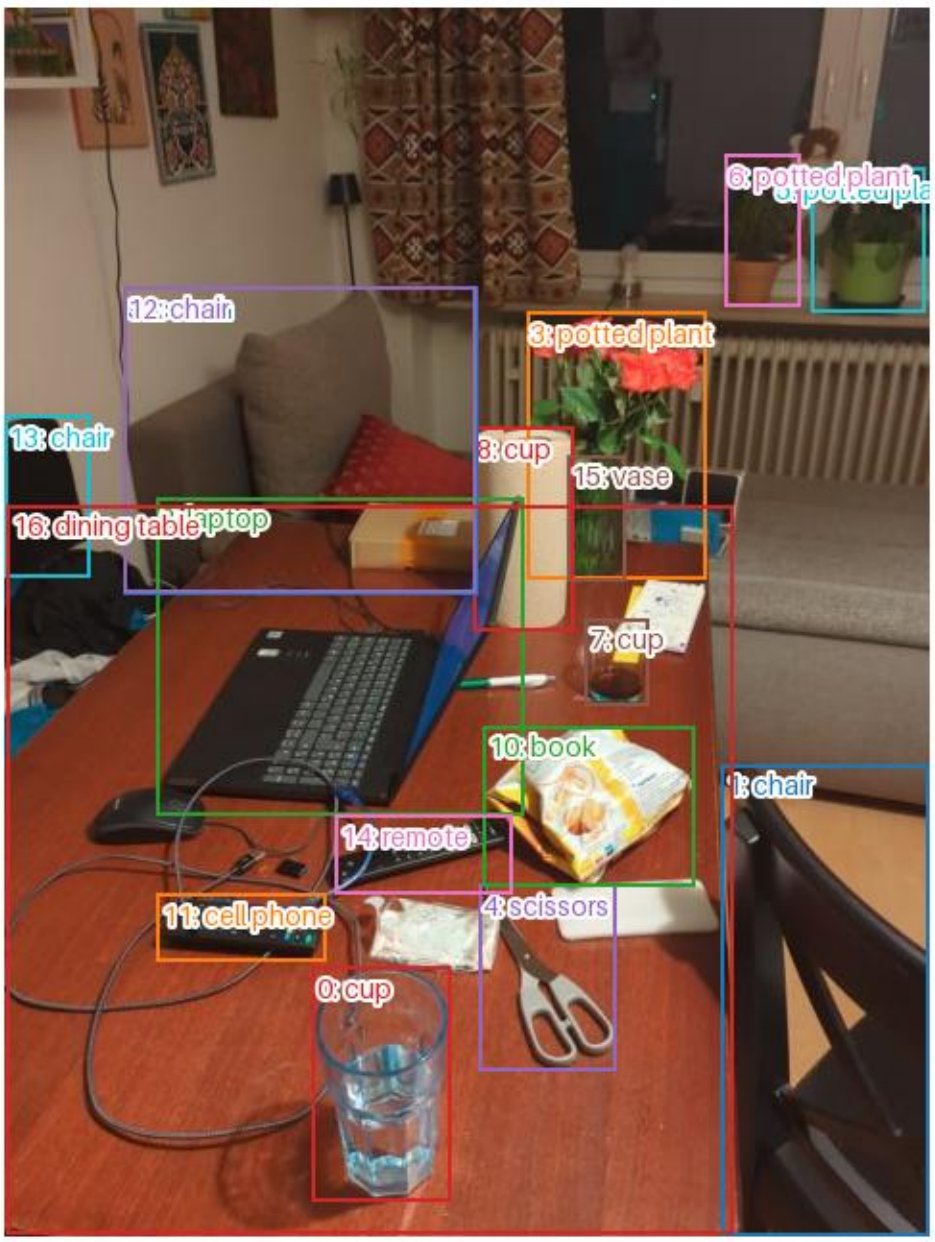

Shown M2 text (relate.context):

M2 relations:
- cup [0] in front of laptop [2].

Raw top-10 M2 rows (score / threshold / pass):

1. cup [5] --(holding)--> potted plant [4] score=0.4935 thr=0.5001 pass=False
2. scissors [3] --(resting on)--> book [8] score=0.4330 thr=0.5001 pass=False
3. potted plant [4] --(beside)--> cup [5] score=0.4066 thr=0.5001 pass=False
4. cup [5] --(resting on)--> laptop [2] score=0.3973 thr=0.5001 pass=False
5. laptop [2] --(resting on)--> book [8] score=0.3971 thr=0.5001 pass=False
6. book [8] --(resting on)--> laptop [2] score=0.3587 thr=0.5001 pass=False
7. cup [0] --(in front of)--> laptop [2] score=0.2911 thr=0.2748 pass=True
8. scissors [3] --(beside)--> cup [0] score=0.2747 thr=0.5001 pass=False
9. chair [1] --(surrounding)--> laptop [2] score=0.2720 thr=0.6228 pass=False
10. laptop [2] --(resting on)--> cup [5] score=0.2689 thr=0.5001 pass=False

Frame: 482x640 | Detections: 17 (M2 boxes: 17) | M2 2x identical: True




Shown M2 text (relate.context):

M2 relations:
- potted plant [3] contained in vase [15].
- laptop [2] resting on dining table [16].
- scissors [4] resting on dining table [16].
- remote [14] resting on dining table [16].
- cup [0] resting on dining table [16].

Raw top-10 M2 rows (score / threshold / pass):

1. potted plant [3] --(contained in)--> vase [15] score=0.7035 thr=0.5697 pass=True
2. laptop [2] --(resting on)--> dining table [16] score=0.6914 thr=0.5001 pass=True
3. scissors [4] --(resting on)--> dining table [16] score=0.6911 thr=0.5001 pass=True
4. remote [14] --(resting on)--> dining table [16] score=0.6790 thr=0.5001 pass=True
5. cup [0] --(resting on)--> dining table [16] score=0.6550 thr=0.5001 pass=True
6. book [10] --(resting on)--> dining table [16] score=0.6543 thr=0.5001 pass=True
7. cup [7] --(resting on)--> dining table [16] score=0.6418 thr=0.5001 pass=True
8. cup [8] --(resting on)--> dining table [16] score=0.6393 thr=0.5001 pass=True
9. cell phone [11] --(resting on)--> dining table [16] score=0.5965 thr=0.5001 pass=True
10. chair [1] --(beside)--> dining table [16] score=0.5729 thr=0.5001 pass=True

## Summary: Photos with < 2 boxes (10) -- Page 61 / 61

- bench_photos/empty_5.jpg: 1 box(es) (bed)
- bench_photos/person_facing_040cm.jpg: 1 box(es) (person)
- bench_photos/person_facing_080cm.jpg: 1 box(es) (person)
- bench_photos/roomsig_bed_1.jpg: 1 box(es) (bed)
- bench_photos/roomsig_toilet_3.jpg: 1 box(es) (toilet)
- bench_photos/roomsig_tv_1.jpg: 1 box(es) (tv)
- bench_photos/roomsig_tv_2.jpg: 1 box(es) (tv)
- test_photos/PXL_20260405_213536608.jpg: 1 box(es) (person)
- test_photos/PXL_20260405_213539353.MP.jpg: 1 box(es) (person)
- test_photos/empty_5.jpg: 1 box(es) (bed)